

## CompTIA Server+

# Server Virtualization Concepts

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  - Lab Topology
  - Exercise 1 - Install and Configure a Server-side Virtualization
  - Exercise 2 - Cloud Models
  - Review
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## Introduction

Server+

Virtual Switches

Virtual Machine

Public Cloud

Private Cloud

Hybrid Cloud

Welcome to the **Server Virtualization Concepts** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

## Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Install and Configure a Server-side Virtualization
- Exercise 2 - Cloud Models

After completing this module, you should be able to:

- Install the Microsoft Hyper-V
- Virtual Switching Concepts
- Create a Virtual Machine
- Install Windows Server on a Virtual Machine

After completing this module, you should have further knowledge of:

- Different Cloud Computing Models

## Exam Objectives

The following exam objectives are covered in this module:

### 2.5 Summarize the purpose and operation of virtualization

- Virtual networking
- Resource allocation and provisioning
- Cloud models

## Lab Duration

It will take approximately **1 hour** to complete this lab.

## Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

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## Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)
- **PLABDM01** - (Windows Server 2022 - Domain Member Server)
- **PLABWIN10** - (Windows 10 - Domain Member Workstation)
- **PLABALMA** - (Alma Linux 8.6 - Stand-alone Linux Workstation)
- **PLABUBUNTU** - (Linux Ubuntu - Stand-alone Linux Server)

Click **Next** to proceed to the first exercise.

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## Exercise 1 - Install and Configure a Server-side Virtualization

In this exercise, the learner will install the Microsoft Hyper-V virtualization technology on a Windows Server 2019 host machine. Microsoft's Hyper-V feature is their flagship virtualization software and can be added as an optional feature as it is not installed by default.

In this exercise, you will create different types of virtual switches. You will also create a virtual machine, configure the virtual machine, and install Windows Server 2019.

### Learning Outcomes

After completing this exercise, you should be able to:

- Install the Microsoft Hyper-V
- Virtual Switching Concepts
- Create a Virtual Machine
- Install Windows Server on a Virtual Machine

### Your Devices

You will be using the following devices in this lab. Please power these on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)
- **PLABDM01** - (Windows Server 2022 - Domain Member Server)

### Task 1 - Install the Microsoft Hyper-V

Microsoft Hyper-V is a virtualization technology used to virtualize physical devices and resources, such as creating a virtual machine or virtual switch.

In this task, you will install the Hyper-V virtualization software on a Windows Server 2019 host machine. A virtual machine with the Hyper-V Manager console will then be created.

### Step 1

Connect to **PLABDM01**.

Click the **Start** charm and type the following:

Hyper-V

Select **Turn Windows features on or off** from the **Best match** pop-up menu.

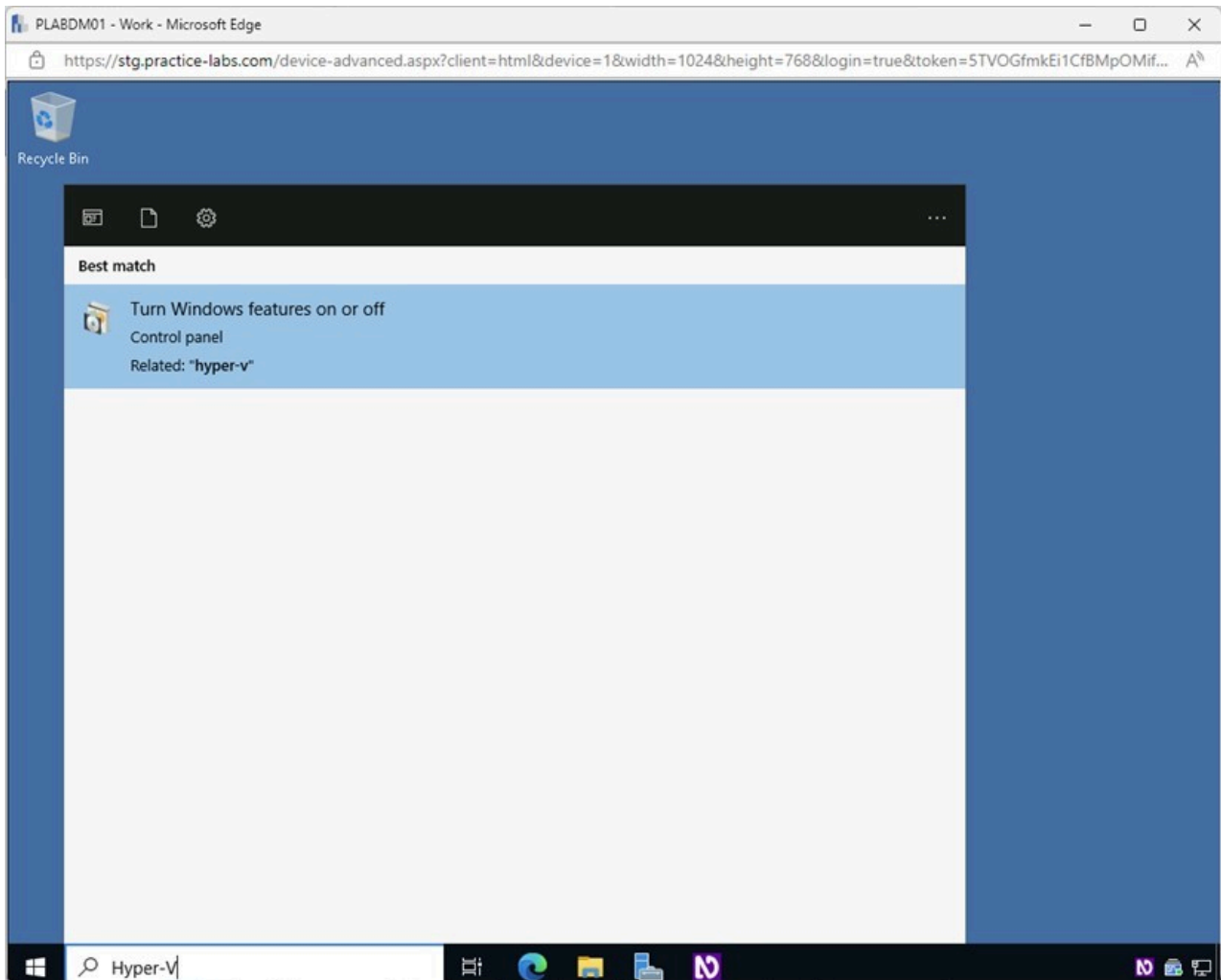


Figure 1.1 Screenshot of PLABDM01: Displaying selecting Turn Windows features on or off from the Best match pop-up menu.

**Note:** Hyper-V is the virtualization technology that is built into Windows servers and client operating systems. However, as a feature, it does require installation to be available.

## Step 2

In the **Add Roles and Features Wizard - Before you begin** page, click **Next**.

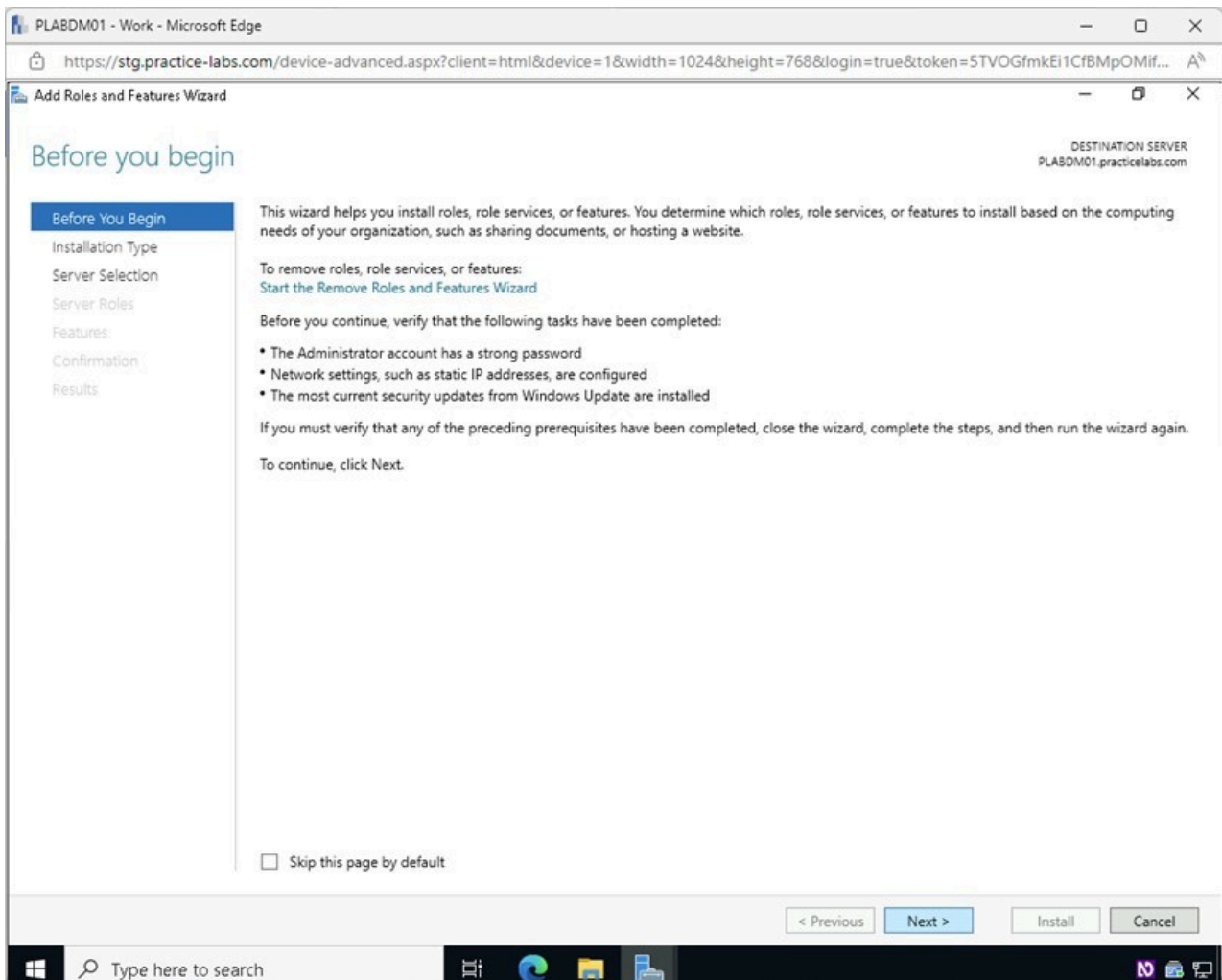


Figure 1.2 Screenshot of PLABDM01: Displaying clicking Next in the Add Roles and Features Wizard - Before you begin page.

### Step 3

In the **Select installation type** page, ensure that **Role-based or feature-based installation** is selected.

Click **Next**.

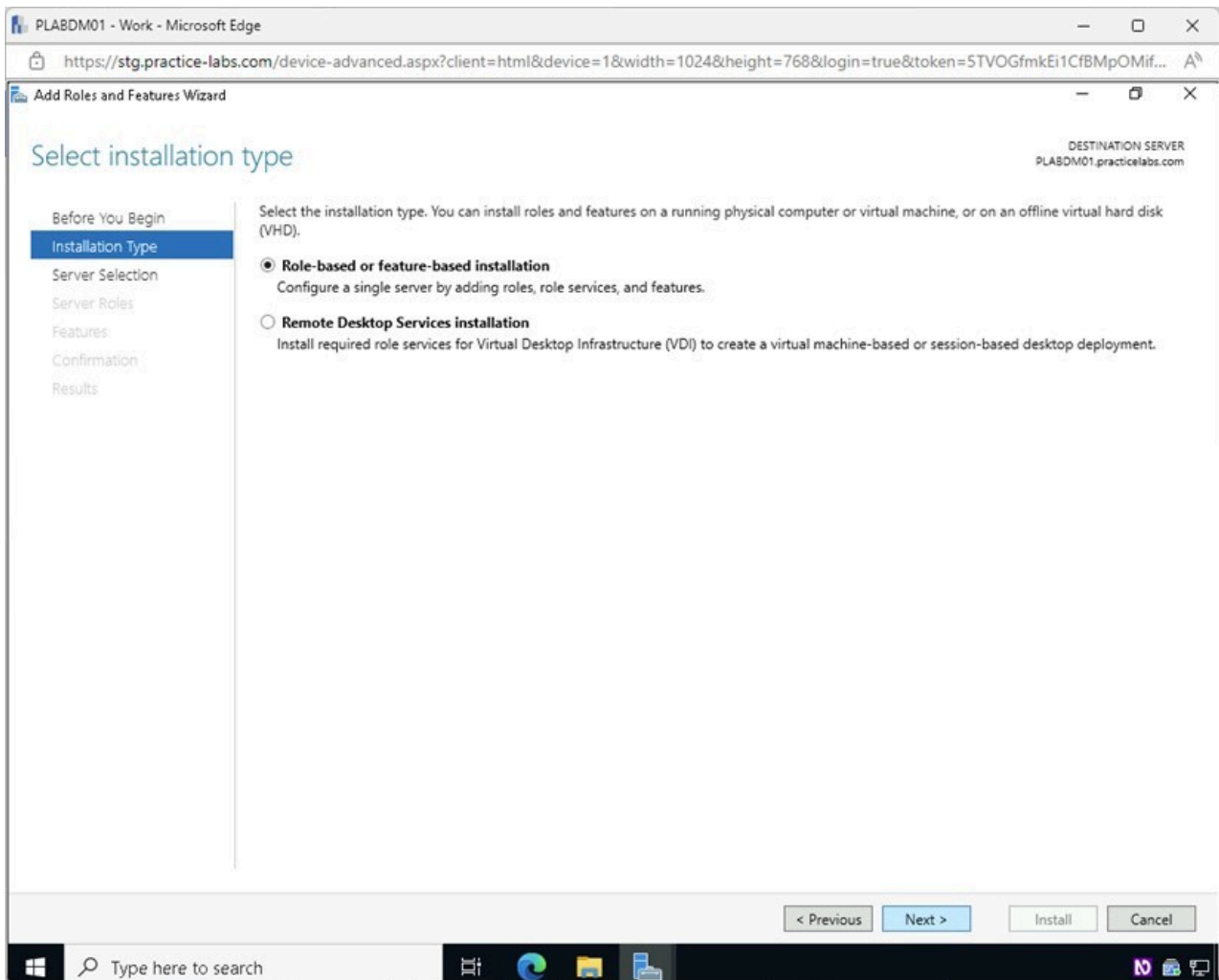


Figure 1.3 Screenshot of PLABDM01: Displaying the Select installation type page with the required option selected and the Next button highlighted.

## Step 4

In the **Select destination server** page, ensure that the **Select a server from the server pool** radio button is selected and that the server **PLABDM01.practicelabs.com** is in the context pane of **Server Pool**.

Click **Next**.

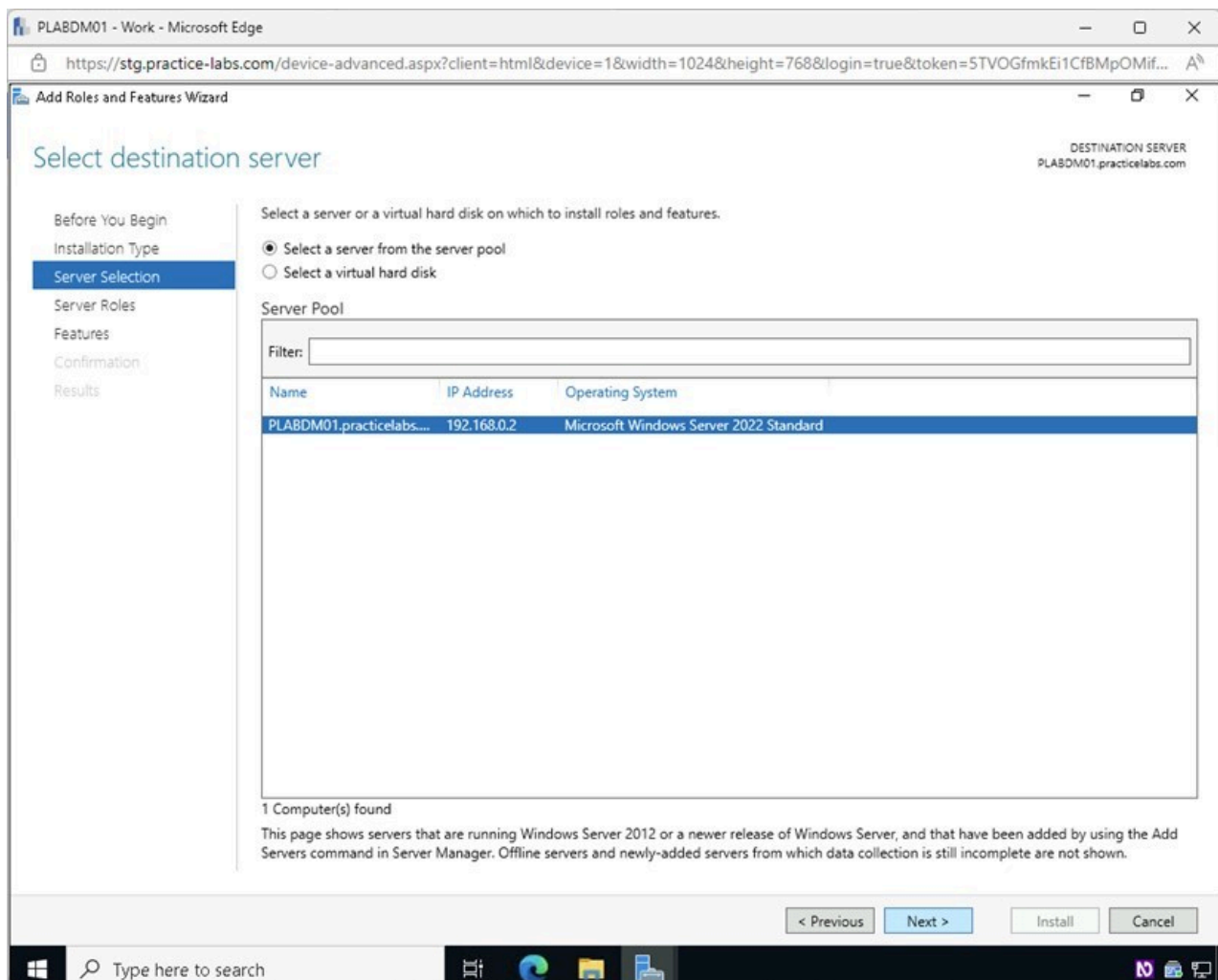


Figure 1.4 Screenshot of PLABDM01: Displaying the Select destination server page with the required option selected and the Next button highlighted.

## Step 5

On the **Select server roles** page, enable the **Hyper-V** checkbox.

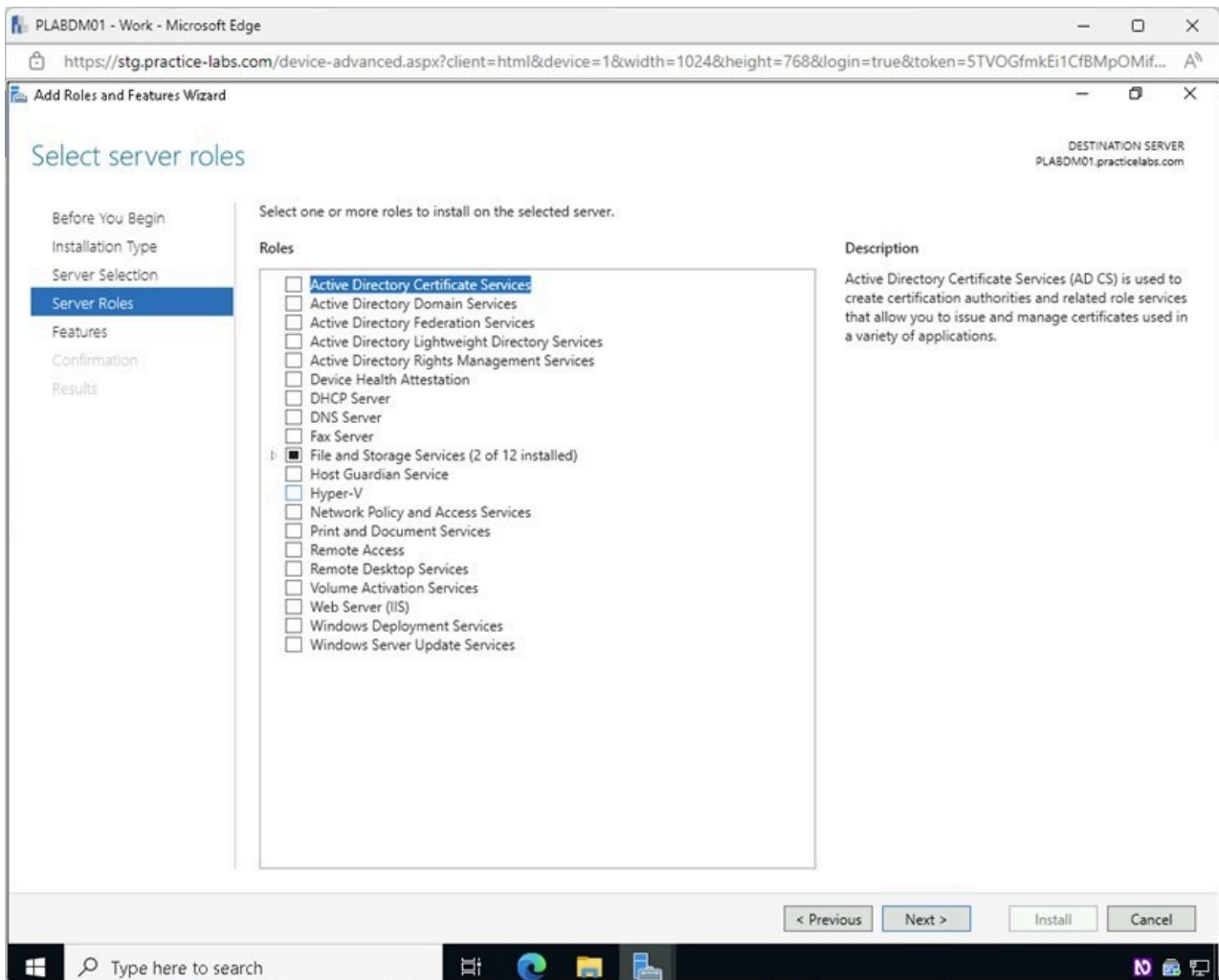


Figure 1.5 Screenshot of PLABDM01: Displaying the Select server roles page with Hyper-V selected.

## Step 6

In the **Add features that are required for Hyper-V?** dialog box, click **Add Features**.

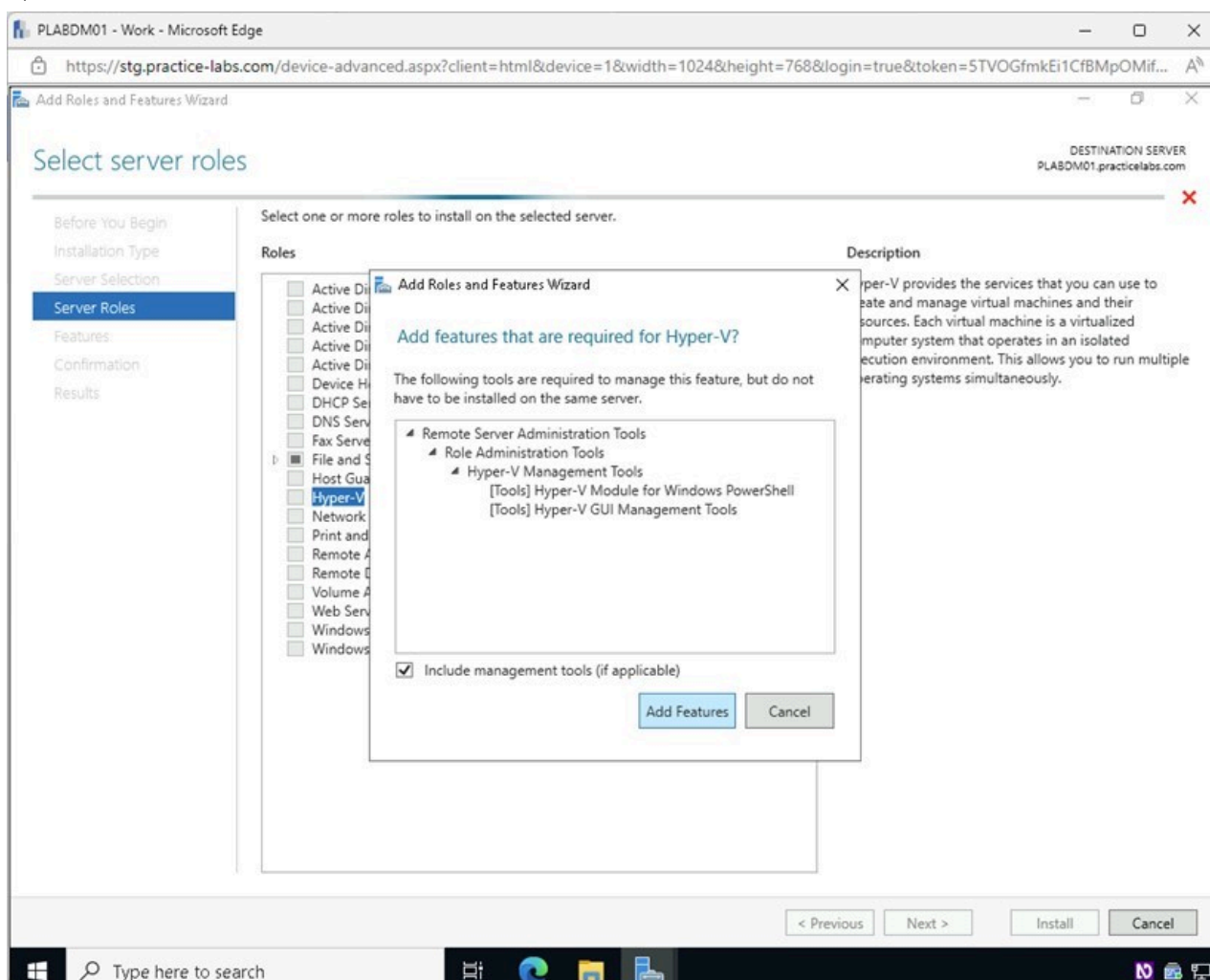


Figure 1.6 Screenshot of PLABDM01: Displaying the Add features that are required for Hyper-V? dialog box with the Add Features button highlighted.

## Step 7

On the **Select server roles** page, click **Next**.

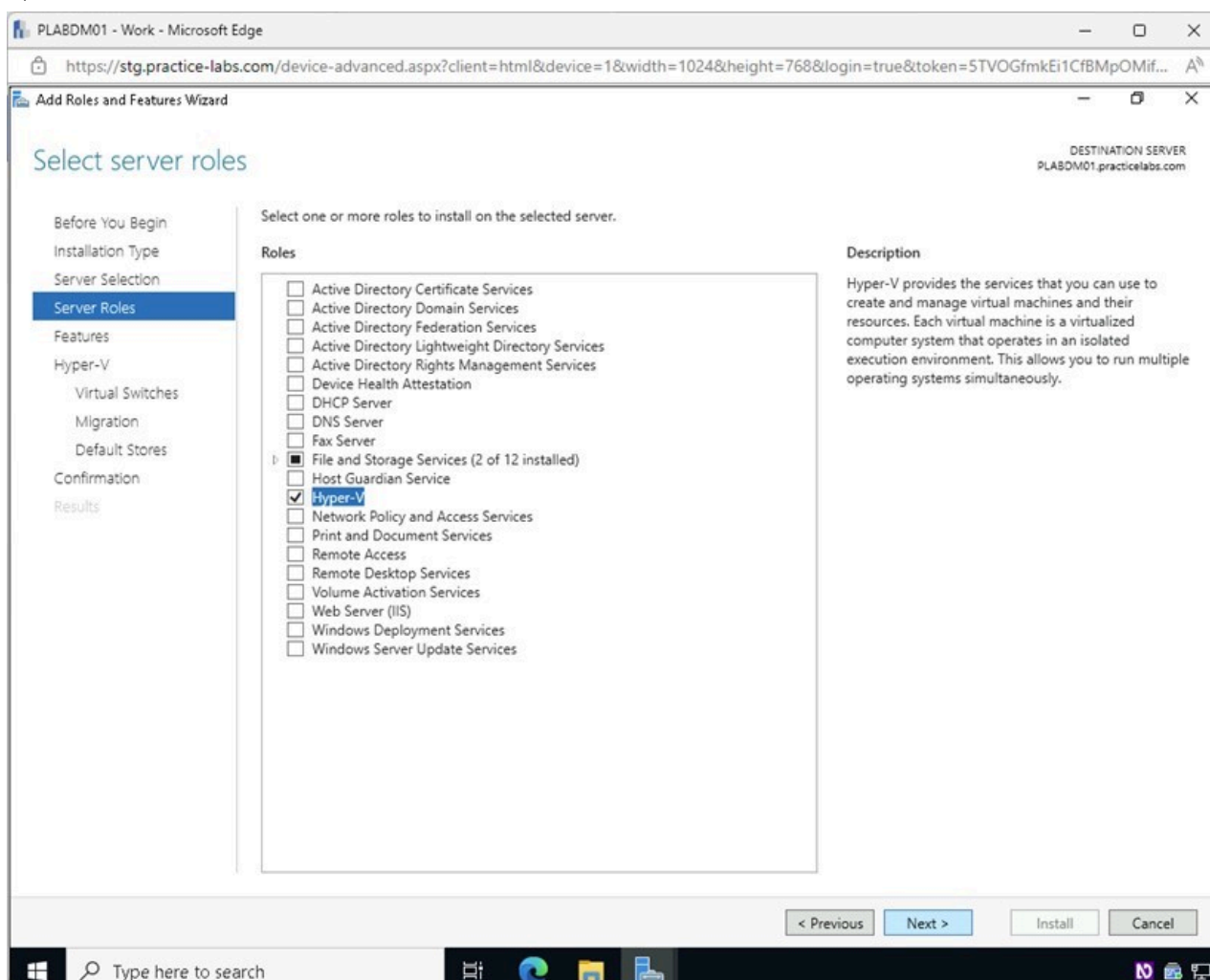


Figure 1.7 Screenshot of PLABDM01: Displaying clicking Next on the Select server roles page.

## Step 8

In the **Select features** page, click **Next**.

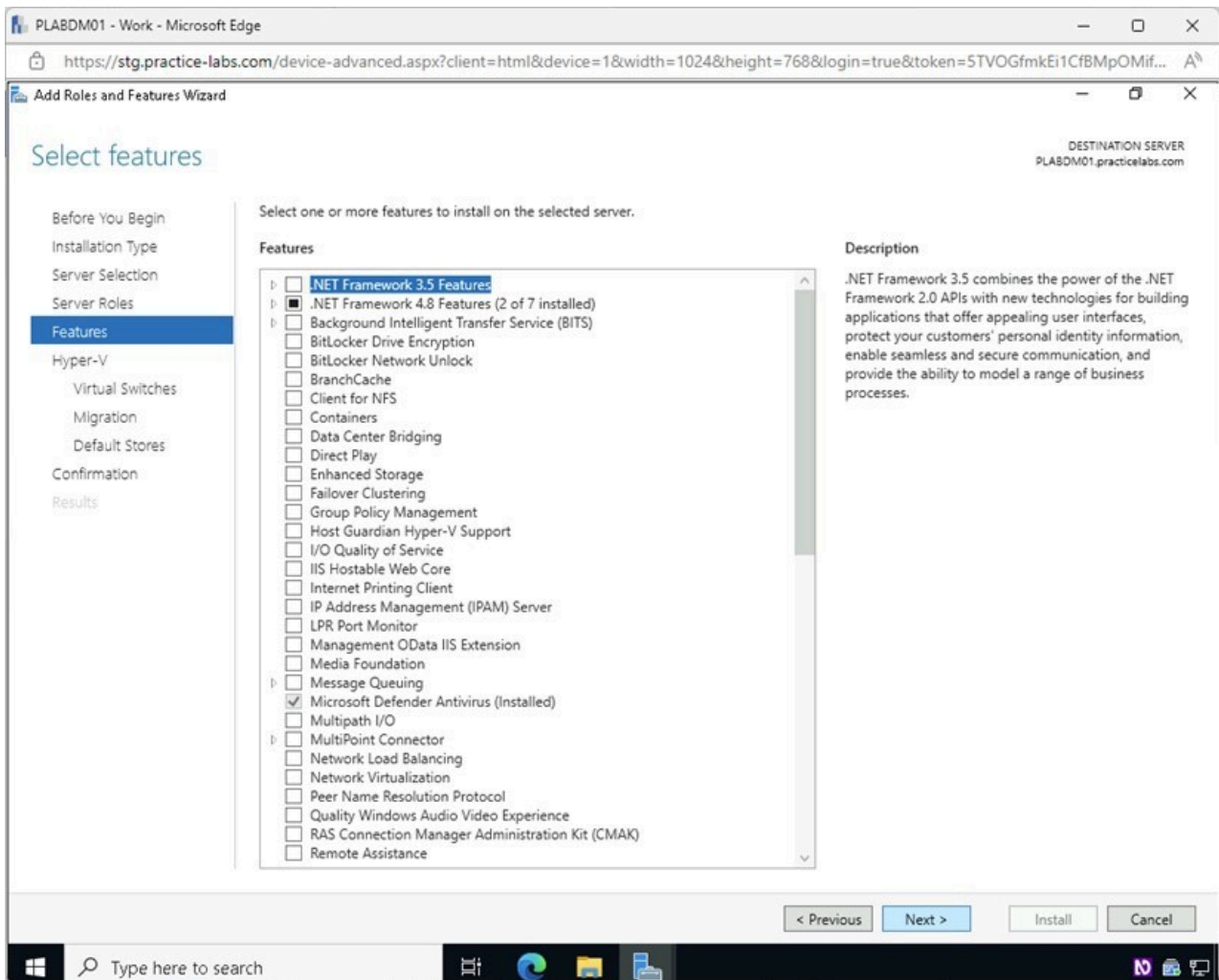


Figure 1.8 Screenshot of PLABDM01: Displaying clicking Next on the Select features page.

## Step 9

On the **Hyper-V** page, click **Next**.

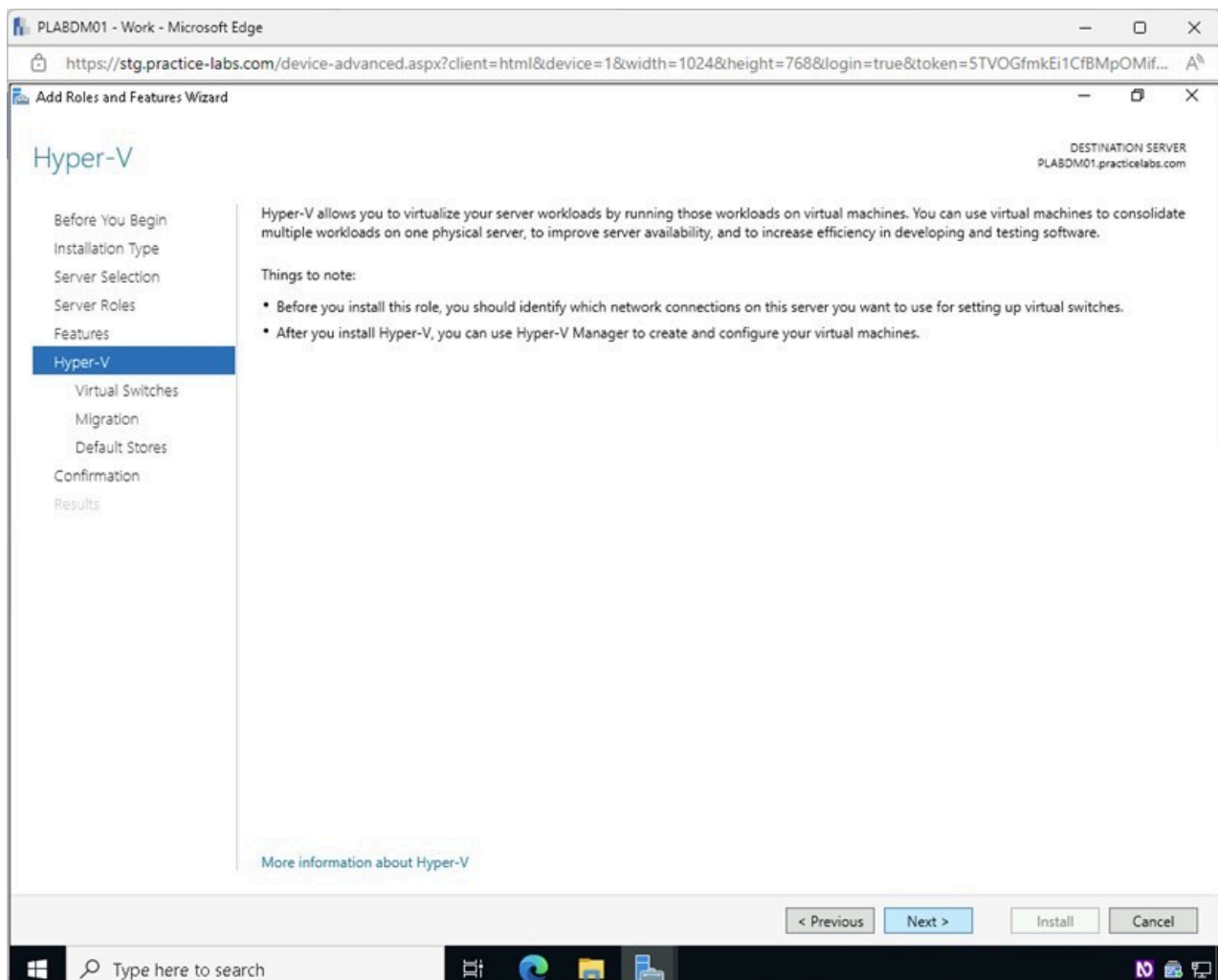


Figure 1.9 Screenshot of PLABDM01: Displaying clicking Next on the Hyper-V page.

## Step 10

In the **Create Virtual Switches** page, click **Next**.

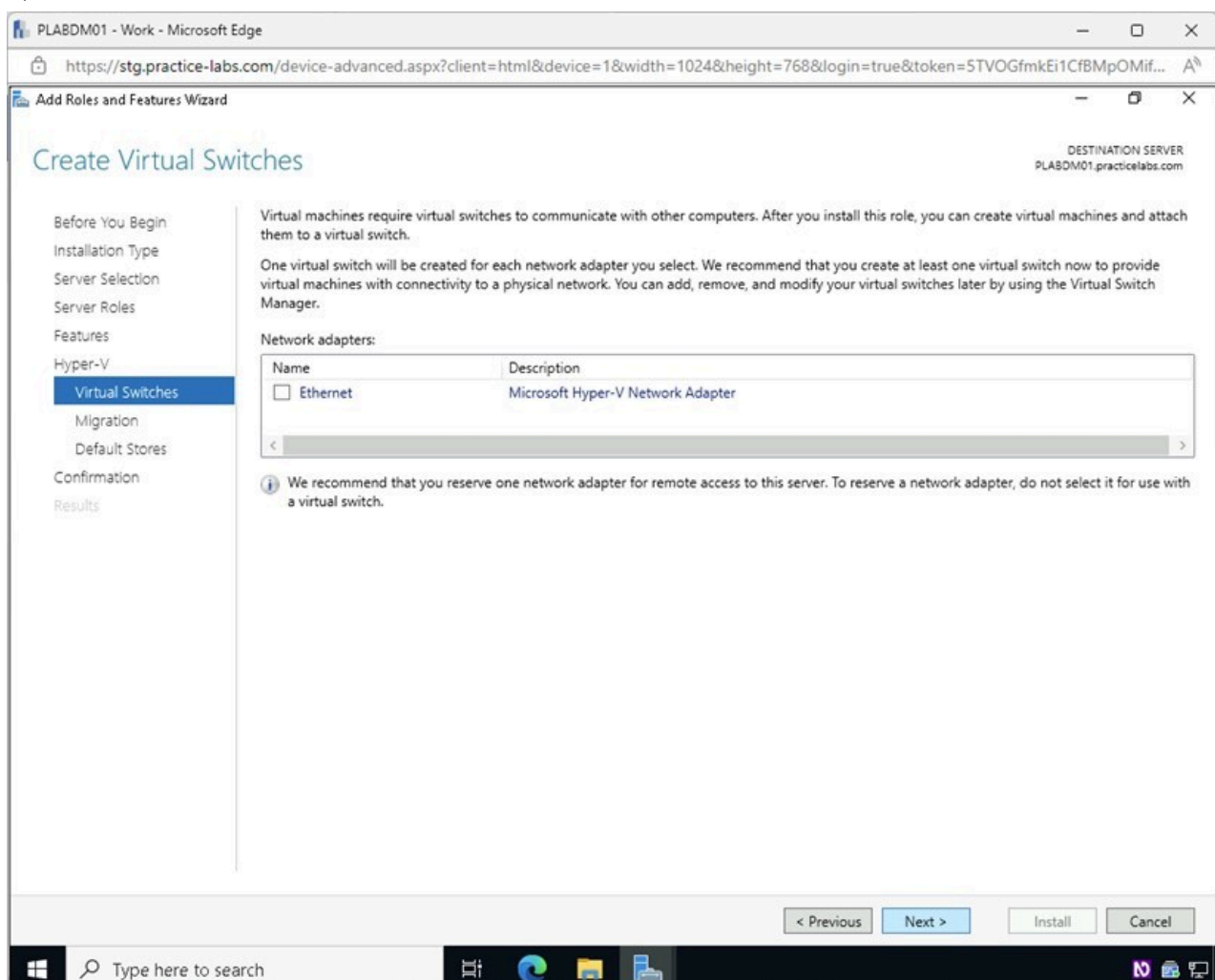


Figure 1.10 Screenshot of PLABDM01: Displaying clicking Next on the Create Virtual Switches page.

## Step 11

On the **Virtual Machine Migration** page, click **Next**.

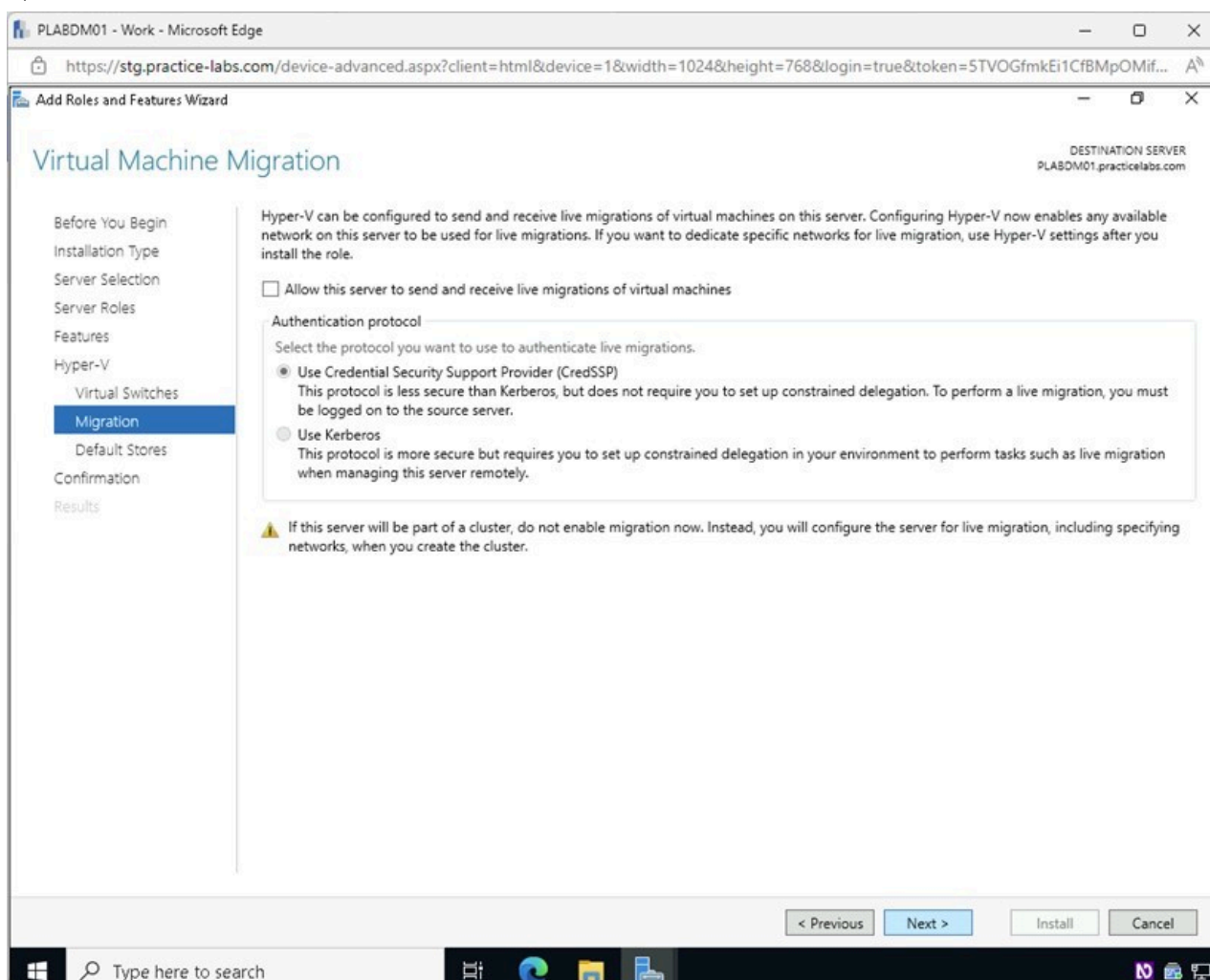


Figure 1.11 Screenshot of PLABDM01: Displaying clicking Next on the Virtual Machine Migration page.

## Step 12

From the **Default Stores** page, click **Next**.

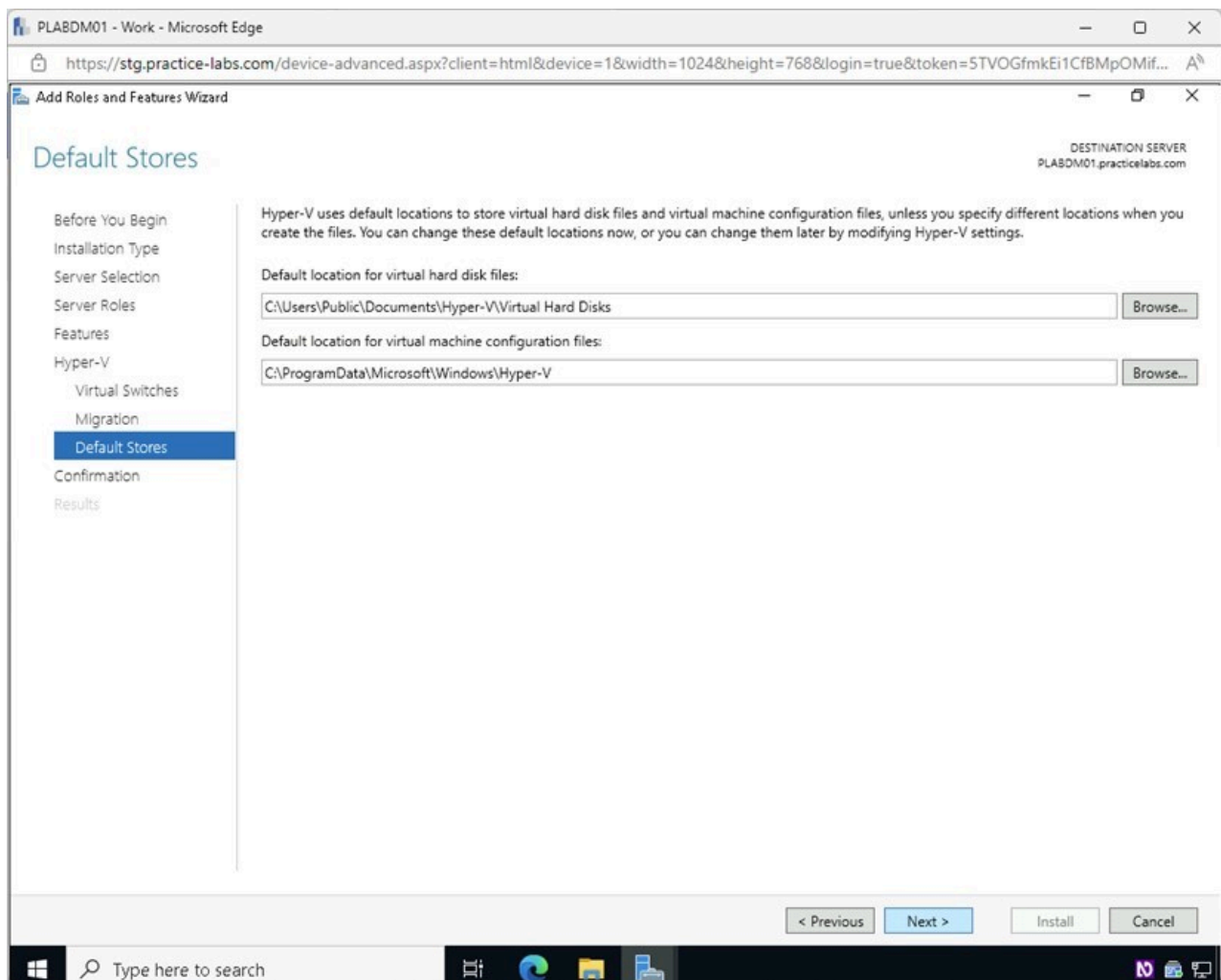


Figure 1.12 Screenshot of PLABDM01: Displaying clicking Next on the Default Stores page.

## Step 13

On the **Confirm installation selections** page, click **Install**.

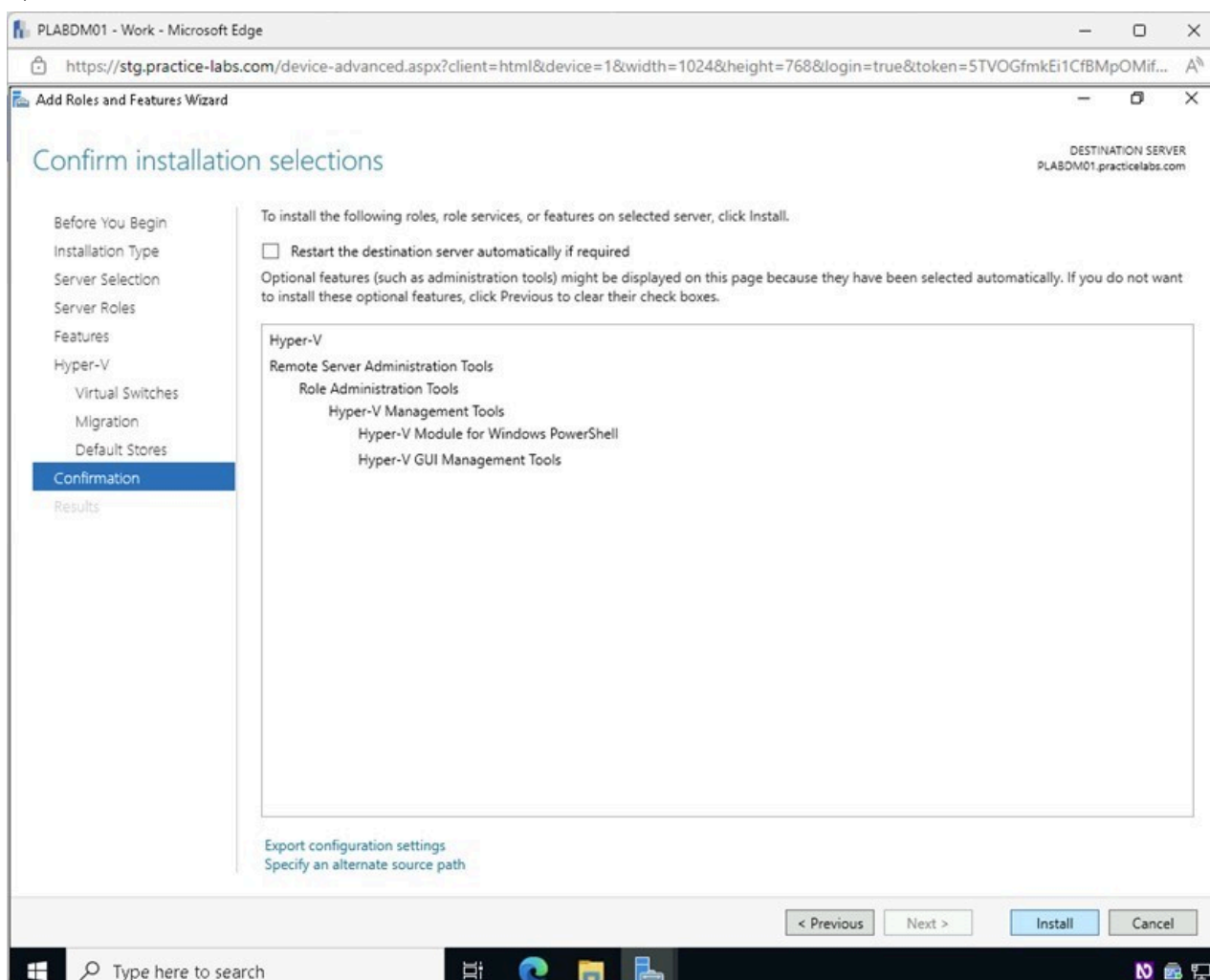


Figure 1.13 Screenshot of PLABDM01: Displaying clicking Install on the Confirm installation selections page.

## Step 14

On the **Installation progress** page, observe the **Feature installation** progress.

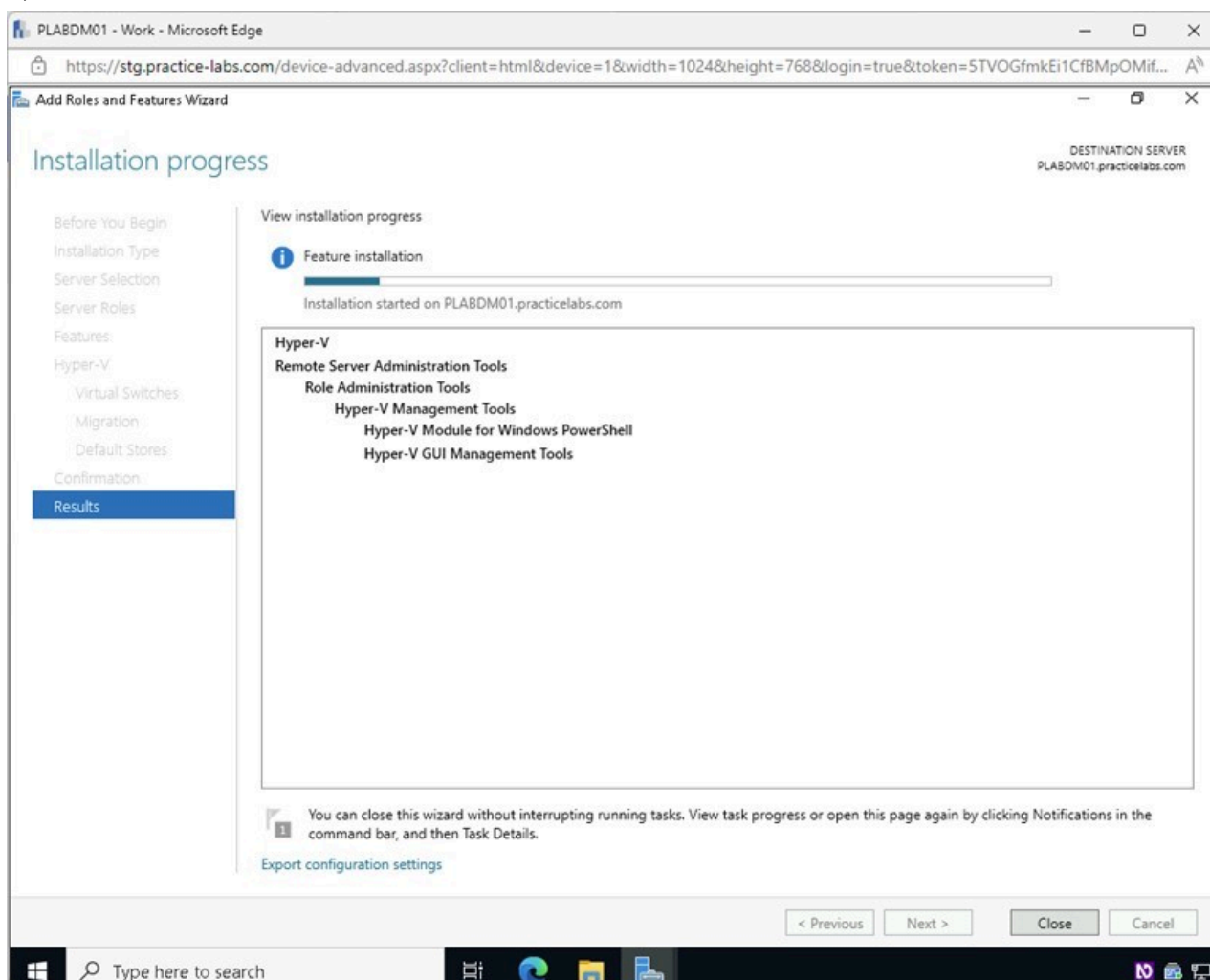


Figure 1.14 Screenshot of PLABDM01: Displaying the feature installation progress on the Installation progress page.

## Step 15

On the **Installation progress** page, click **Close** once the feature installation is complete.

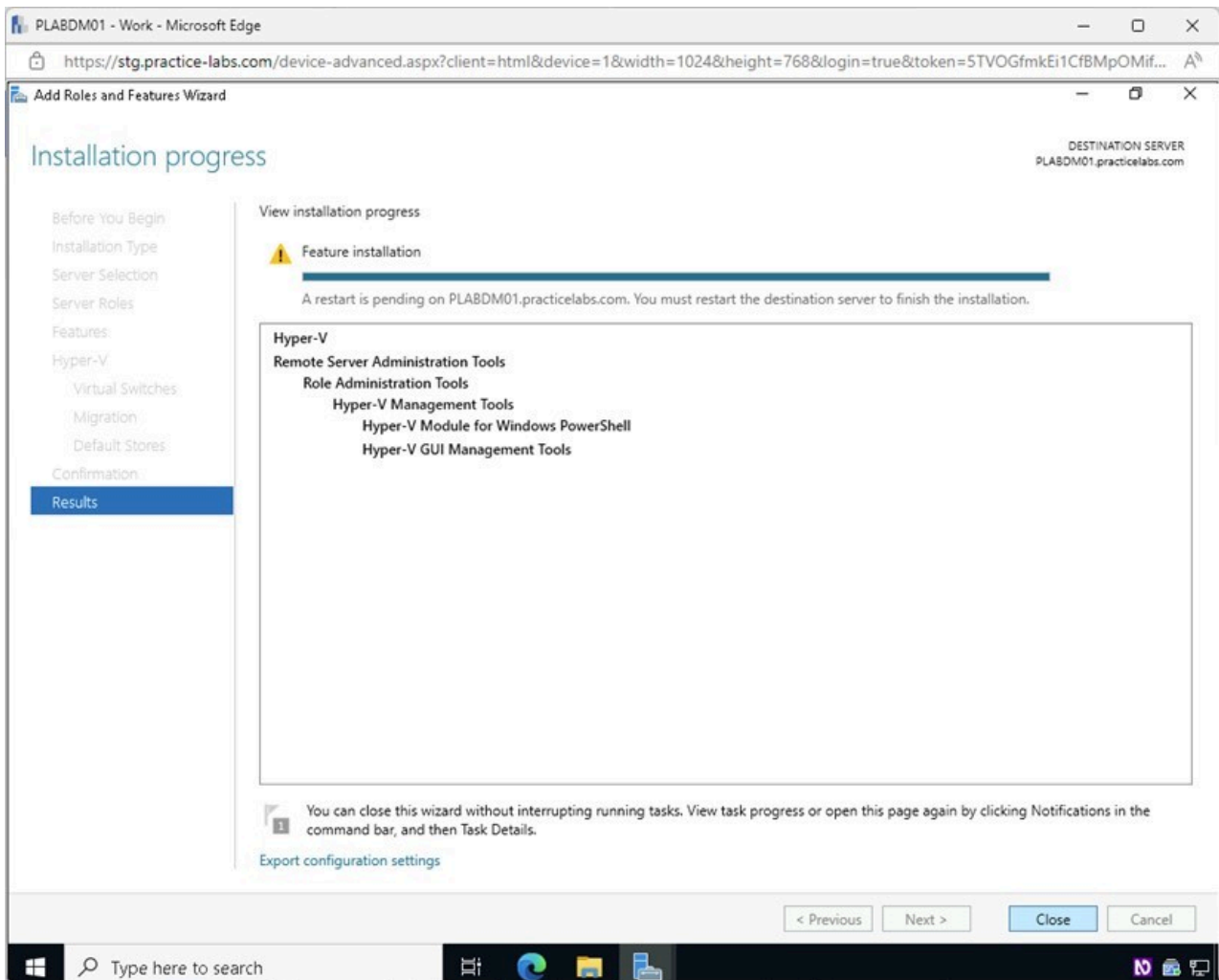


Figure 1.15 Screenshot of PLABDM01: Displaying clicking Close on the Installation progress page.

## Step 16

Minimize the **Server Manager** window.

On the **Desktop**, right-click the **Start** charm and select **Shut down or sign out > Restart** menu options.

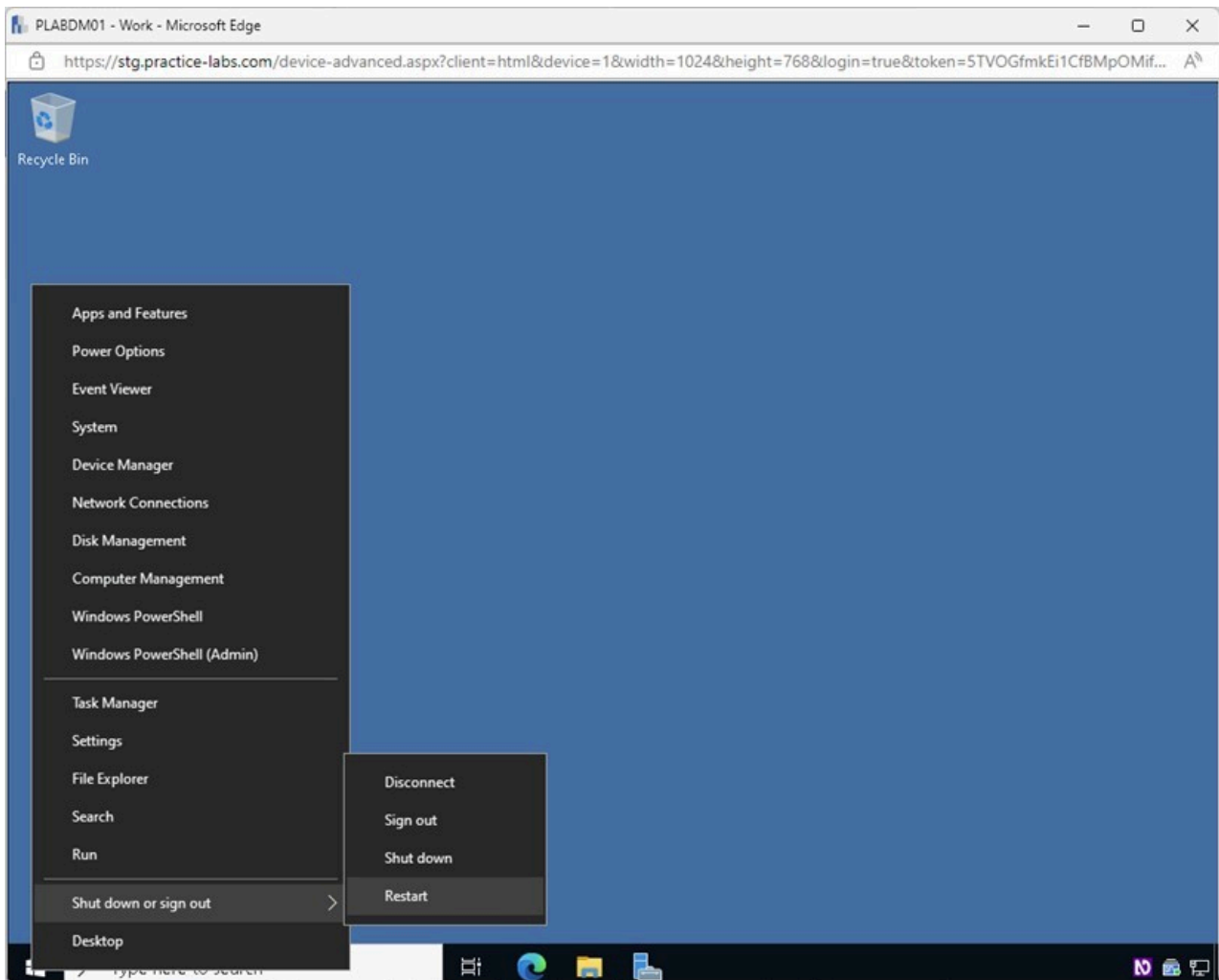


Figure 1.16 Screenshot of PLABDM01: Displaying right-clicking the Start charm and selecting Shut down or sign out > Restart.

## Step 17

In the **Choose a reason that best describes why you want to shut down this computer** dialog box, select **Operating System: Recovery (Planned)** from the drop-down menu.

Click **Continue**.

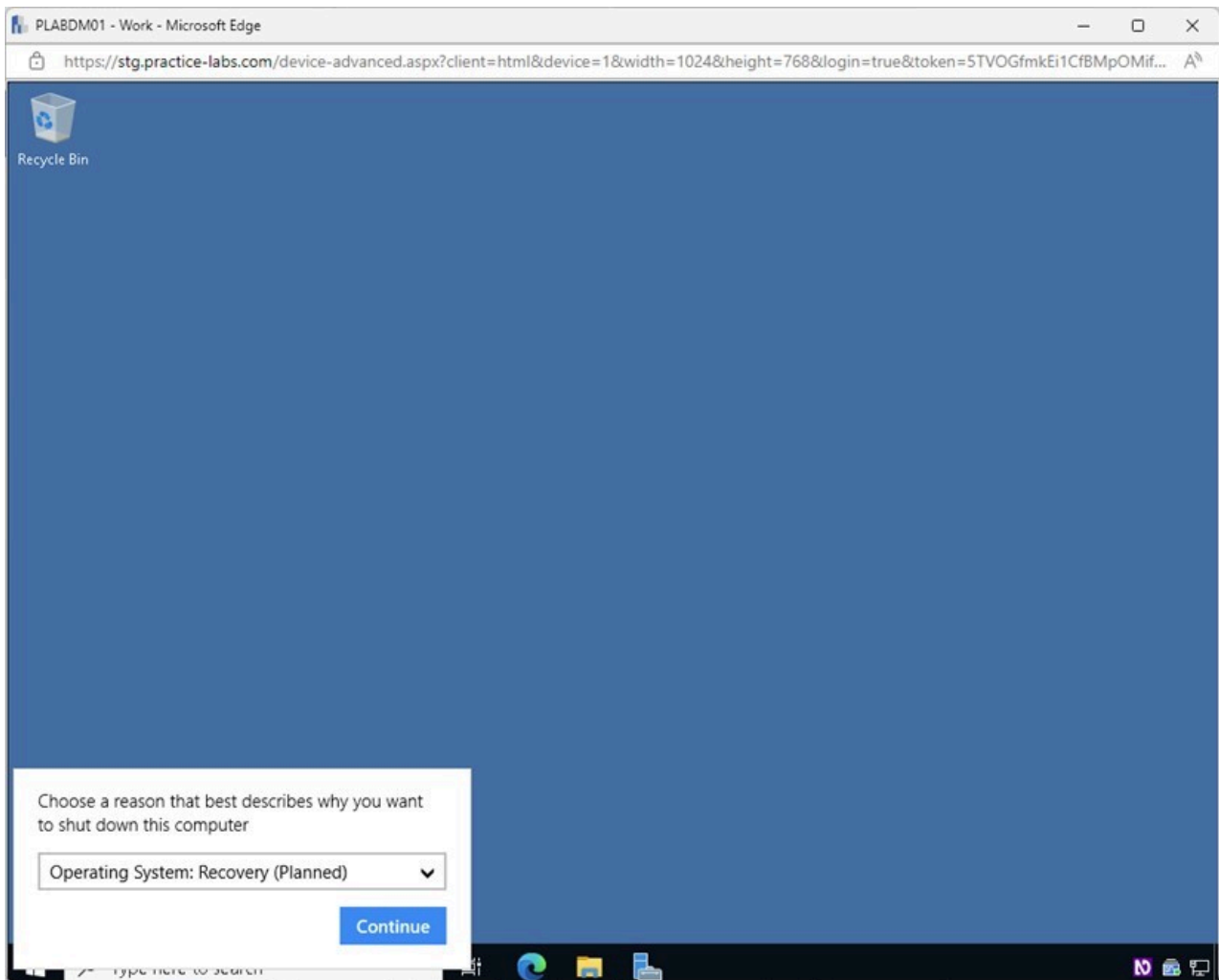


Figure 1.17 Screenshot of PLABDM01: Displaying the Choose a reason that best describes why you want to shut down this computer dialog box with the required option selected and the Continue button highlighted.

**Note:** You might have to close and re-open the **PLABDM01** virtual machine window. When you re-open the window, you will see the installation in progress.

## Task 2 - Virtual Switching Concepts

The Microsoft Hyper-V application has the ability to virtualize a virtual switch. These virtual switches can be used to connect physical and virtual networking devices.

In this task, different types of virtual switches will be explored.

### Step 1

Ensure you are connected to **PLABDM01**.

Click the **Start** charm and type the following:

# Hyper-V Manager

Select **Hyper-V Manager** from the **Best match** pop-up menu.

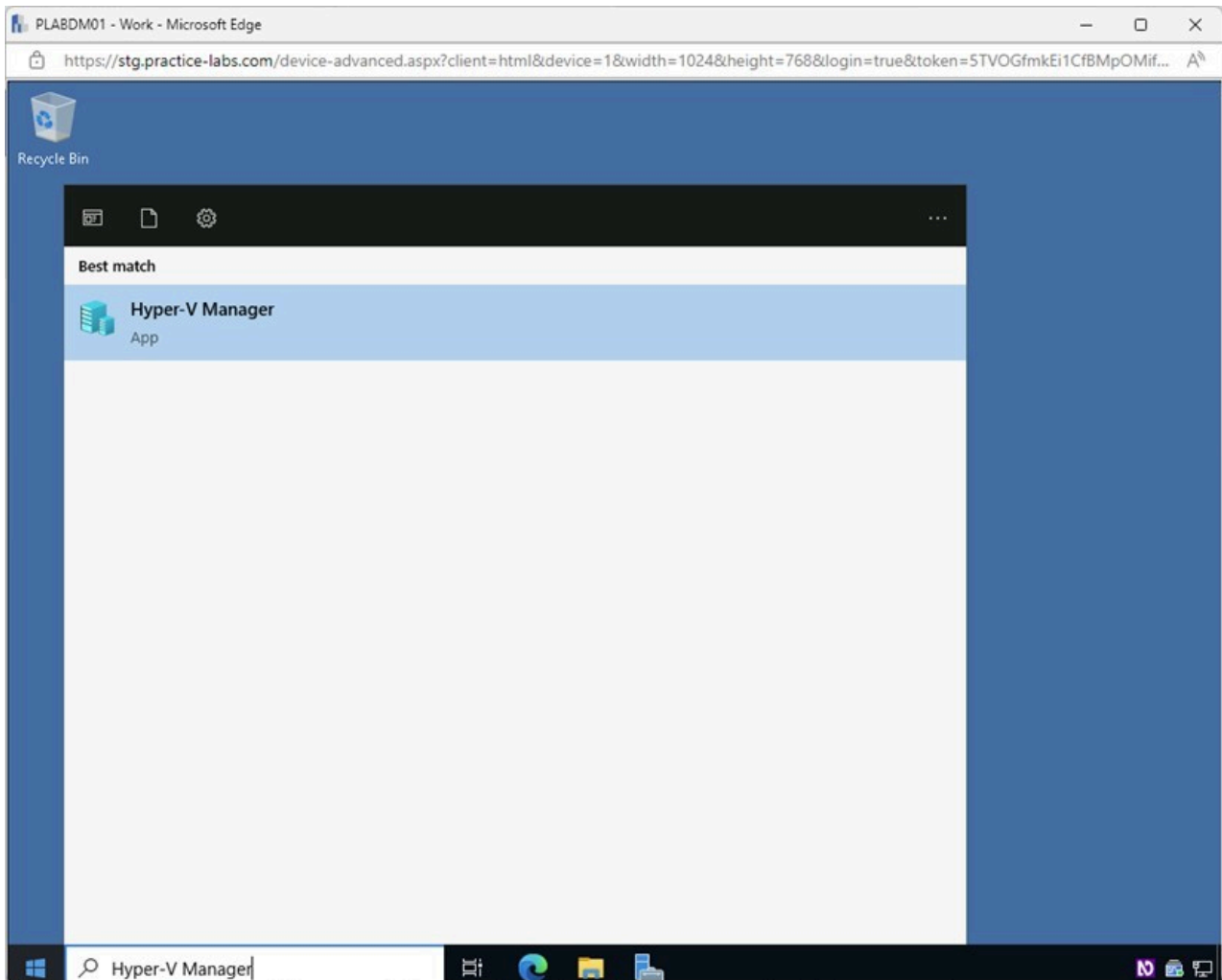


Figure 1.18 Screenshot of PLABDM01: Displaying opening the Hyper-V Manager application.

## Step 2

On the **Hyper-V Manager** window, click **PLABM01** on the left pane.

Select **Virtual Switch Manager** on the right-most **Actions** pane.

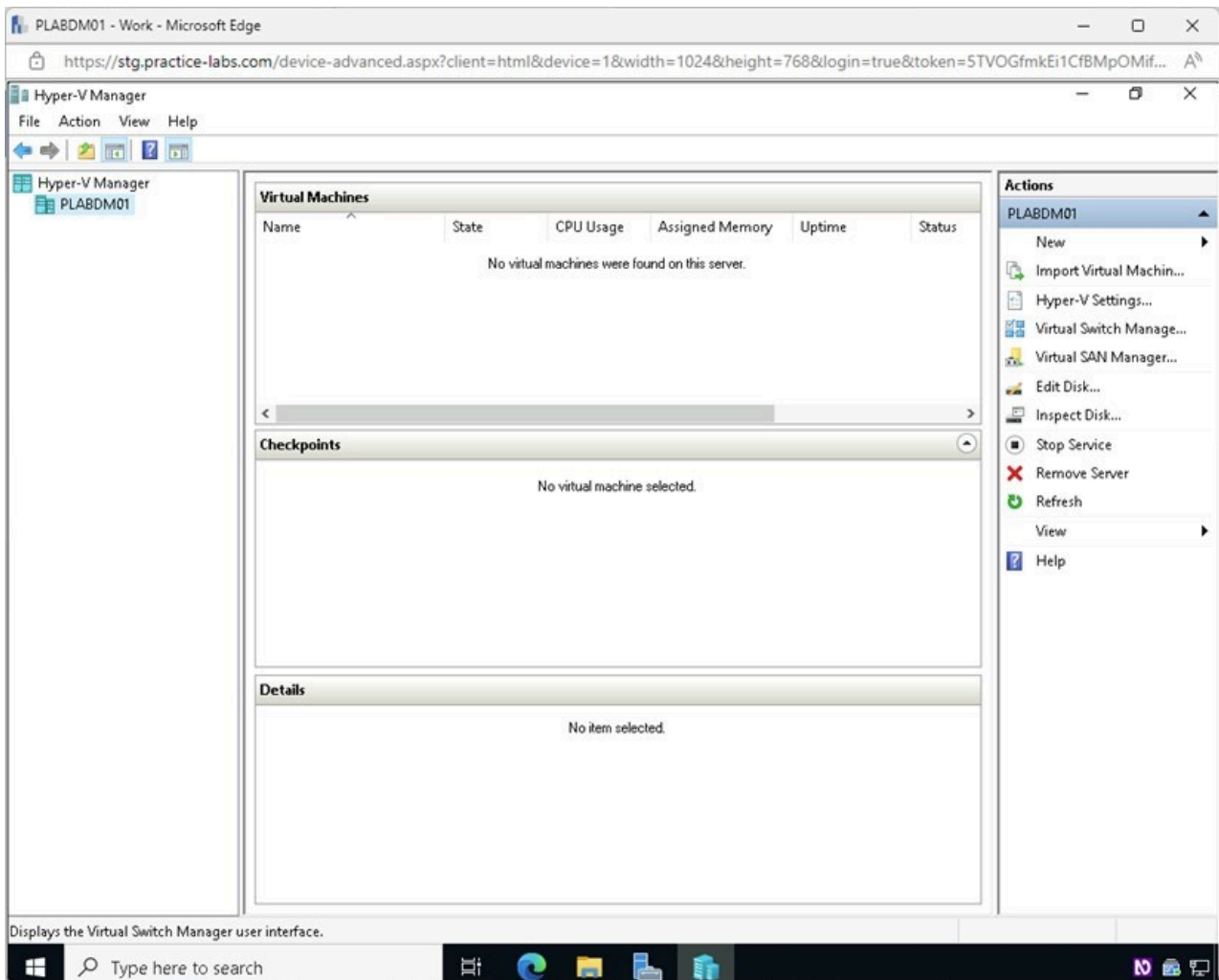


Figure 1.19 Screenshot of PLABDM01: Displaying selecting Virtual Switch Manager in Hyper-V Manager.

### Step 3

In the **Virtual Switch Manager for PLABDM01** window, ensure **External** is selected in the **Create virtual switch** pane.

Click **Create Virtual Switch**.

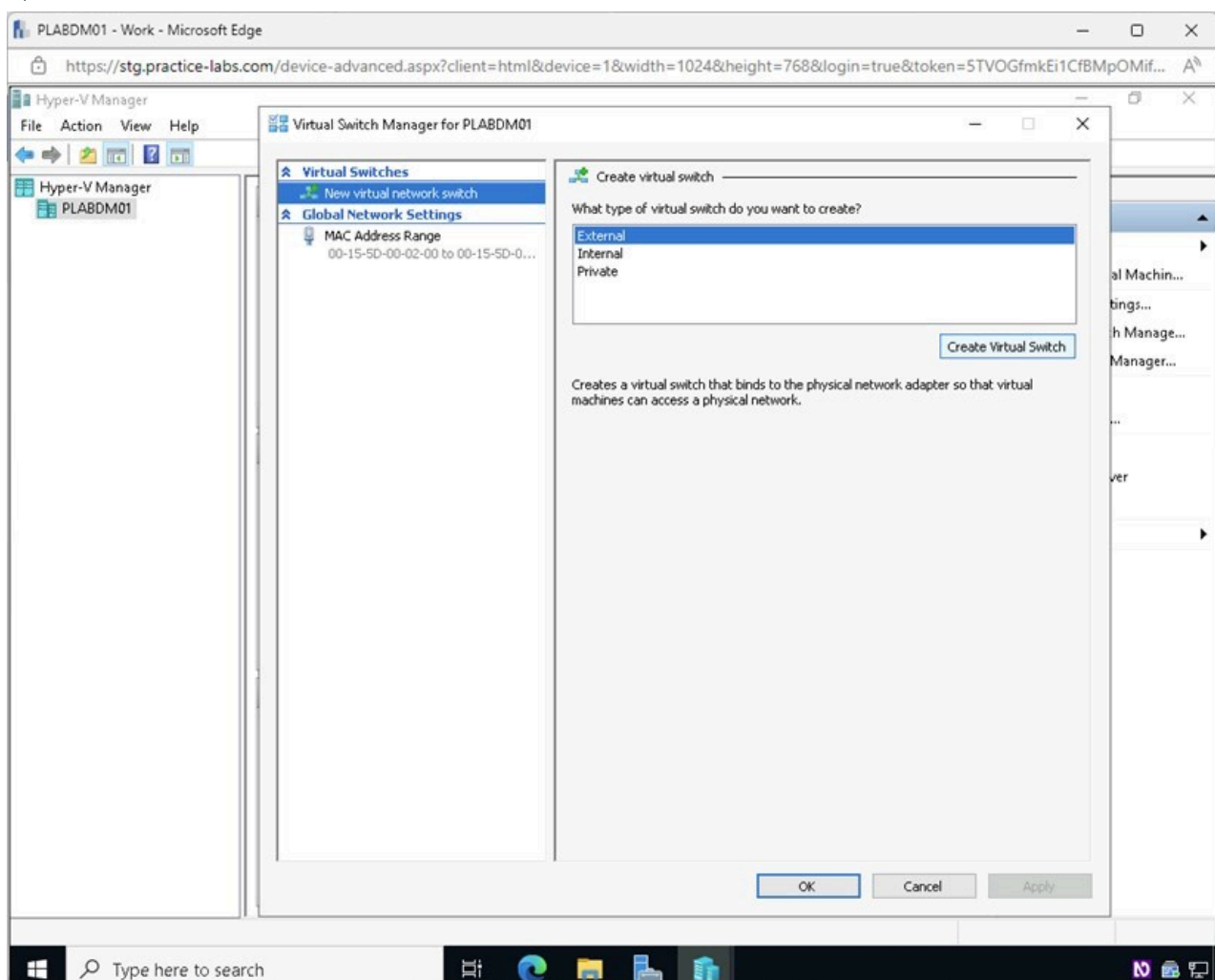


Figure 1.20 Screenshot of PLABDM01: Displaying creating an external virtual switch in Virtual Switch Manager.

## Step 4

In the **Virtual Switch Properties** pane, change the **Name** to the following and click **OK**:

PLAB External

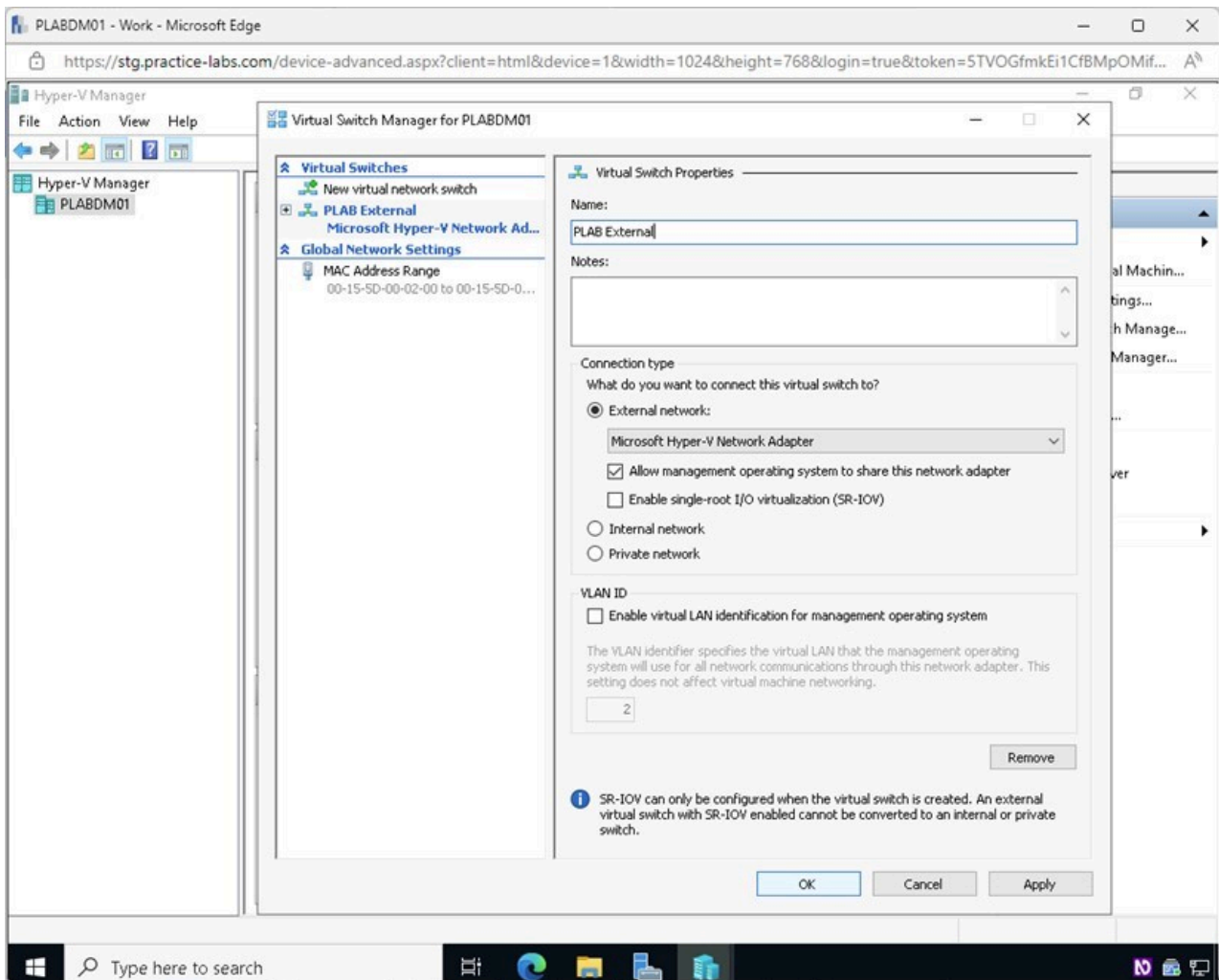


Figure 1.21 Screenshot of PLABDM01: Displaying the Virtual Switch properties pane with the required Name entered and the OK button selected.

**Note:** Creating an external switch using Hyper-V Manager will ensure the virtual devices created in Hyper-V Manager will connect to the physical device and other virtual or physical devices on the network. When creating an external switch, a physical network adapter must be associated with the virtual switch to facilitate communication.

## Step 5

Click **Yes** in the **Apply Networking Changes** pop-up window.

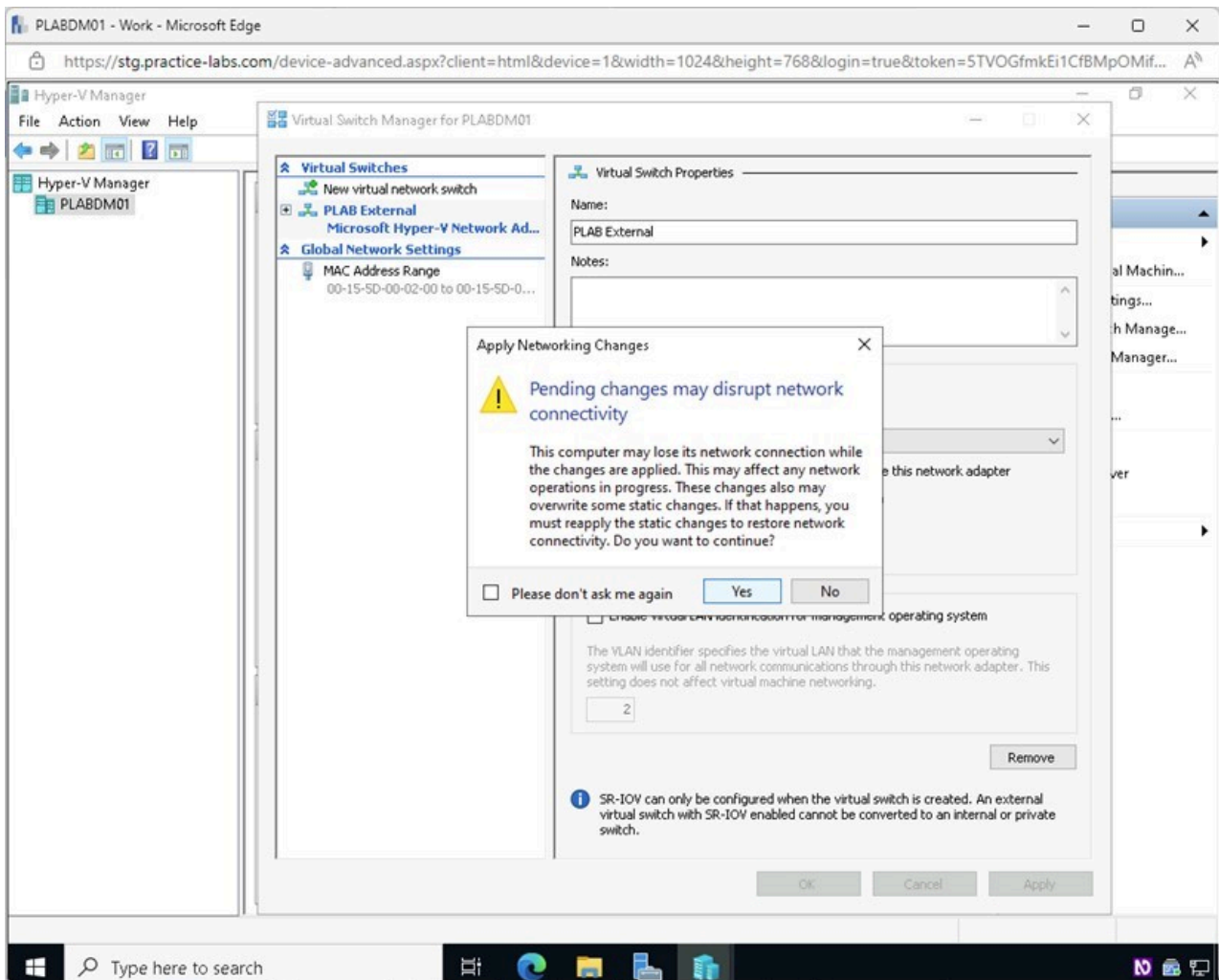


Figure 1.22 Screenshot of PLABDM01: Displaying clicking Yes in the Apply Networking Changes pop-up window.

**Note:** After applying the Network changes, the device might disconnect. Please reconnect to the device.

## Step 6

Back on the **Hyper-V Manager** window, click **Virtual Switch Manager** on the right pane.

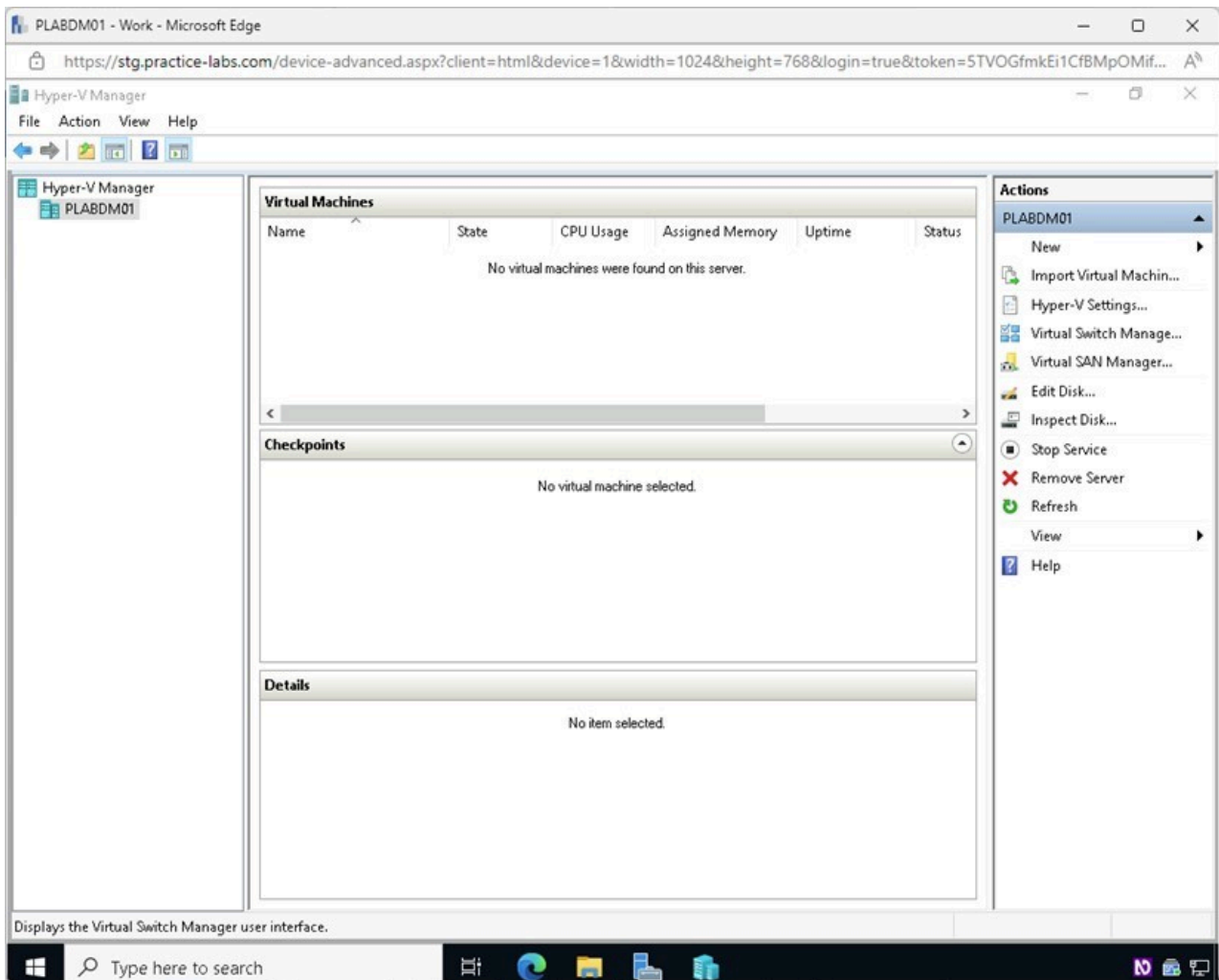


Figure 1.23 Screenshot of PLABDM01: Displaying opening Virtual Switch Manager in Hyper-V.

## Step 7

In the **Virtual Switch Manager for PLABDM01** window, in the **Create virtual switch** pane, select **Internal**.

Click **Create Virtual Switch**.

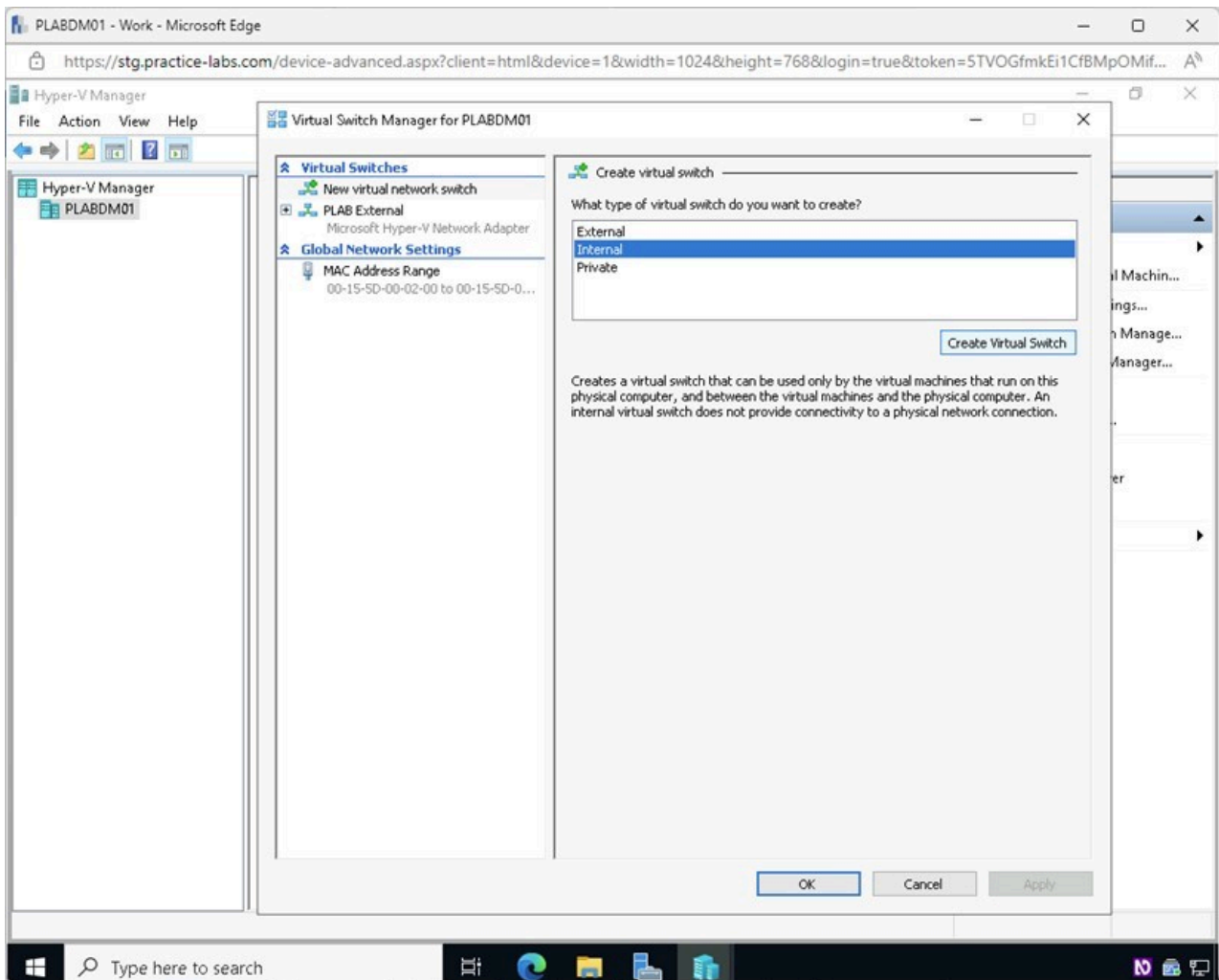


Figure 1.24 Screenshot of PLABDM01: Displaying creating an Internal Virtual switch using Virtual Switch Manager.

**Note:** An Internal Virtual Switch in Hyper-V is used to connect the specific host's virtual devices. This type of connection will be restricted to only other virtual machines on the device and the physical machine. No interaction with the physical network device is possible when using this type of connection.

## Step 8

On the **Virtual Switch Properties** window, change the **Name** of the Internal switch to the following:

PLAB Internal

Click **OK**.

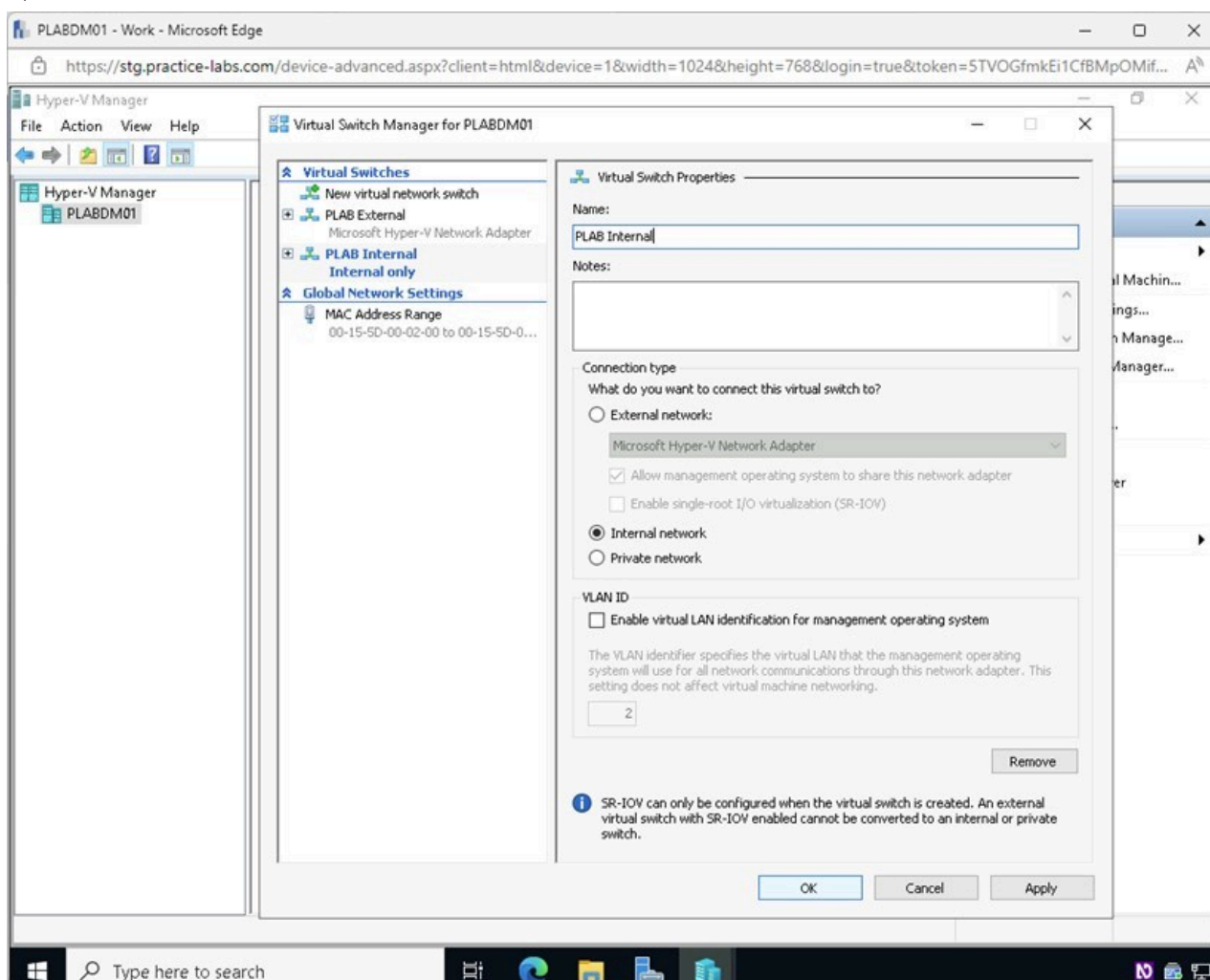


Figure 1.25 Screenshot of PLABDM01: Displaying the Virtual Switch Properties pane with the required Name entered and the OK button selected.

## Step 9

Back on the **Hyper-V Manager** window, click **Virtual Switch Manager** on the right pane.

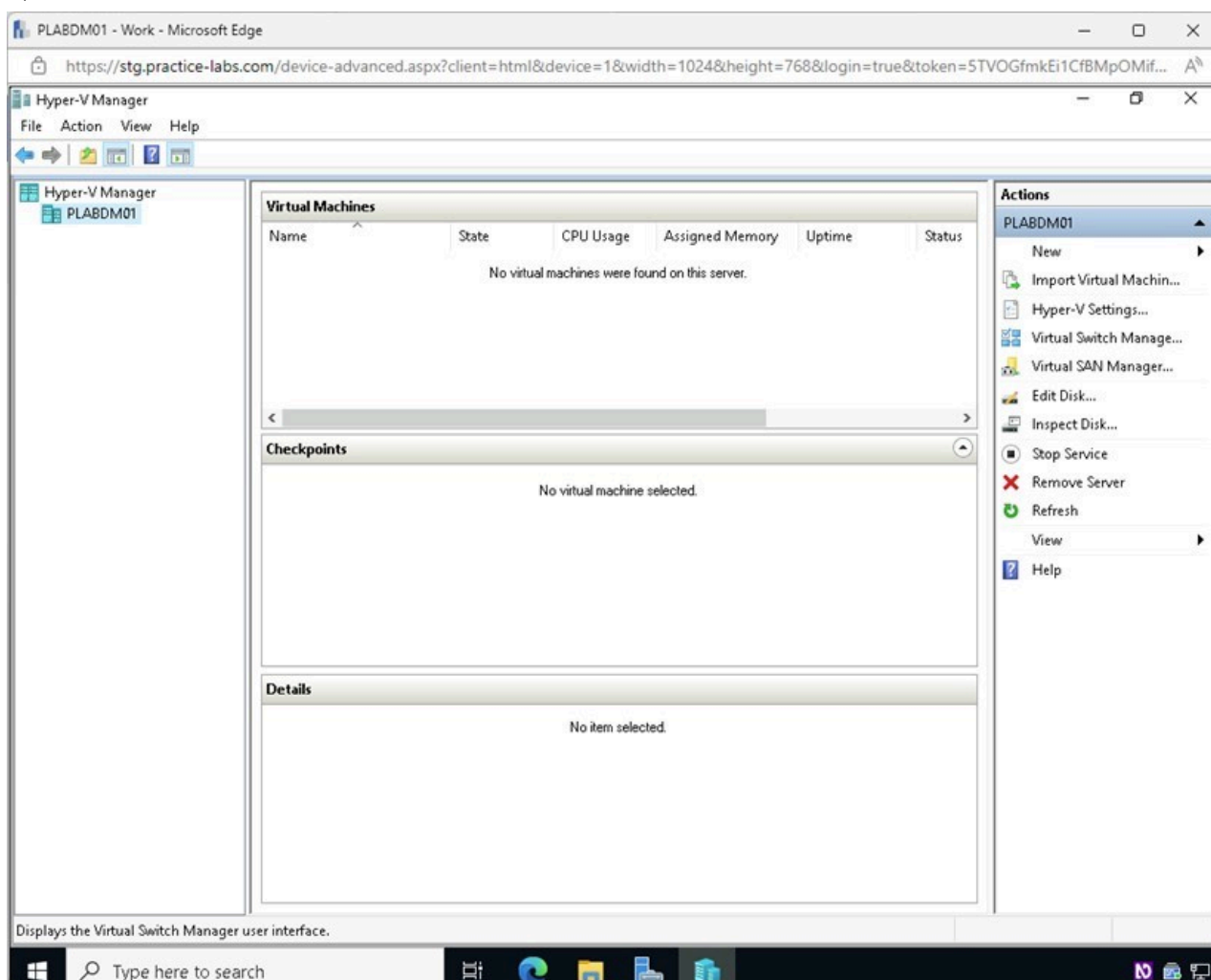


Figure 1.26 Screenshot of PLABDM01: Displaying opening Virtual Switch Manager in Hyper-V.

## Step 10

In the **Virtual Switch Manager for PLABDM01** window, in the **Create virtual switch** pane, select **Private**.

Click **Create Virtual Switch**.

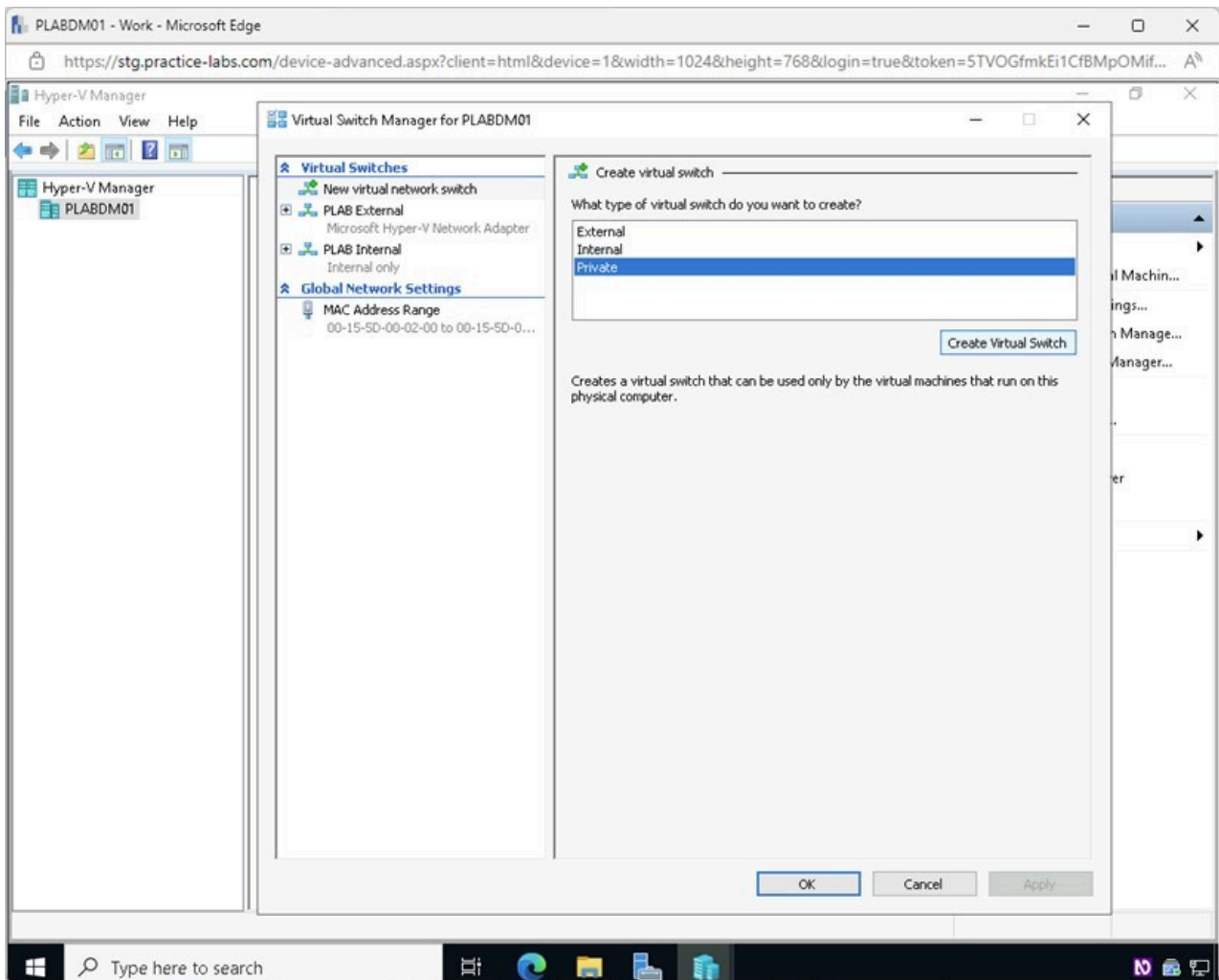


Figure 1.27 Screenshot of PLABDM01: Displaying creating a Private Virtual Switch using Virtual Switch Manager.

## Step 11

On the **Virtual Switch Properties** window, change the **Name** of the Private switch to the following:

PLAB Private

Click **OK**.

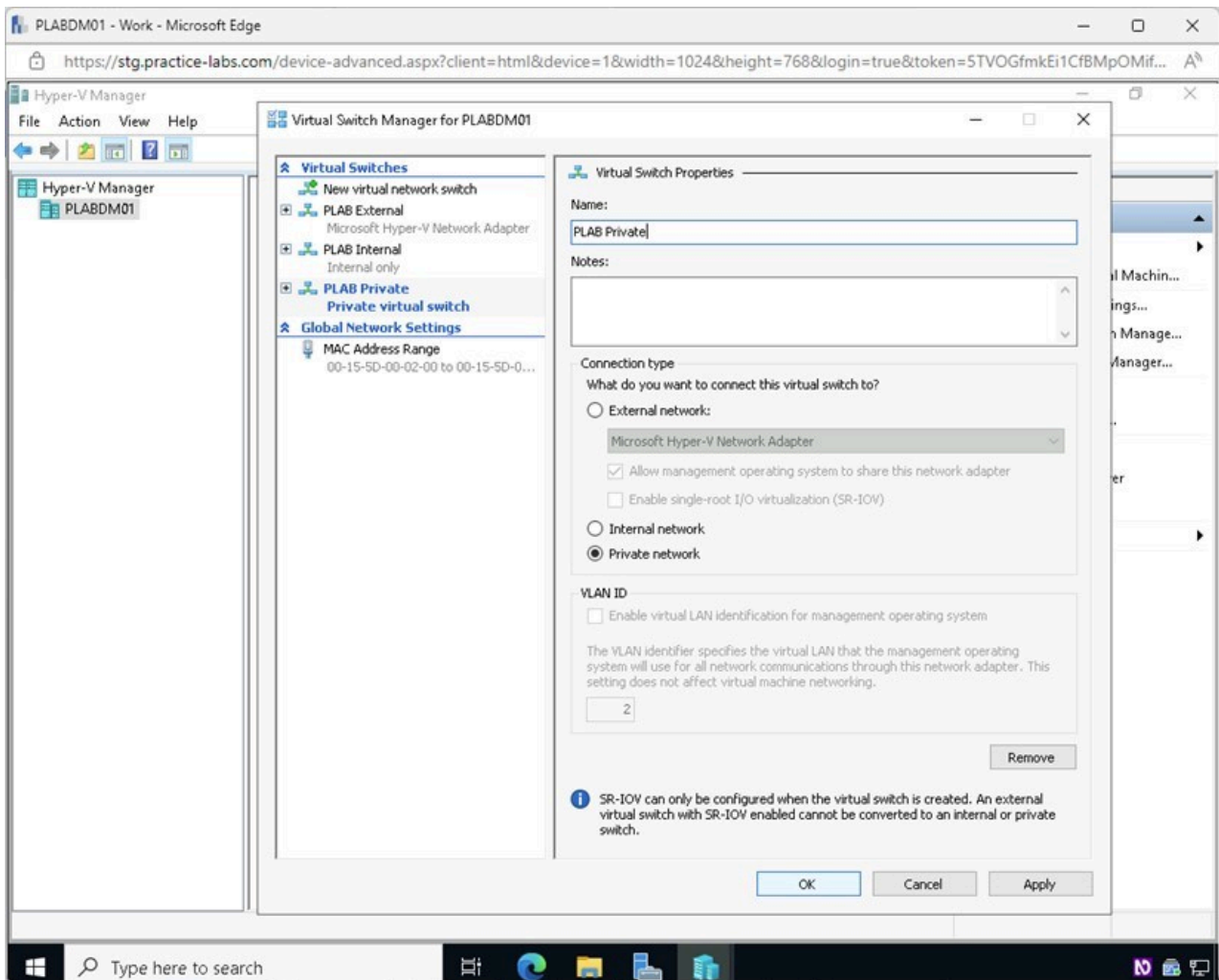


Figure 1.28 Screenshot of PLABDM01: Displaying the Virtual Switch Properties pane with the required Name entered and the OK button selected.

**Note:** A Private virtual switch in Hyper-V is used to connect the physical machine's virtual devices. When these devices are connected with a Private switch, they will only be able to communicate with each other and not with the physical host or the physical network adapter. This type of switch can be used for a sandbox solution.

Keep the **Hyper-V Manager** window open.

### Task 3 - Create a Virtual Machine

In this task, you will create a virtual machine.

#### Step 1

Ensure you are connected to **PLABDM01**, and the **Hyper-V Manager** window is open.

Select and right-click on **PLABDM01** and select **New > Virtual Machine** from the context menu.

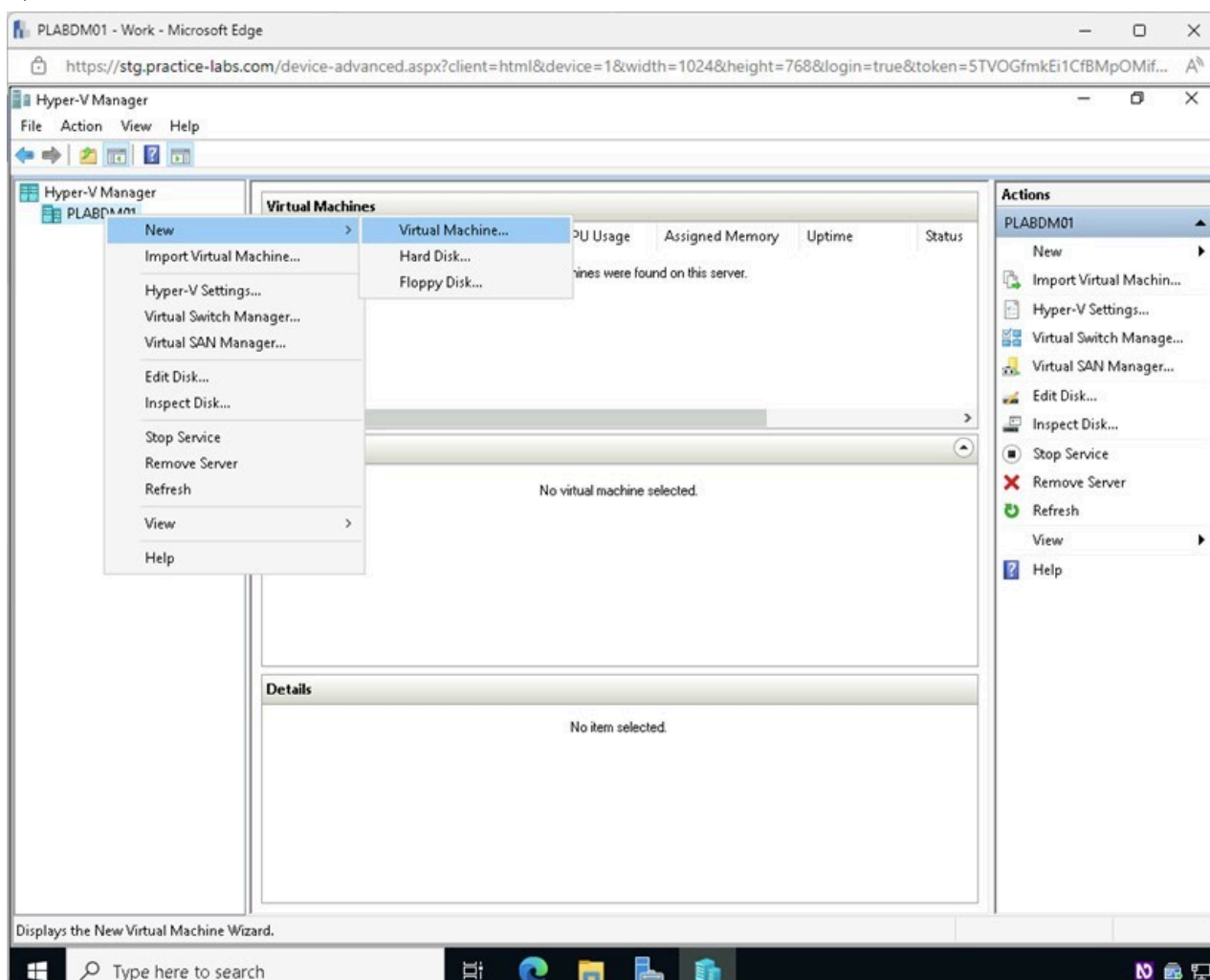


Figure 1.29 Screenshot of PLABDM01: Displaying the Hyper-V Manager window with PLABDM01 right-clicked and New > Virtual Machine selected from the context menu.

## Step 2

In the **New Virtual Machine Wizard - Before You Begin** page, click **Next**.

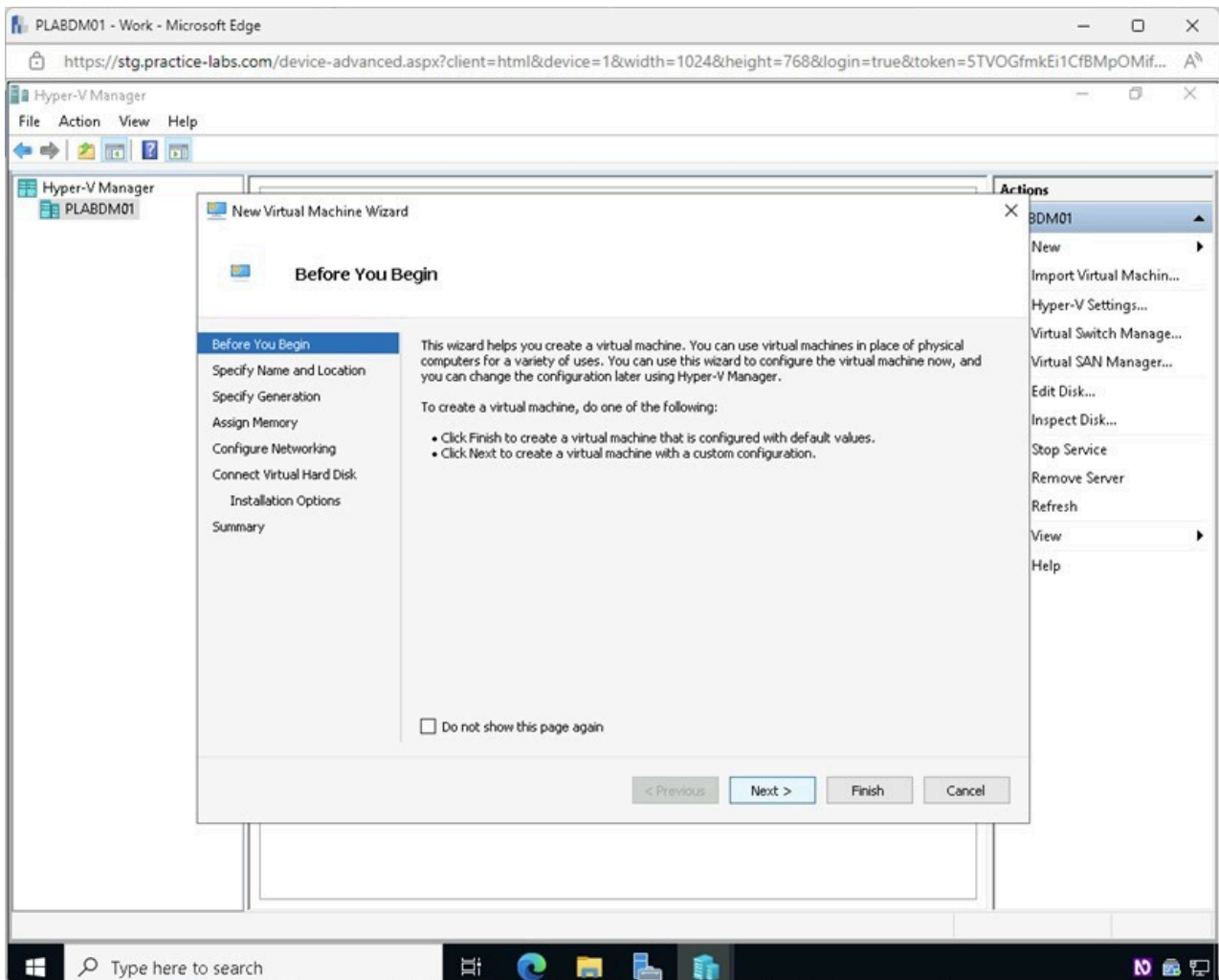


Figure 1.30 Screenshot of PLABDM01: Displaying clicking Next in the New Virtual Machine Wizard - Before You Begin page.

### Step 3

On the **Specify Name and Location** page, type the following in the **Name** field:

PLABVM01

Click **Next**.

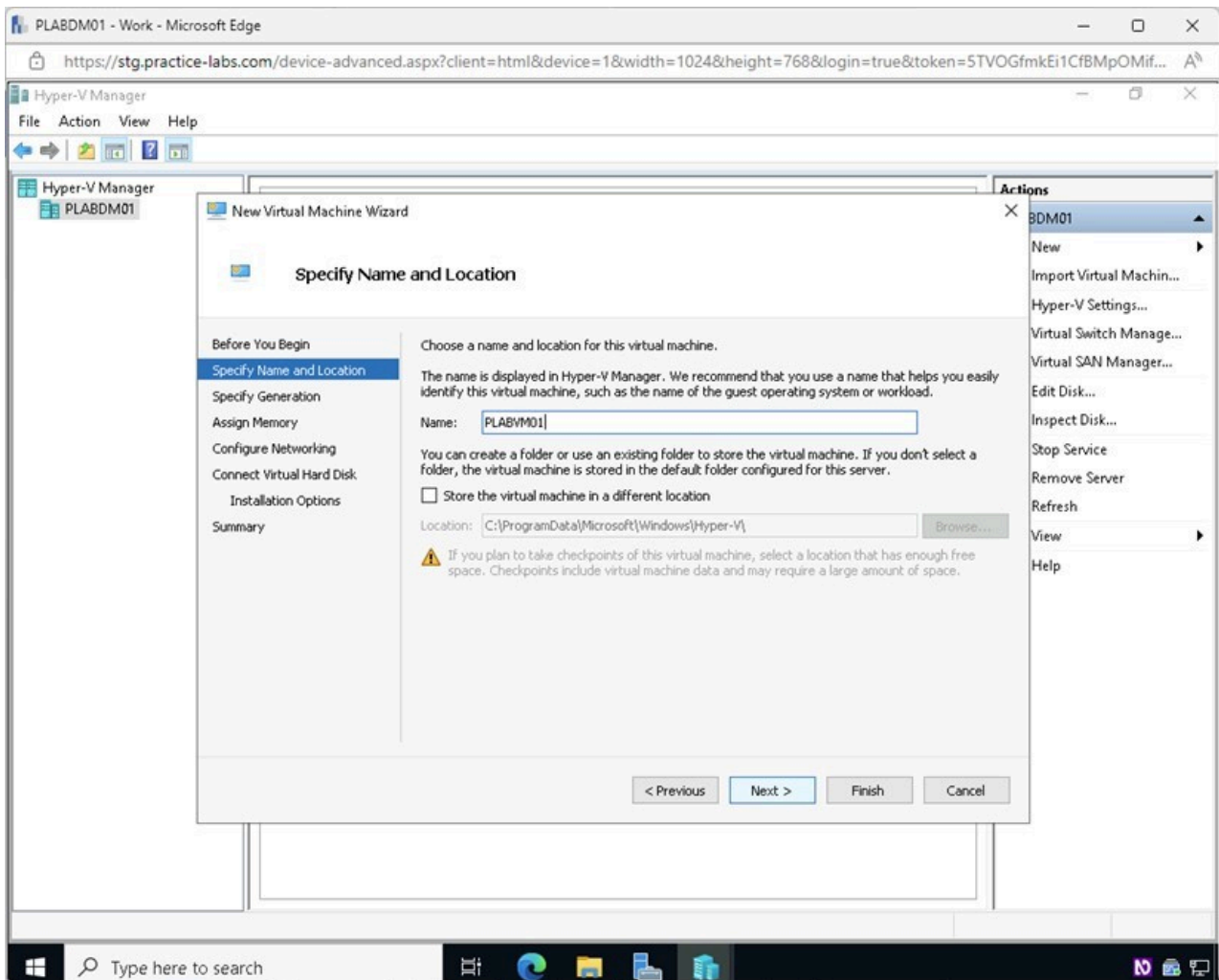


Figure 1.31 Screenshot of PLABDM01: Displaying the Specify Name and Location page with the required Name entered and the Next button highlighted.

## Step 4

In the **Specify Generation** page, select the **Generation 2** radio button.

Click **Next**.

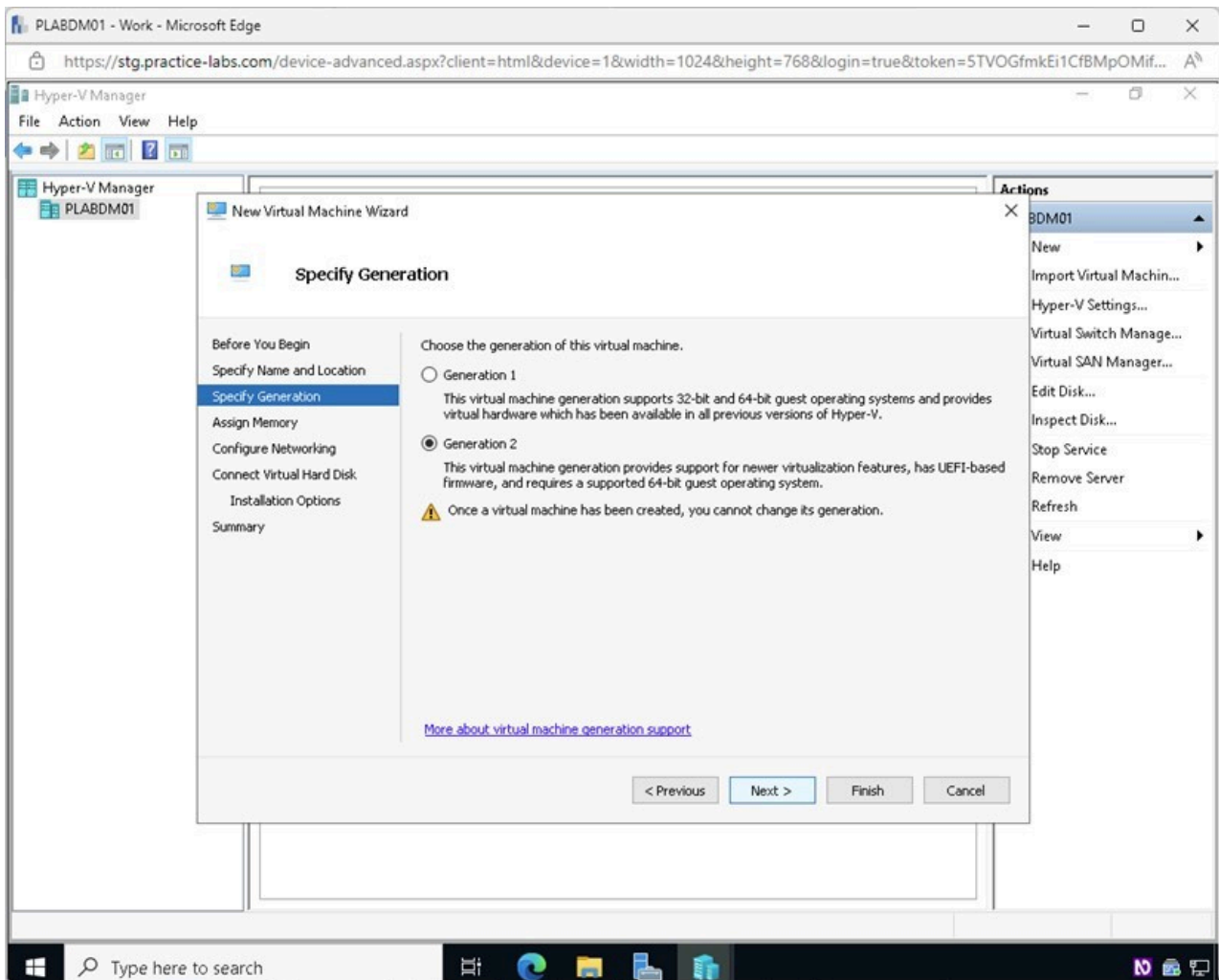


Figure 1.32 Screenshot of PLABDM01: Displaying the Specify Generation page with the required option selected and the Next button highlighted.

**Note:** The generation that is chosen here cannot be changed later. If your company is supporting legacy systems, choose Generation 1. However, for all other scenarios, you should select Generation 2 for the latest features offered in Microsoft's Hyper-V platform.

## Step 5

In the **Assign Memory** page, click **Next**.

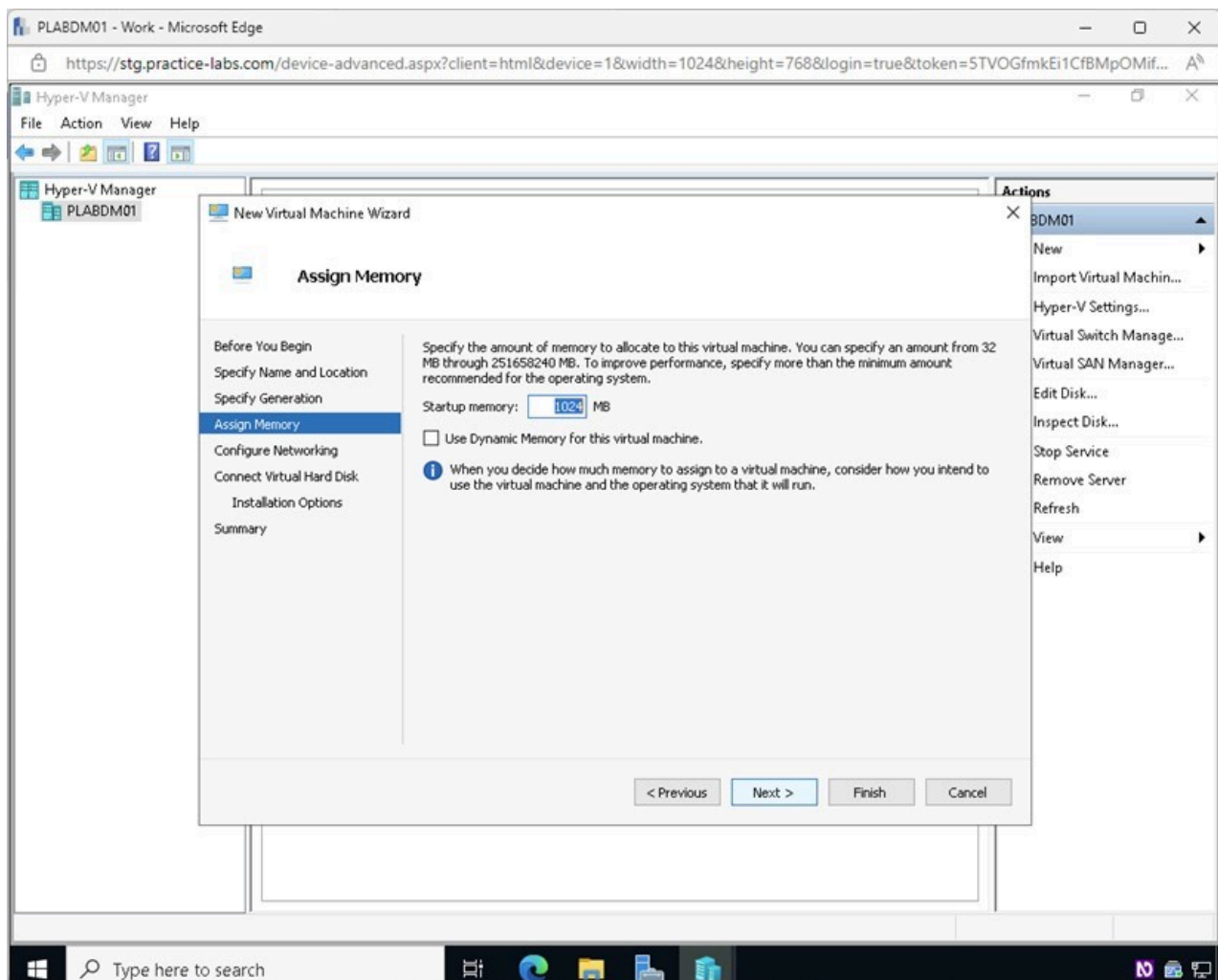


Figure 1.33 Screenshot of PLABDM01: Displaying clicking Next on the Assign Memory page.

## Step 6

In the **Configure Networking** page, click **Next**.

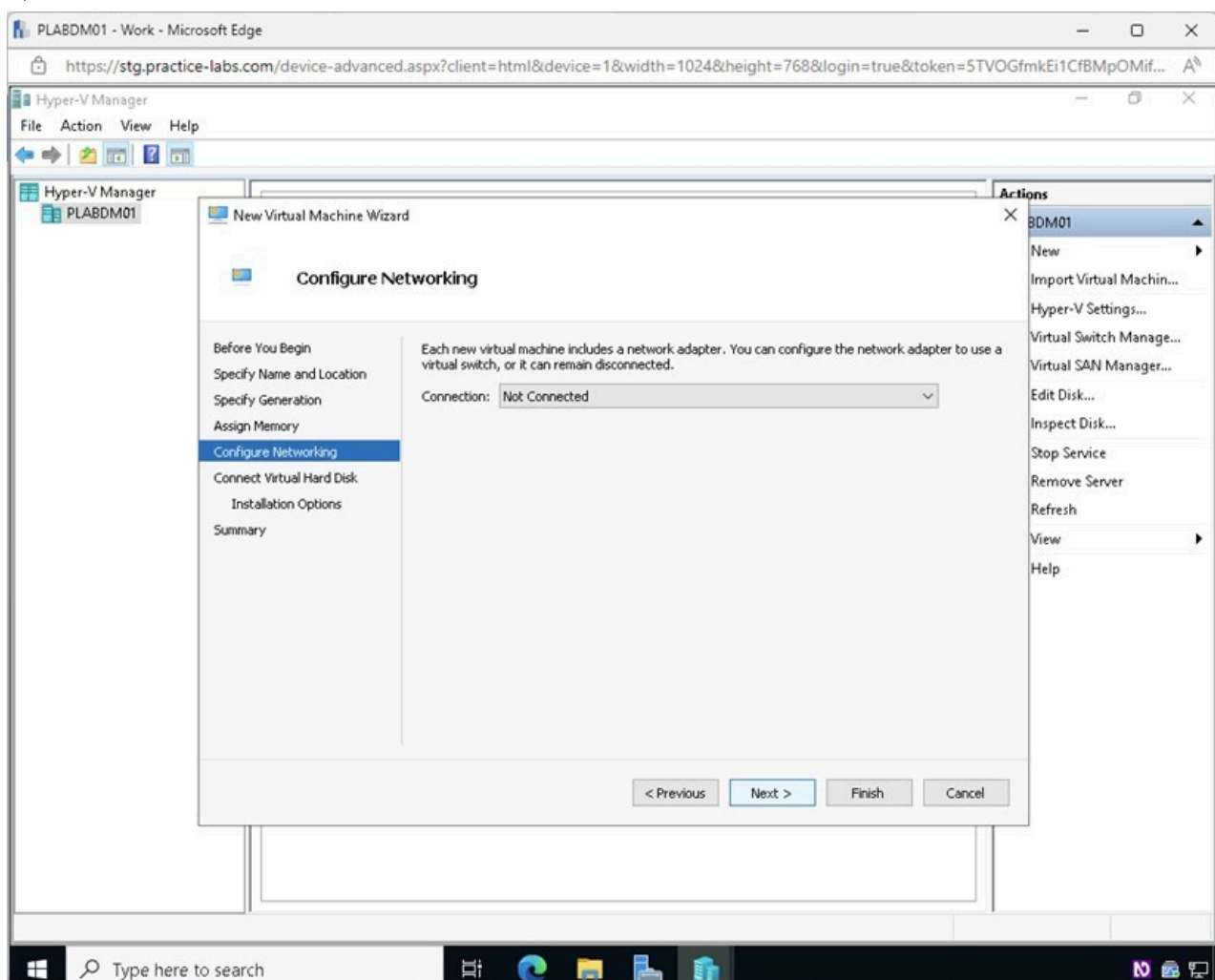


Figure 1.34 Screenshot of PLABDM01: Displaying clicking Next on the Configure Networking page.

## Step 7

In the **Connect Virtual Hard Disk** page, type the following in the **Size** field:

60

Click **Next**.

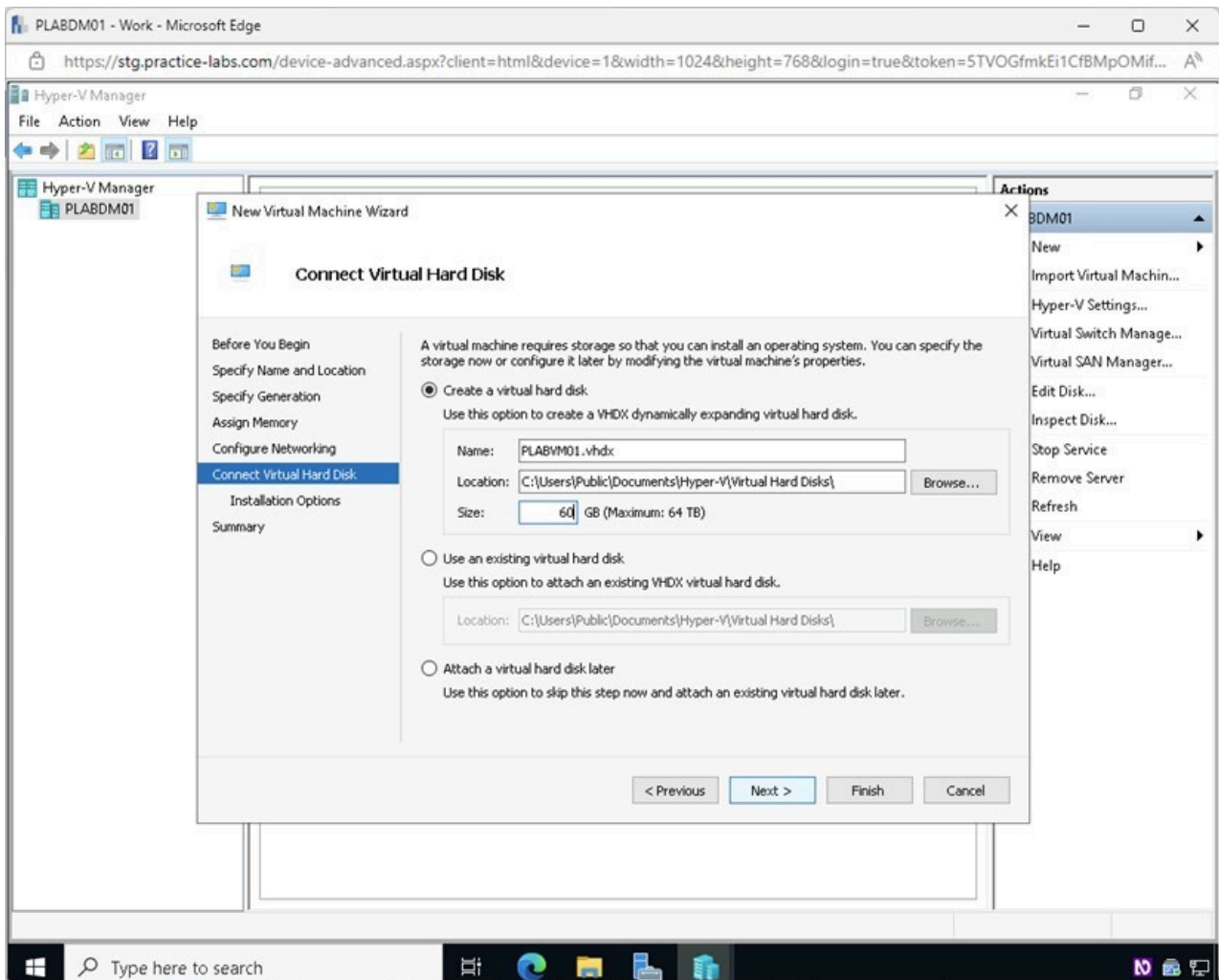


Figure 1.35 Screenshot of PLABDM01: Displaying the Connect Virtual Hard Disk page with the required Size entered and the Next button highlighted.

## Step 8

In the **Installation Options** page, select the **Install an operating system later** radio button.

Click **Next**.

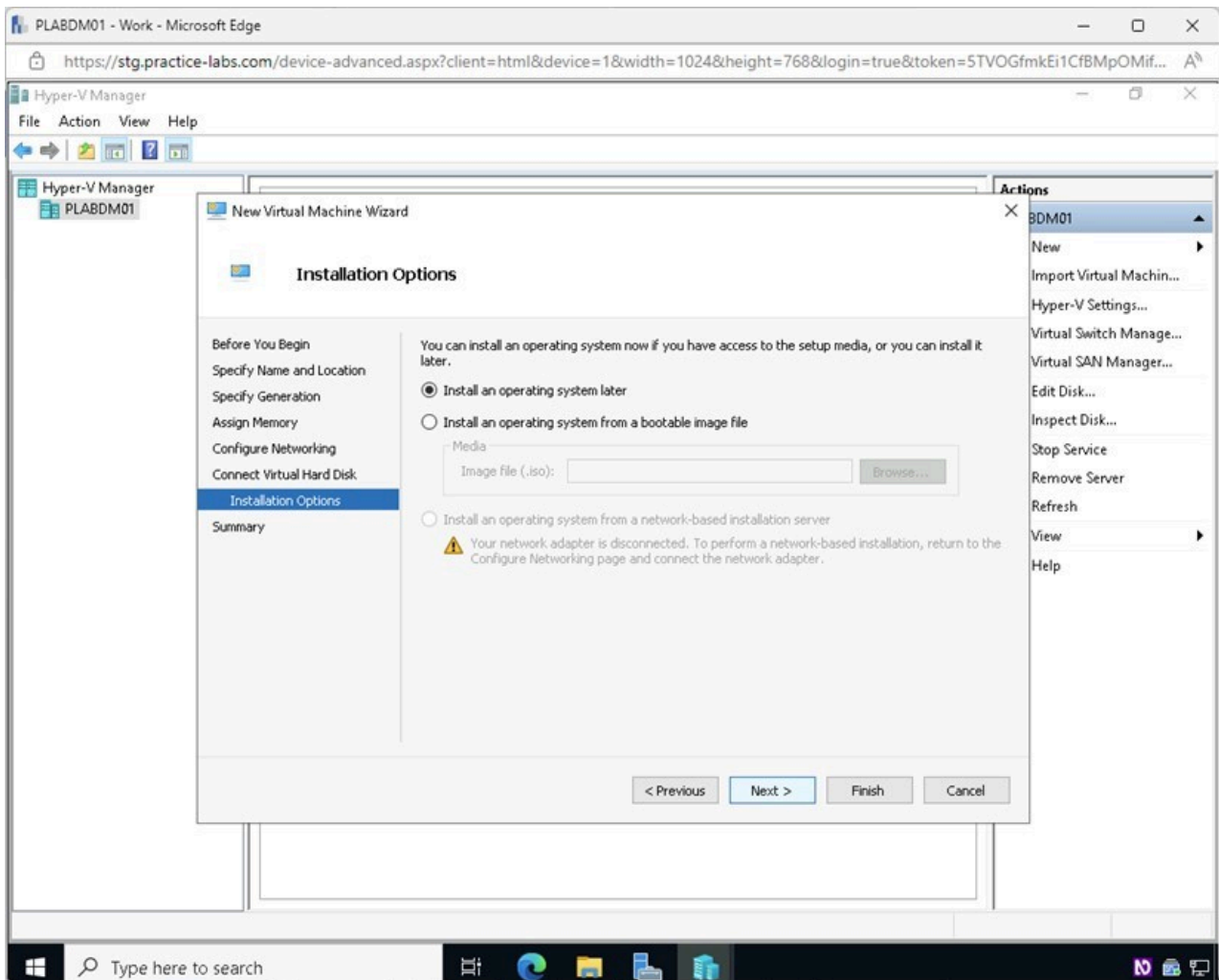


Figure 1.36 Screenshot of PLABDM01: Displaying the Installation Options page with the required option selected and the Next button highlighted.

## Step 9

On the **Completing the New Virtual Machine Wizard** page, review the selections and click **Finish**.

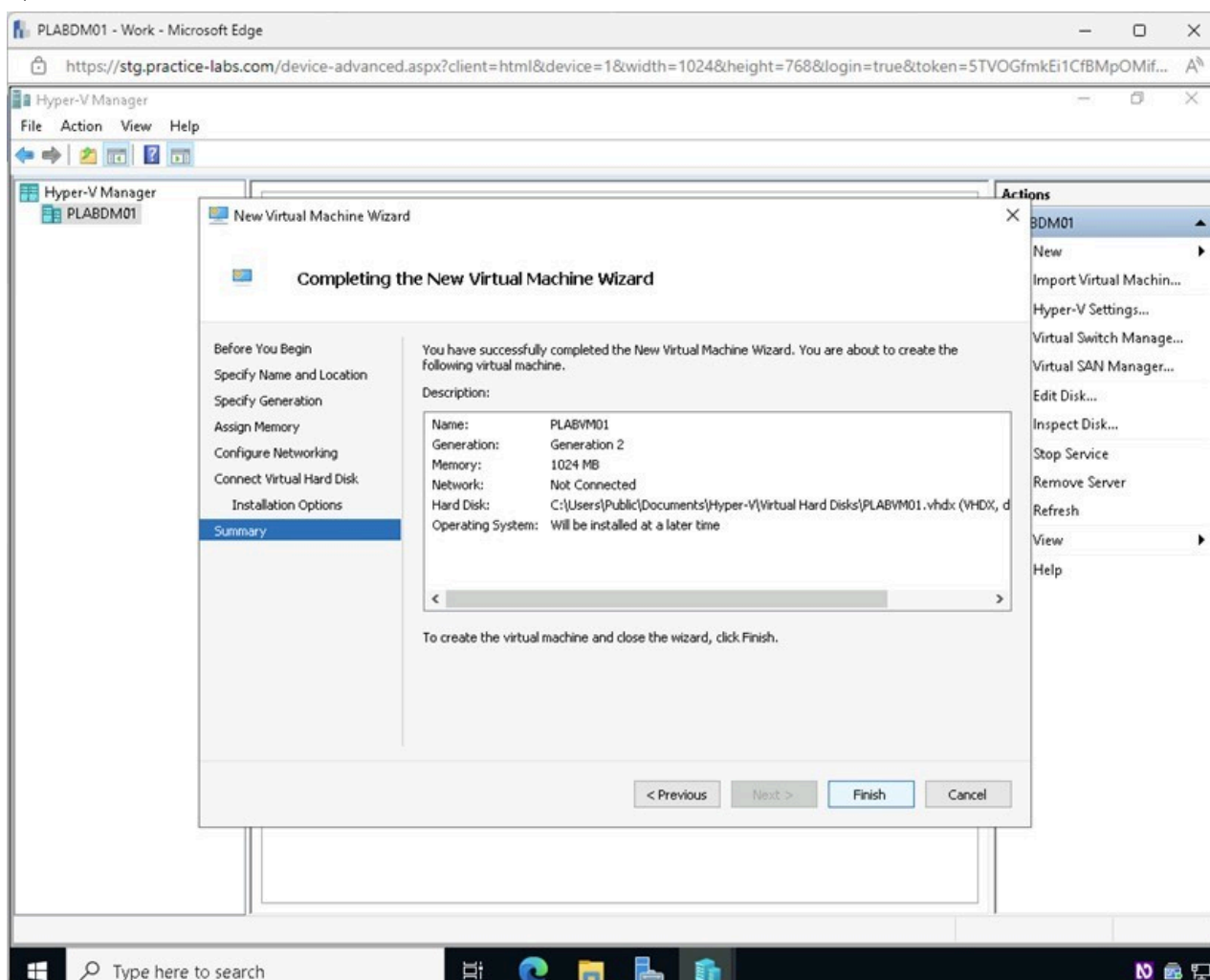


Figure 1.37 Screenshot of PLABDM01: Displaying clicking Finish on the Completing the New Virtual Machine Wizard page.

## Step 10

On the **Completing the New Virtual Machine Wizard** page, observe the virtual machine creation process.

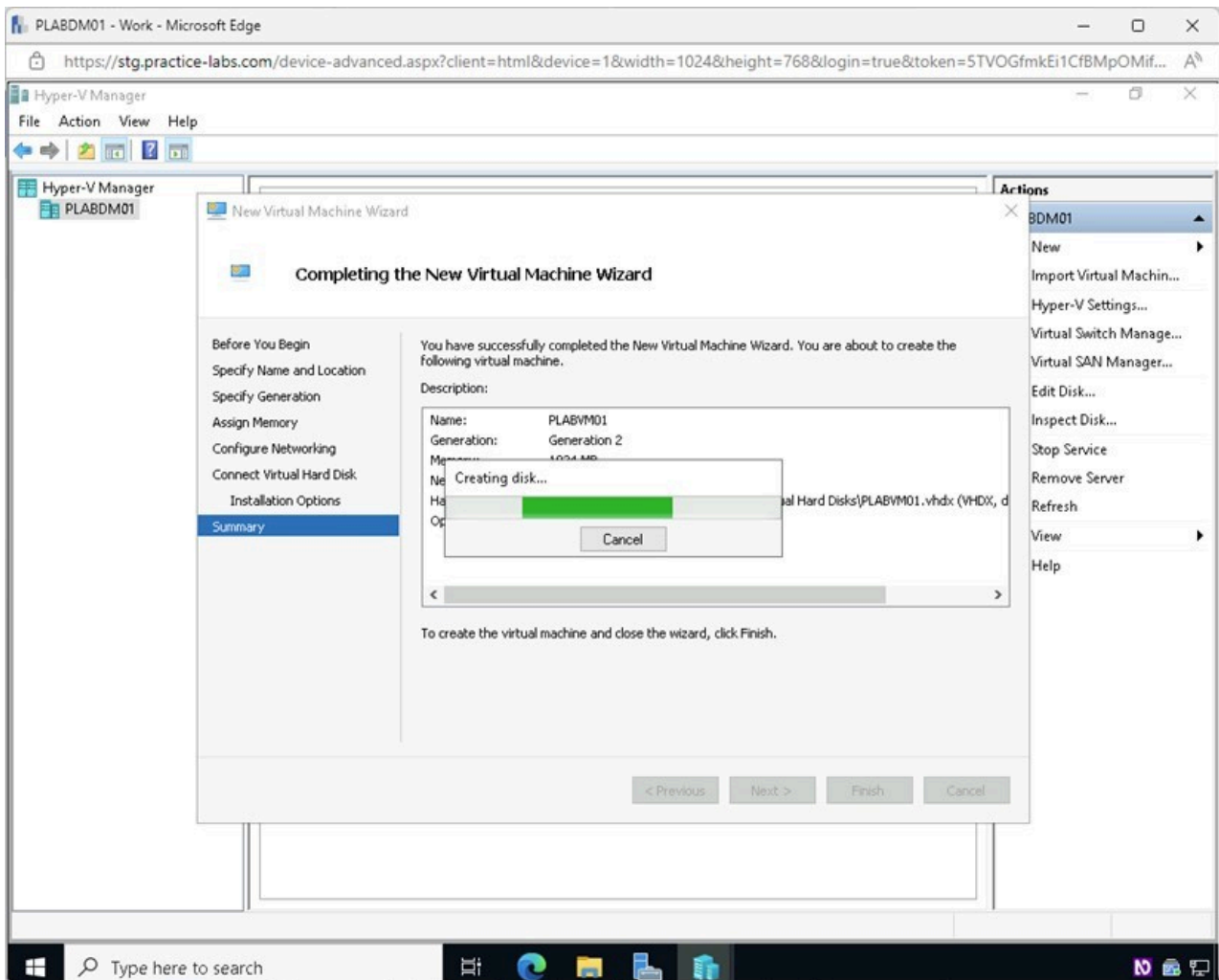


Figure 1.38 Screenshot of PLABDM01: Displaying the virtual machine creation process on the Completing the New Virtual Machine Wizard page.

Keep the **Hyper-V Manager** window open.

## Task 4 - Install Windows Server on a Virtual Machine

In this task, you will build upon the last task by installing the Windows Server 2019 operating system on the newly created virtual machine.

### Step 1

In **PLABDM01**, the **Hyper-V Manager** window is open.

In the **Virtual Machines** pane, select and right-click on **PLABVM01** and select **Settings**.

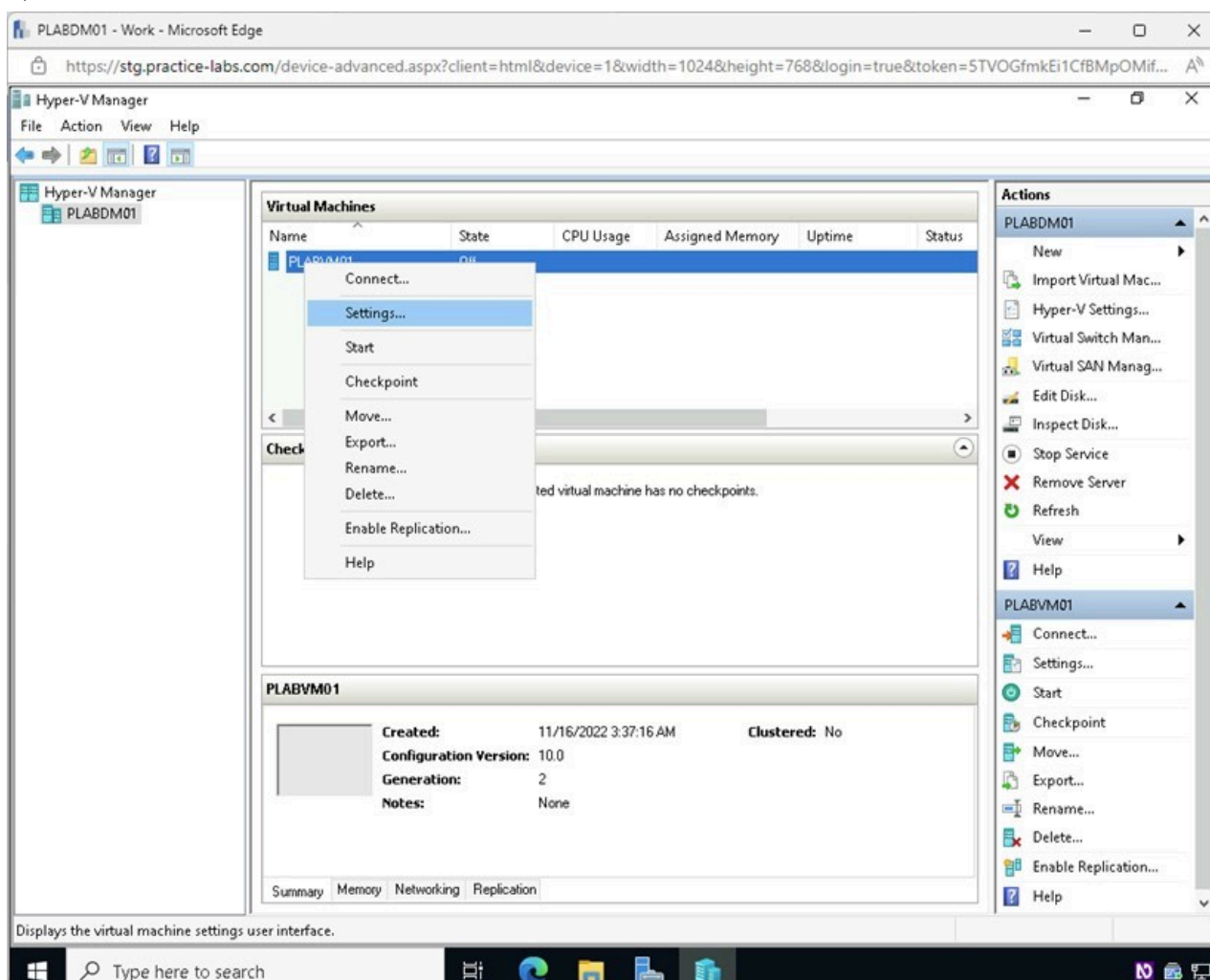


Figure 1.39 Screenshot of PLABDM01: Displaying the Hyper-V Manager window. PLABVM01 virtual machine is right-clicked and Setting selected from the context menu.

## Step 2

In the **Settings for PLABVM01 on PLABDM01** window, review the virtual hardware available to the virtual machine.

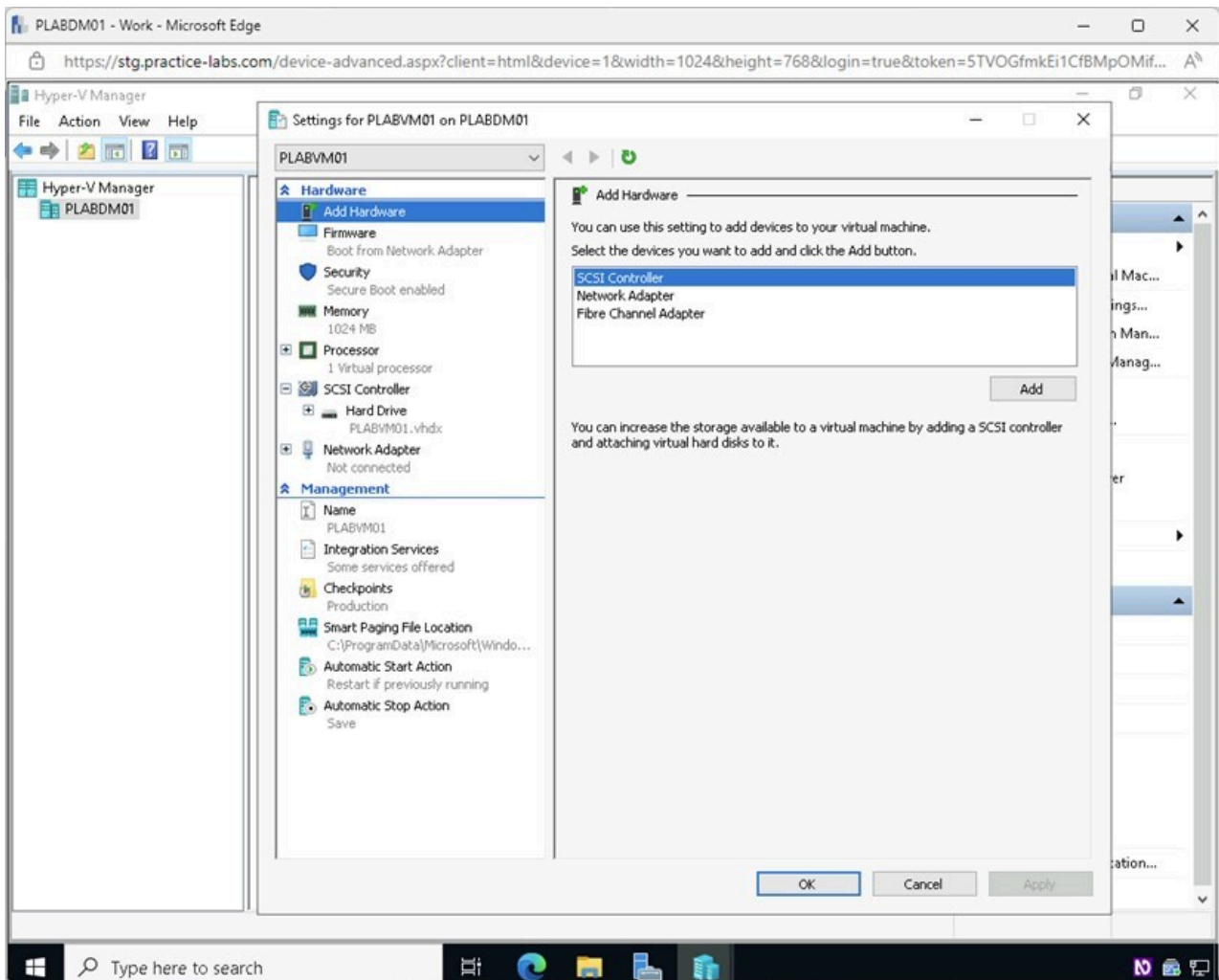


Figure 1.40 Screenshot of PLABDM01: Displaying the Settings for PLABVM01 on PLABDM01 window.

### Step 3

In the **Settings for PLABVM01 on PLABDM01** window, select **SCSI Controller** in the **Hardware** section.

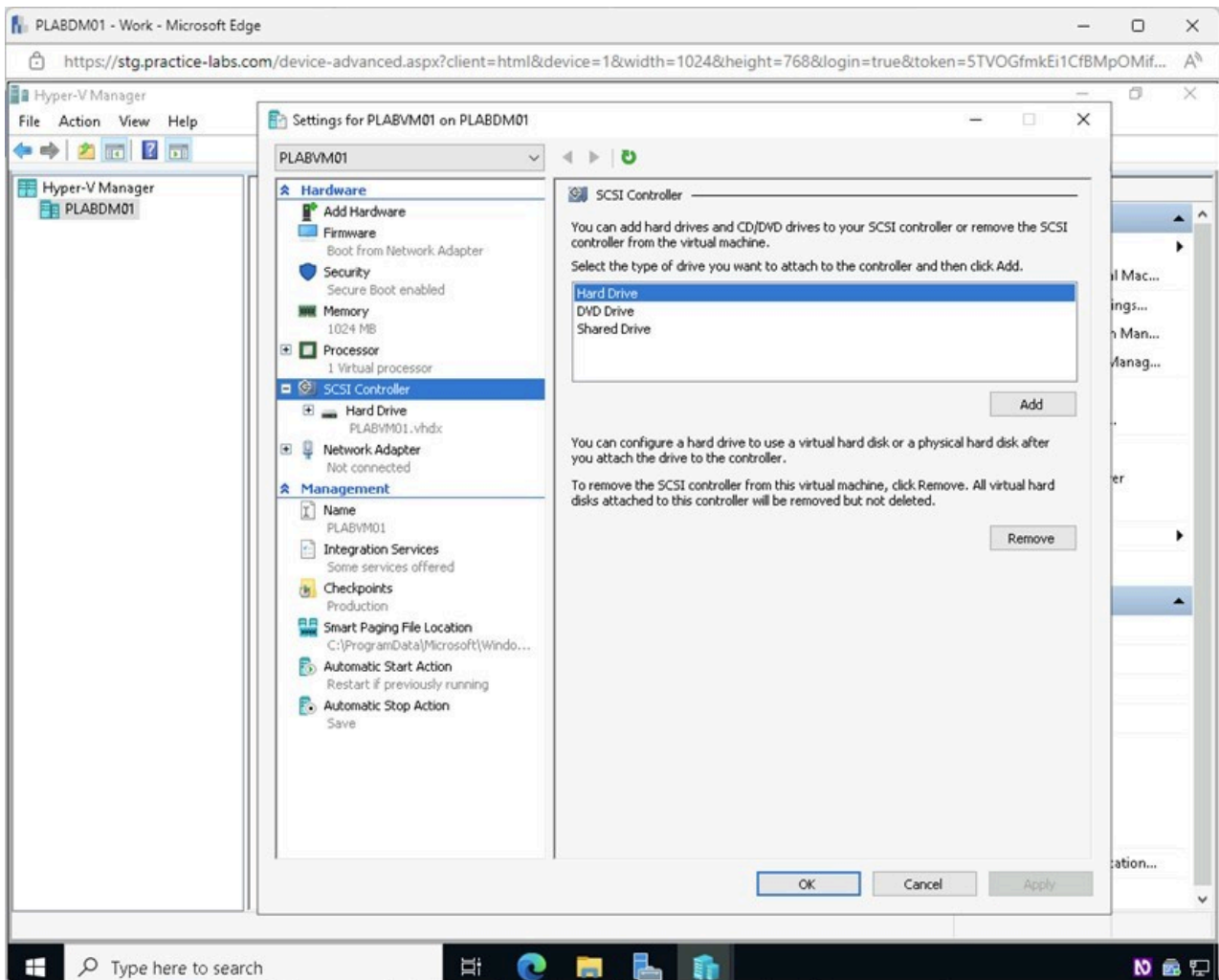


Figure 1.41 Screenshot of PLABDM01: Displaying the Settings for PLABVM01 on PLABDM01 window with SCSI Controller in the Hardware section selected.

## Step 4

In the **Settings for PLABVM01 on PLABDM01** window, select **DVD Drive** under **SCSI Controller**.

Click **Add**.

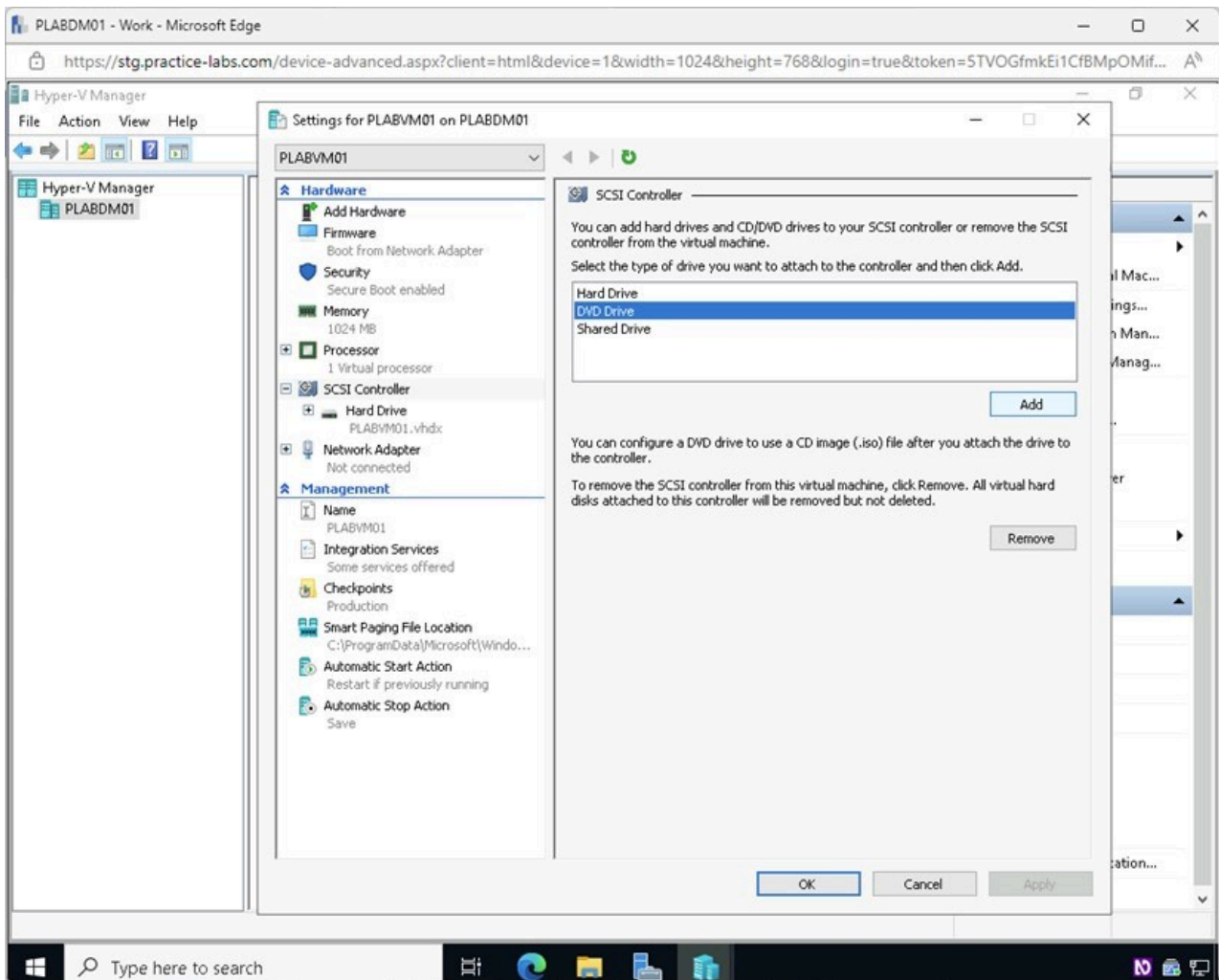


Figure 1.42 Screenshot of PLABDM01: Displaying the Settings for PLABVM01 on PLABDM01 window with DVD Drive selected and the Add button highlighted.

## Step 5

In the **Settings for PLABVM01 on PLABDM01** window, on the **DVD Drive** pane, select the radio button next to **Image File** in the context pane under **Media**.

Click **Browse**.

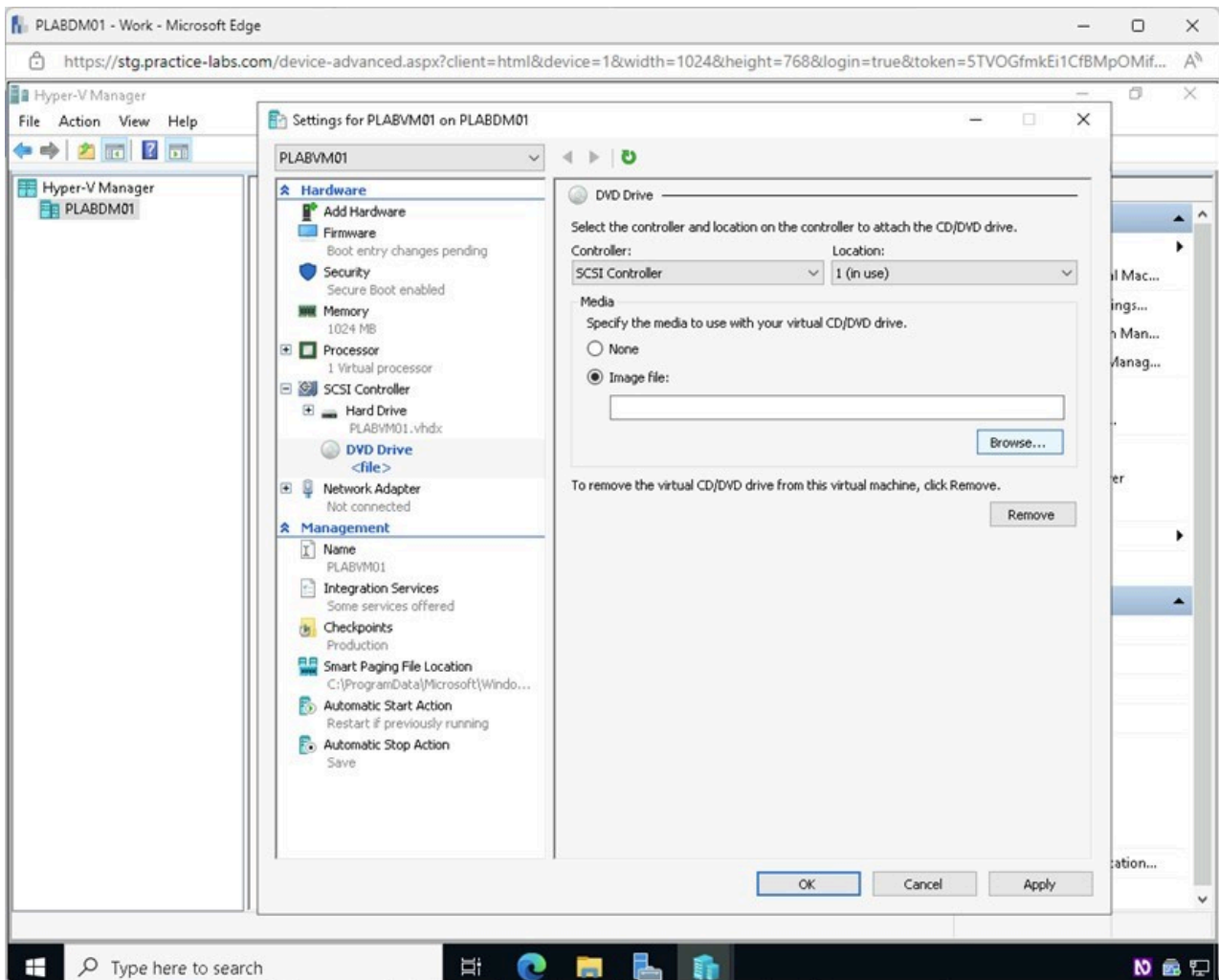


Figure 1.43 Screenshot of PLABDM01: Displaying the Settings for PLABVM01 on PLABDM01 window with the Image File option selected and the Browse button highlighted.

## Step 6

In the **Open** dialog box, select the **ISO folder** on the left pane.

Select the **en-us\_windows\_server\_2019\_updated\_jul\_2021\_x64\_dvd\_01319f3c** ISO Image file and click **Open**.

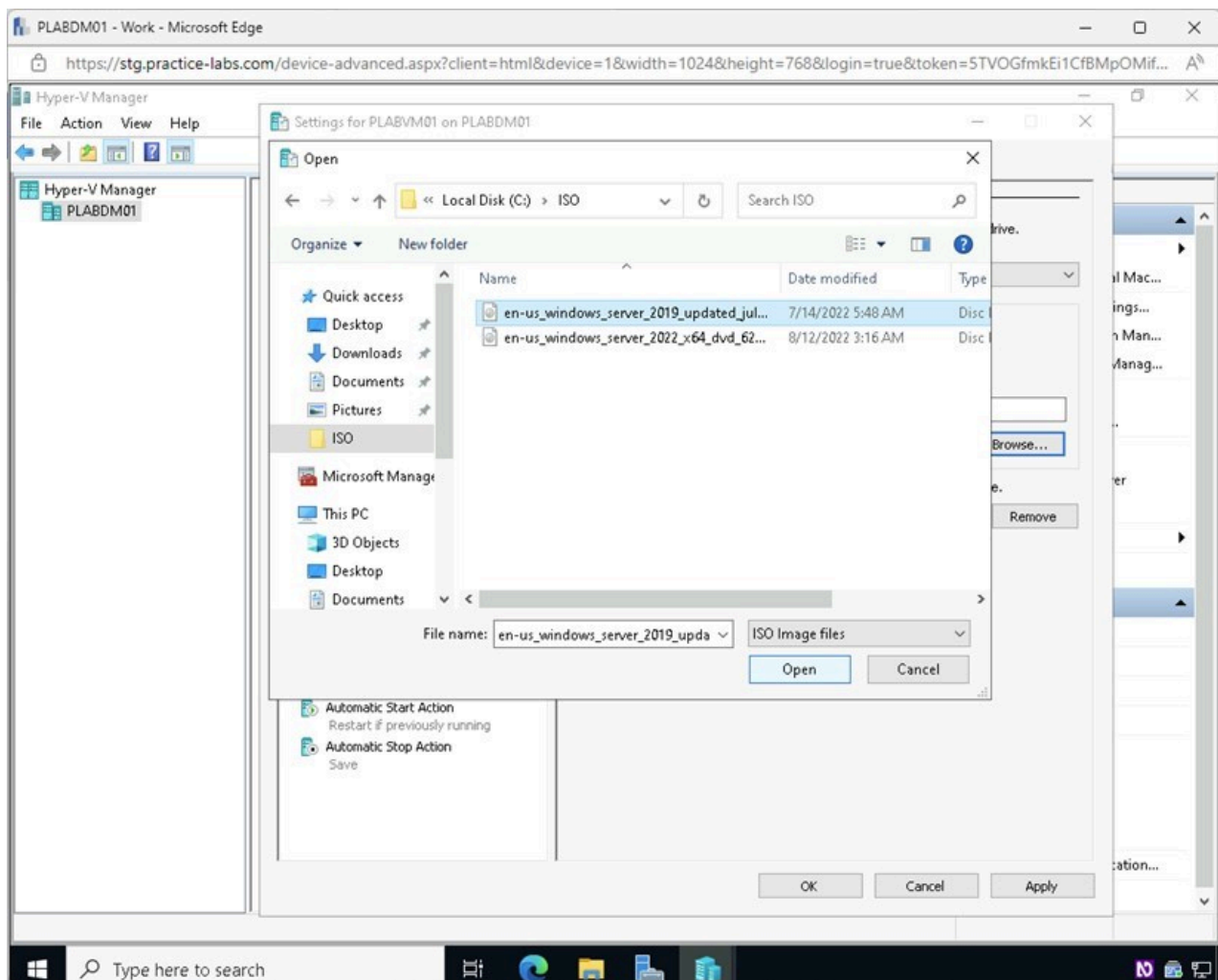


Figure 1.44 Screenshot of PLABDM01: Displaying the Open dialog box with the required ISO Image file selected and the Open button highlighted.

## Step 7

In the **Settings for PLABVM01 on PLABDM01** window, click **Apply**.

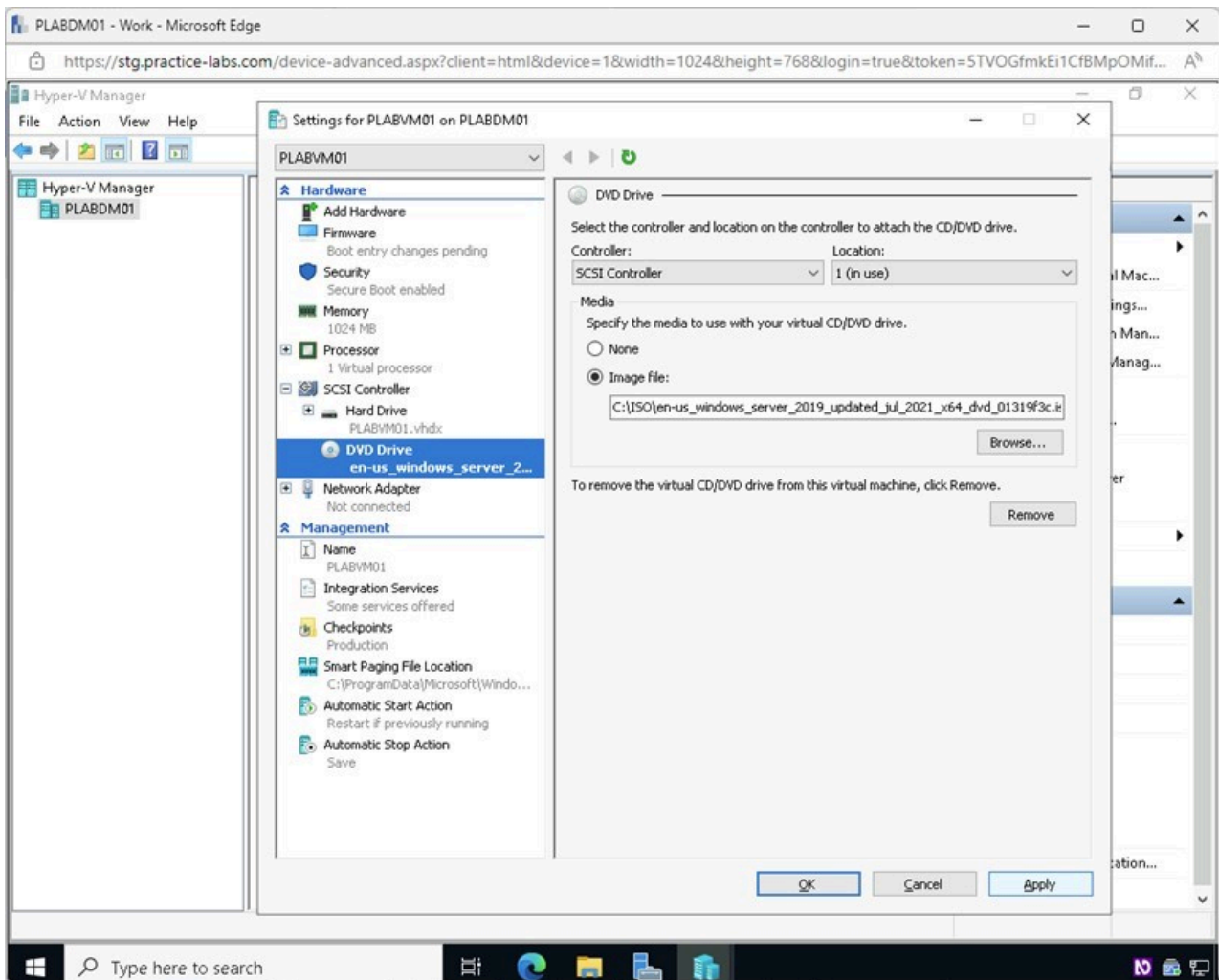


Figure 1.45 Screenshot of PLABDM01: Displaying clicking Apply in the Settings for PLABVM01 on PLABDM01 window.

## Step 8

In the **Settings for PLABVM01 on PLABDM01** window, select **Firmware** under the **Hardware** section.

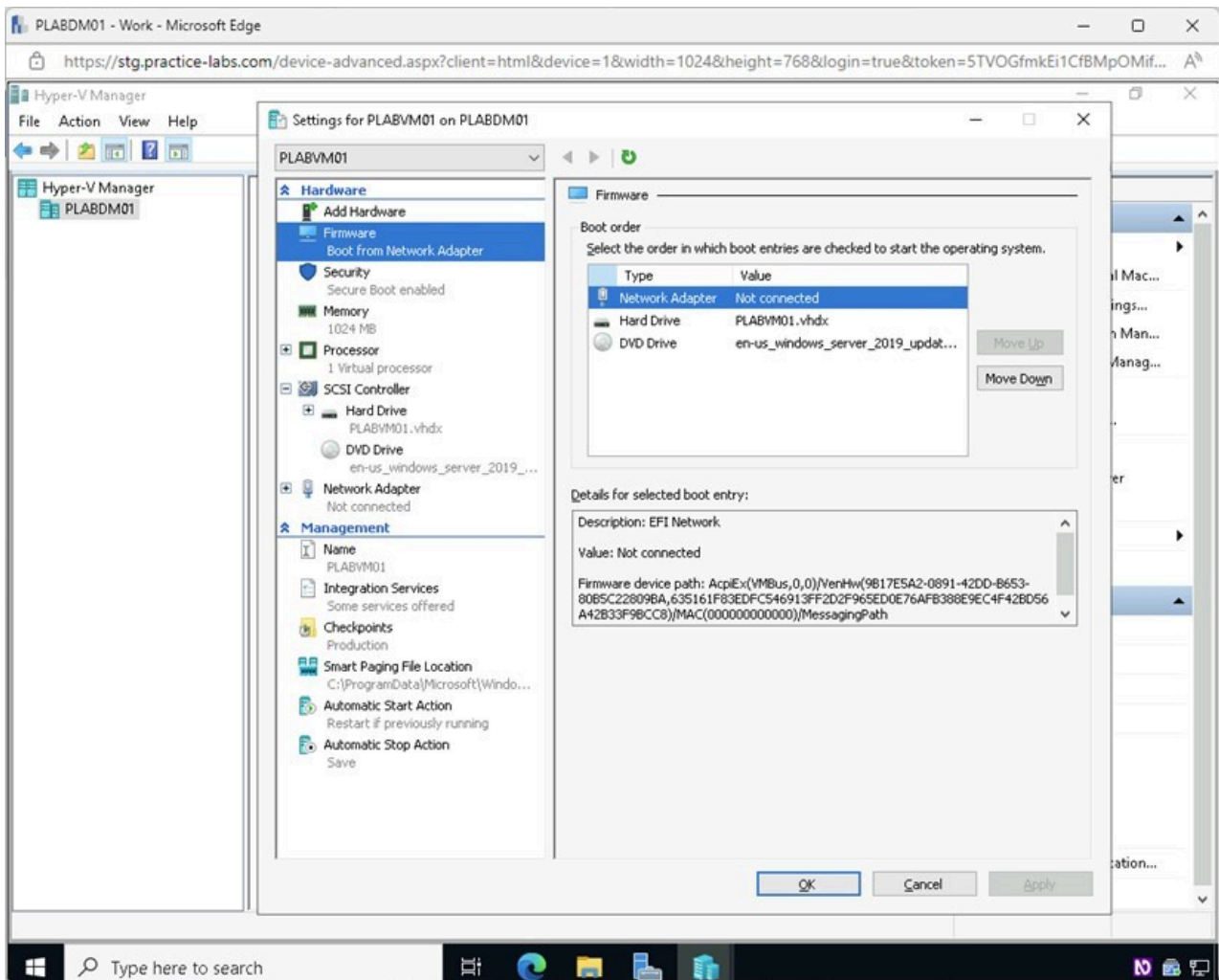


Figure 1.46 Screenshot of PLABDM01: Displaying the Settings for PLABVM01 on PLABDM01 window with Firmware in the Hardware section selected.

## Step 9

In the **Settings for PLABVM01 on PLABDM01** window, in the **Firmware** context pane under **Boot Order**, select the **DVD Drive: en-us\_windows\_server\_2019\_update...**

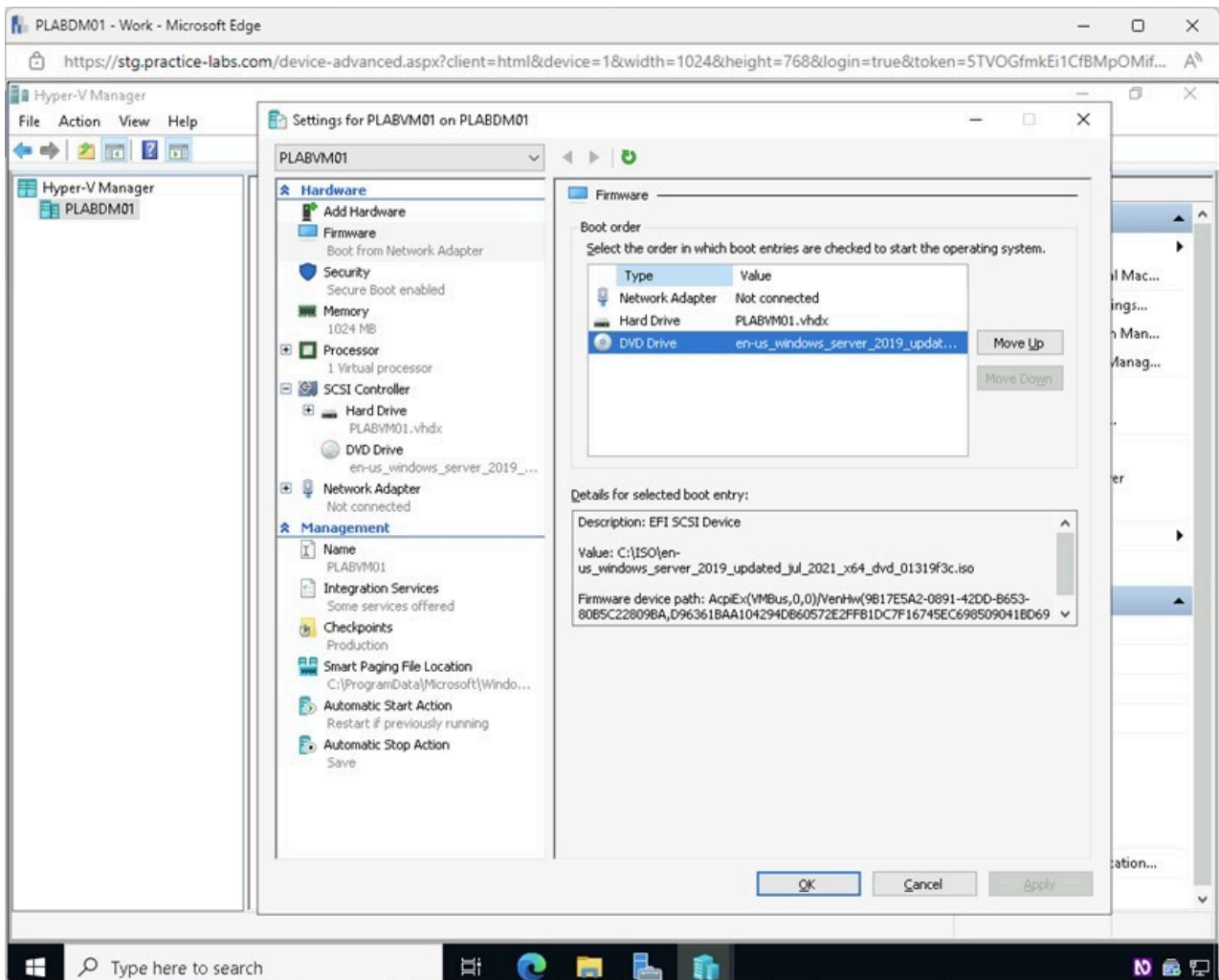


Figure 1.47 Screenshot of PLABDM01: Displaying the Settings for PLABVM01 on PLABDM01 window with DVD Drive under Boot Order selected.

## Step 10

Click the **Move Up** button **twice** until the **DVD Drive** is on top.

Click **OK**.

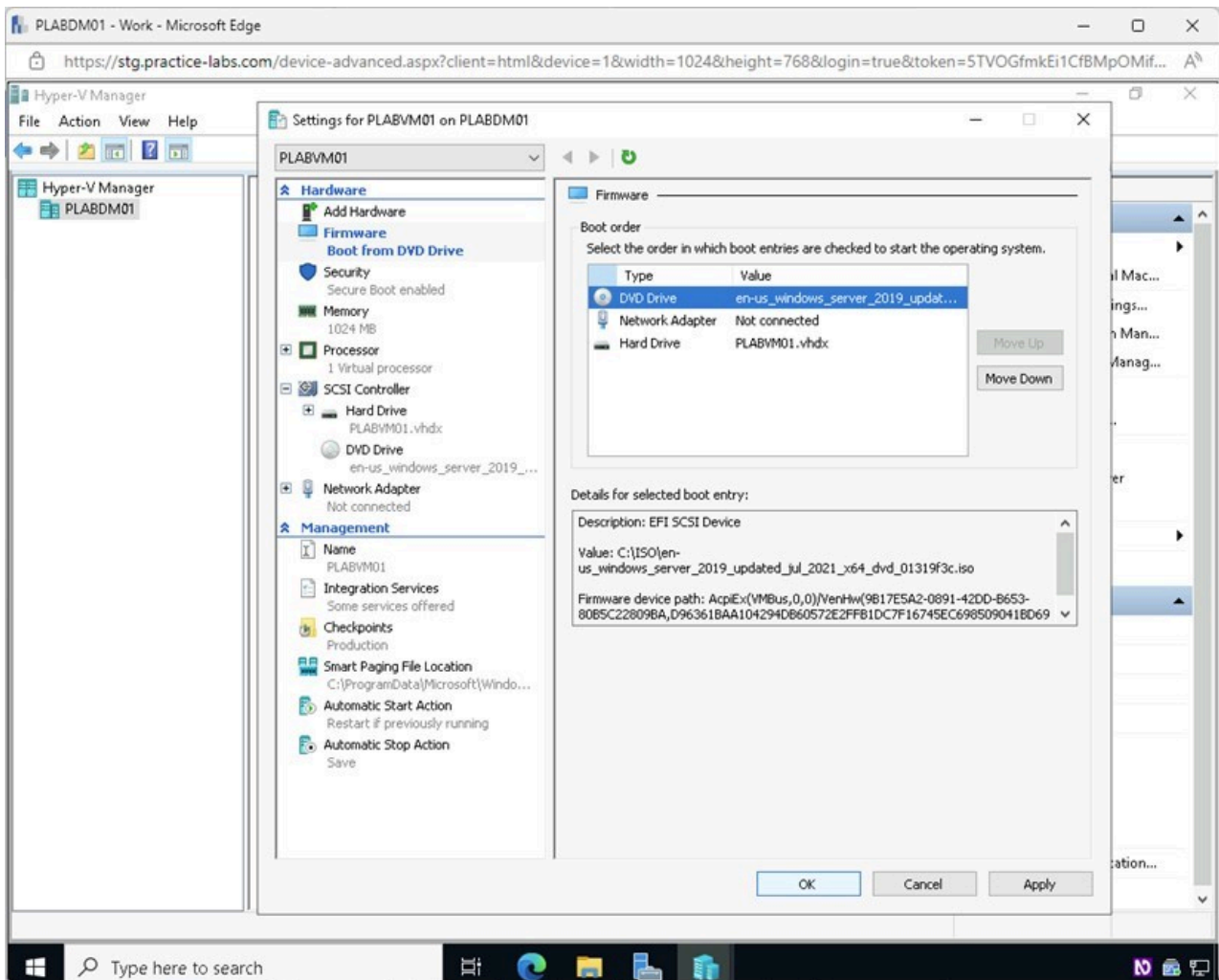


Figure 1.48 Screenshot of PLABDM01: Displaying the Settings for PLABVM01 on PLABDM01 window with the OK button selected.

## Step 11

In the **Hyper-V Manager** window, under the **Virtual Machines** pane, select and right-click on **PLABVM01** and select **Connect**.

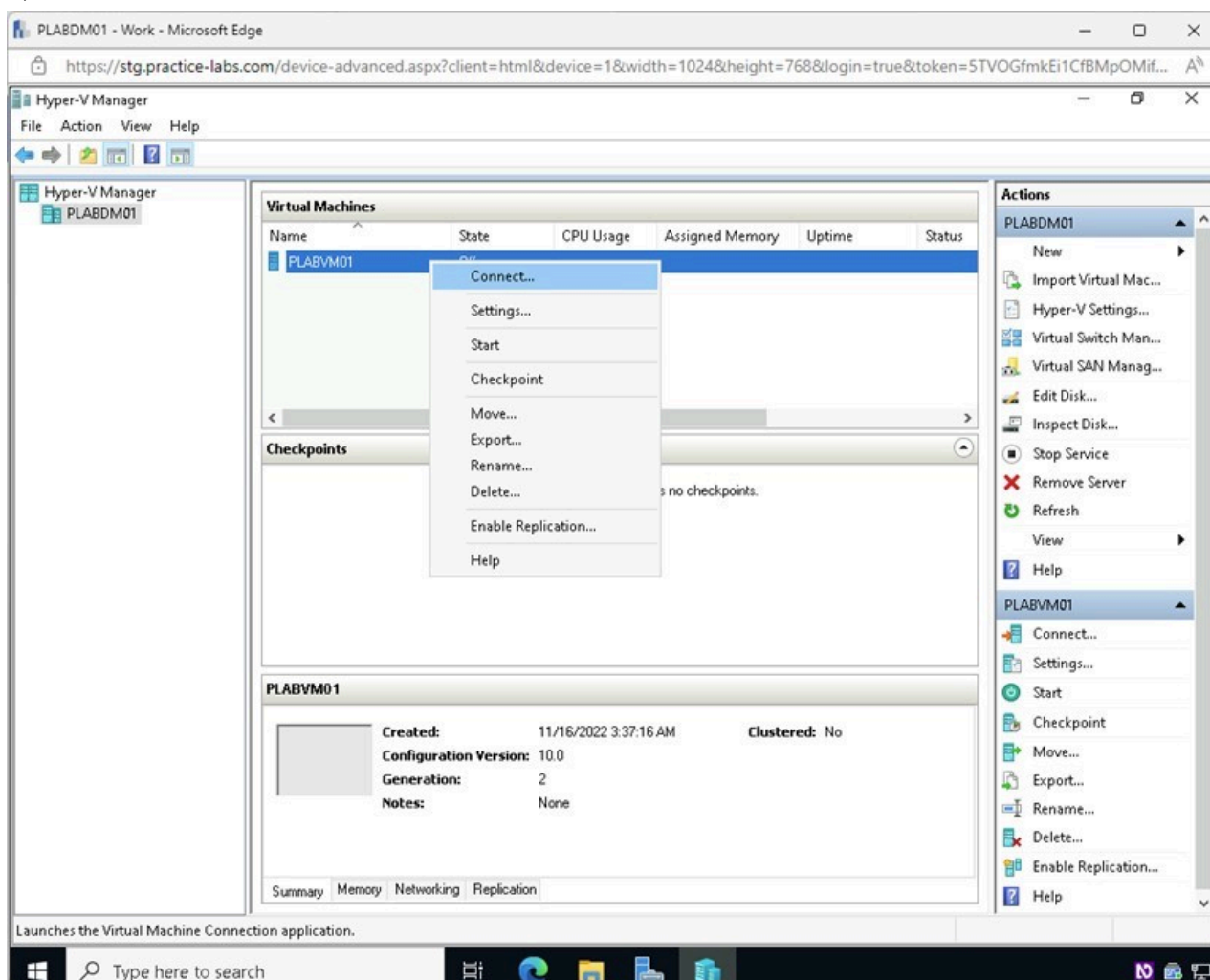


Figure 1.49 Screenshot of PLABDM01: Displaying the Hyper-V Manager window. PLABVM01 virtual machine is right-clicked, and Connect selected from the context menu.

## Step 12

In the **PLABVM01 on PLABDM01 - Virtual Machine Connection** window, click the **Start** button.

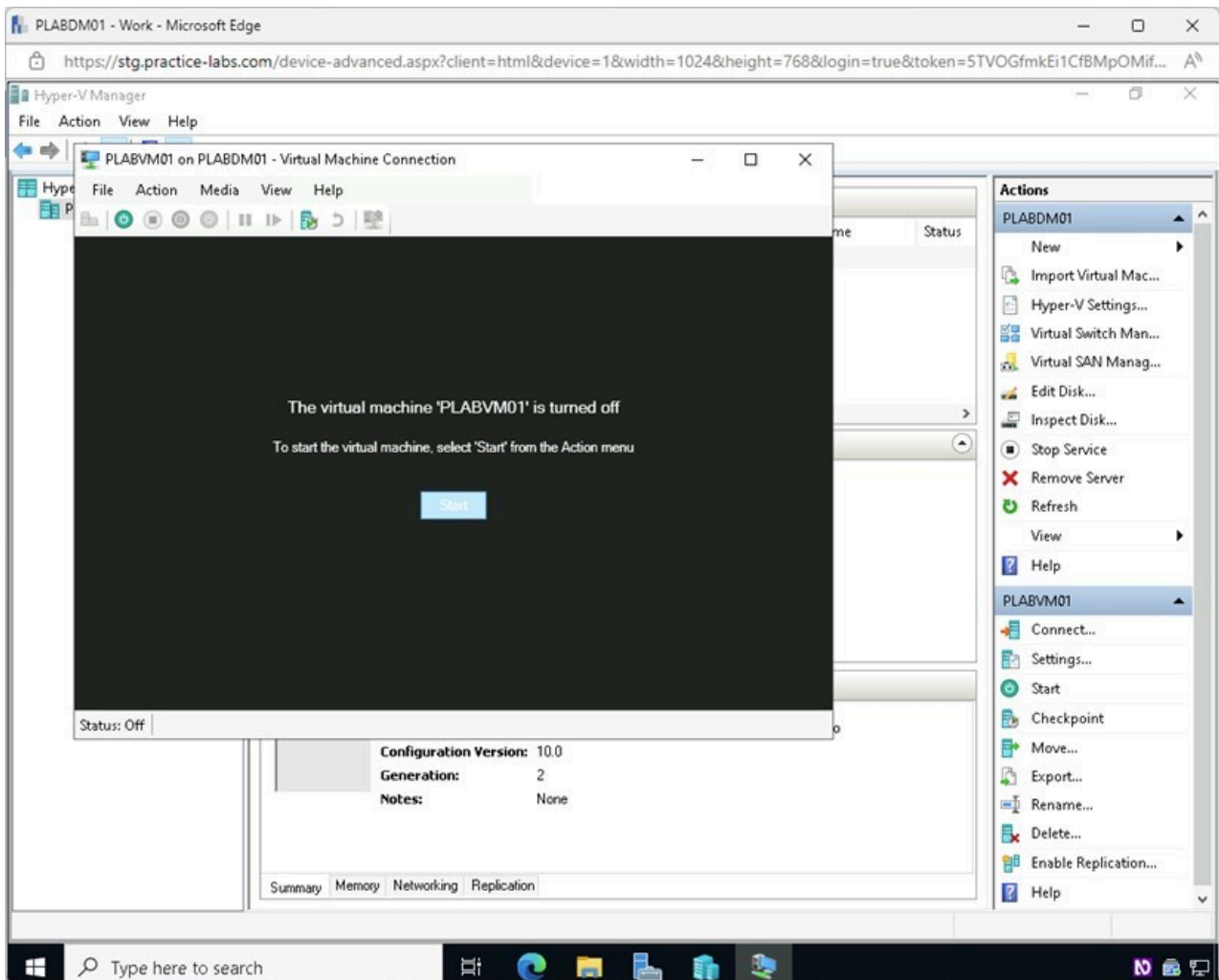


Figure 1.50 Screenshot of PLABDM01: Displaying the PLABVM01 on PLABDM01 - Virtual Machine Connection window with the Start button selected.

## Step 13

In the **PLABVM01 on PLABDM01 - Virtual Machine Connection** window, on the **Press any key to boot from CD or DVD...** screen, press the spacebar on the keyboard to start the installation.

Ensure the spacebar is pressed as soon as you see the **Press any key to boot from CD or DVD...** screen. If you do not press the spacebar in time, click the **Ctrl+Alt+Delete** icon (the first icon in the upper left corner) on the menu bar. Once the device restarts again, press the spacebar in time.

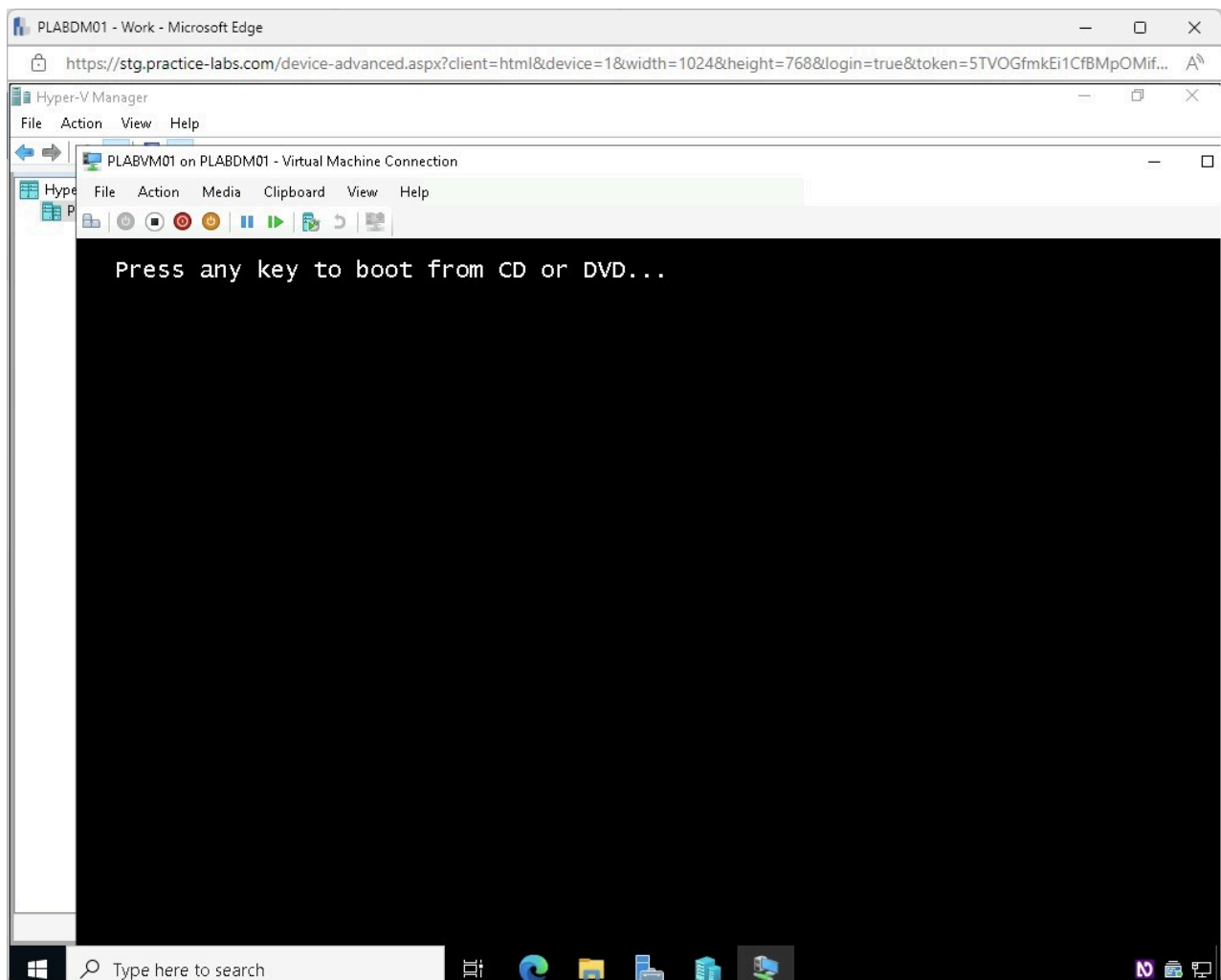


Figure 1.51 Screenshot of PLABDM01: Displaying the Press any key to boot from CD or DVD... screen in the PLABVM01 on PLABDM01 - Virtual Machine Connection window.

## Step 14

In the **PLABVM01 on PLABDM01 - Virtual Machine Connection** window, select the **Maximize** button.

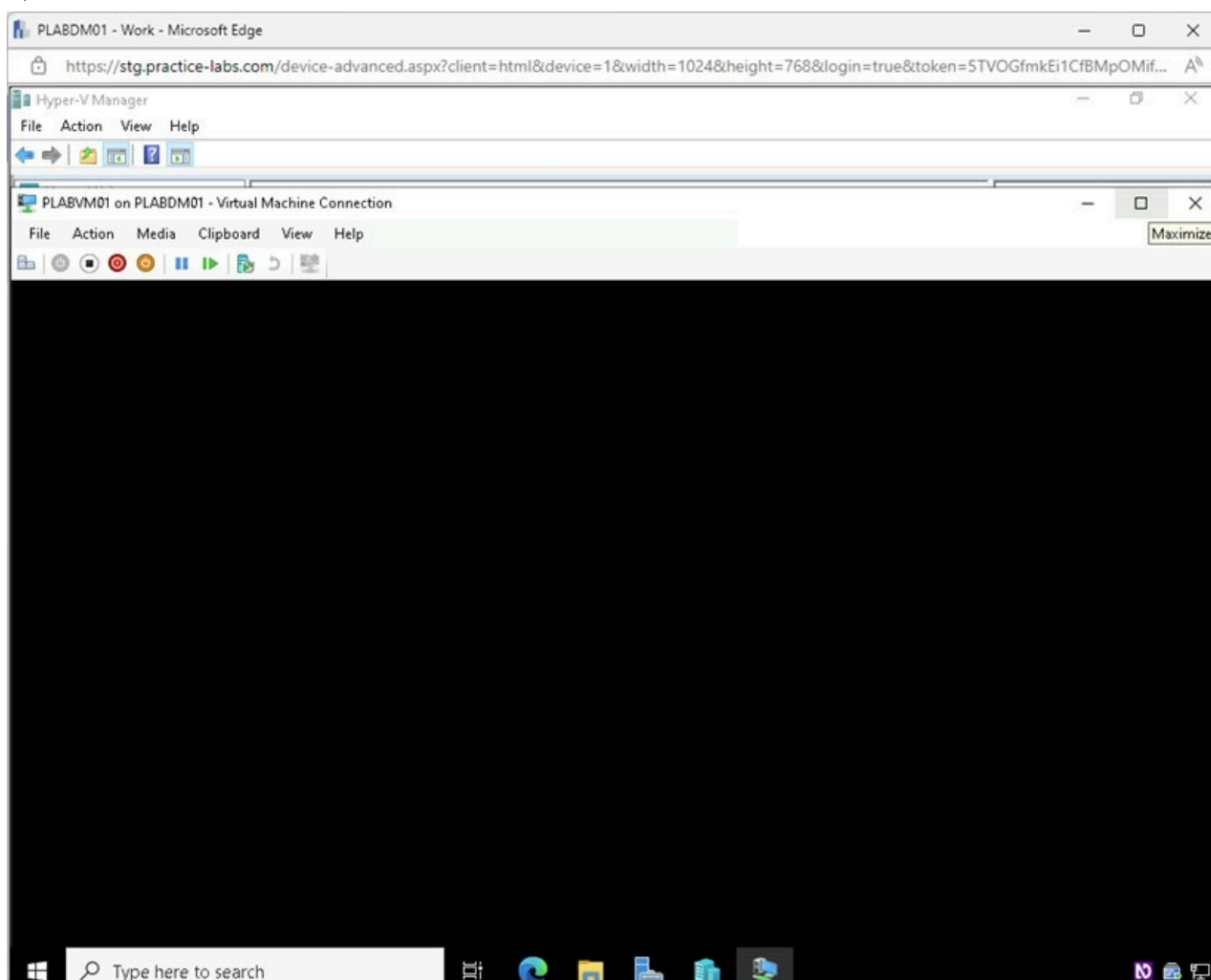


Figure 1.52 Screenshot of PLABDM01: Displaying clicking the Maximize icon on the PLABVM01 on PLABDM01 - Virtual Machine Connection window.

## Step 15

In the **Windows Setup** installation window, review the regional options and click **Next**.

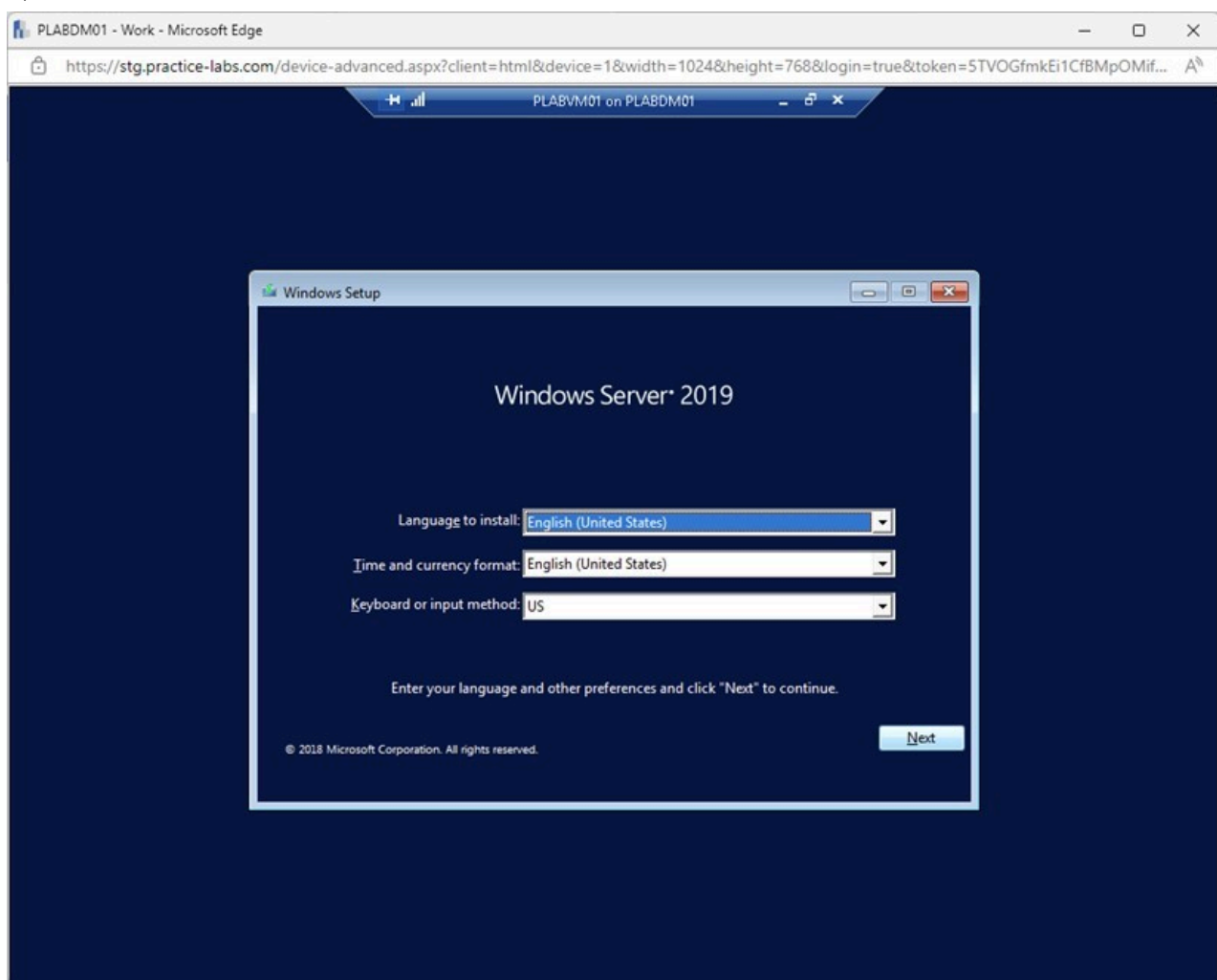


Figure 1.53 Screenshot of PLABDM01: Displaying clicking Next in the Windows Setup installation window.

## Step 16

In the **Windows Setup** installation window, click **Install now**.

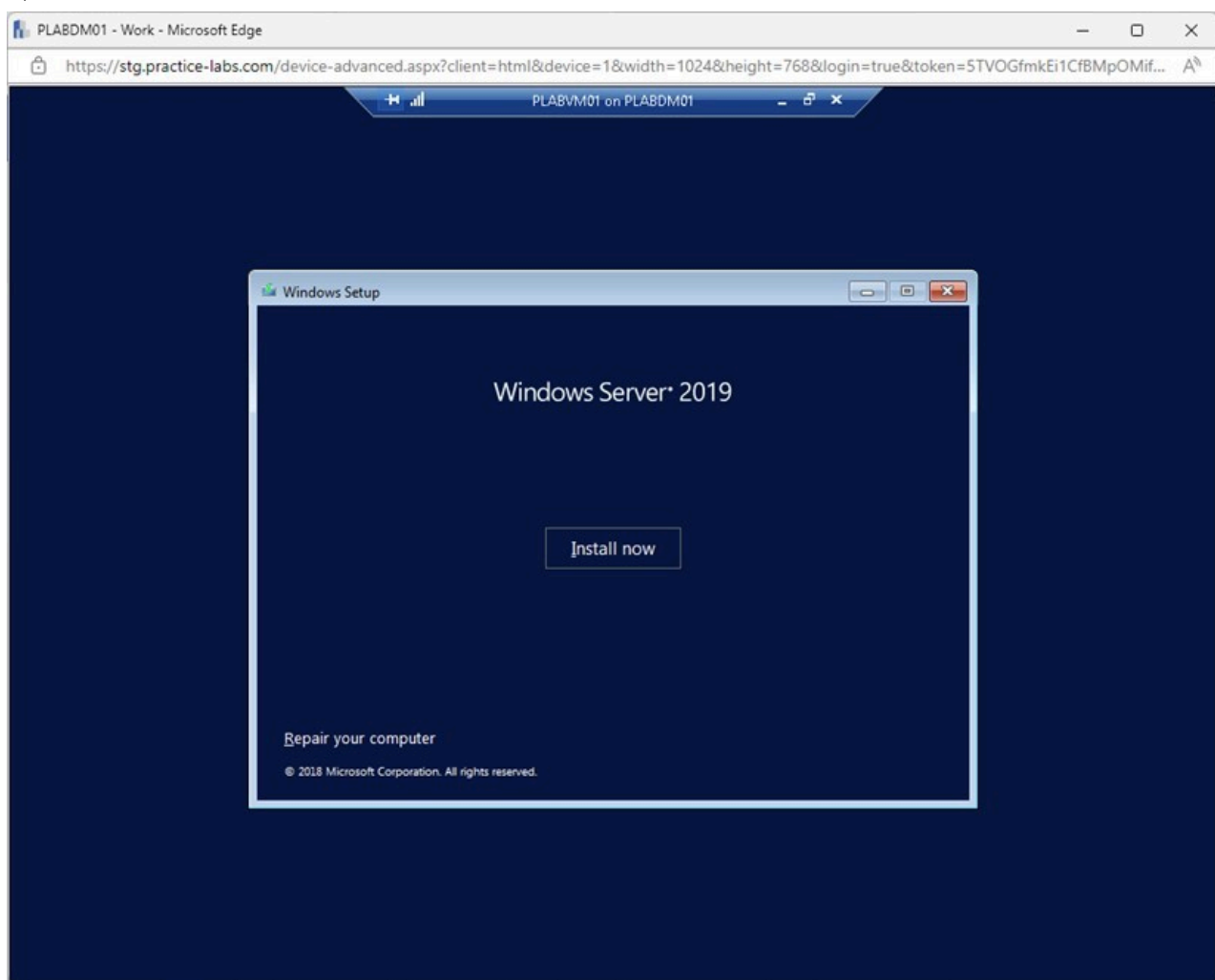


Figure 1.54 Screenshot of PLABDM01: Displaying clicking Install now in the Windows Setup installation window.

## Step 17

In the **Windows Setup** installation window, observe the start of the installation process.

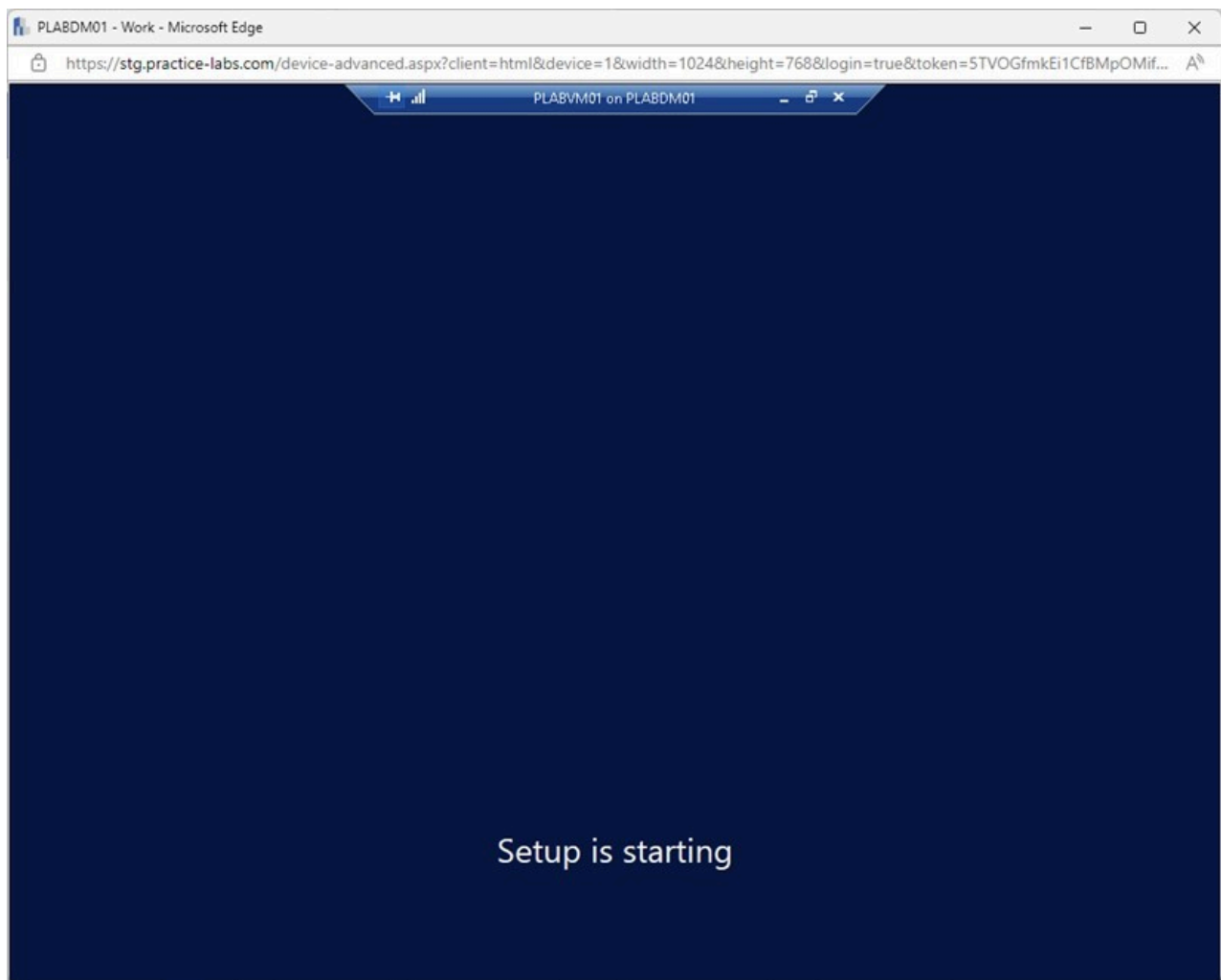


Figure 1.55 Screenshot of PLABDM01: Displaying the Windows installation process.

## Step 18

In the **Windows Setup - Select the operating system you want to install** page, select **Windows Server 2019 Standard (Desktop Experience)**.

Click **Next**.

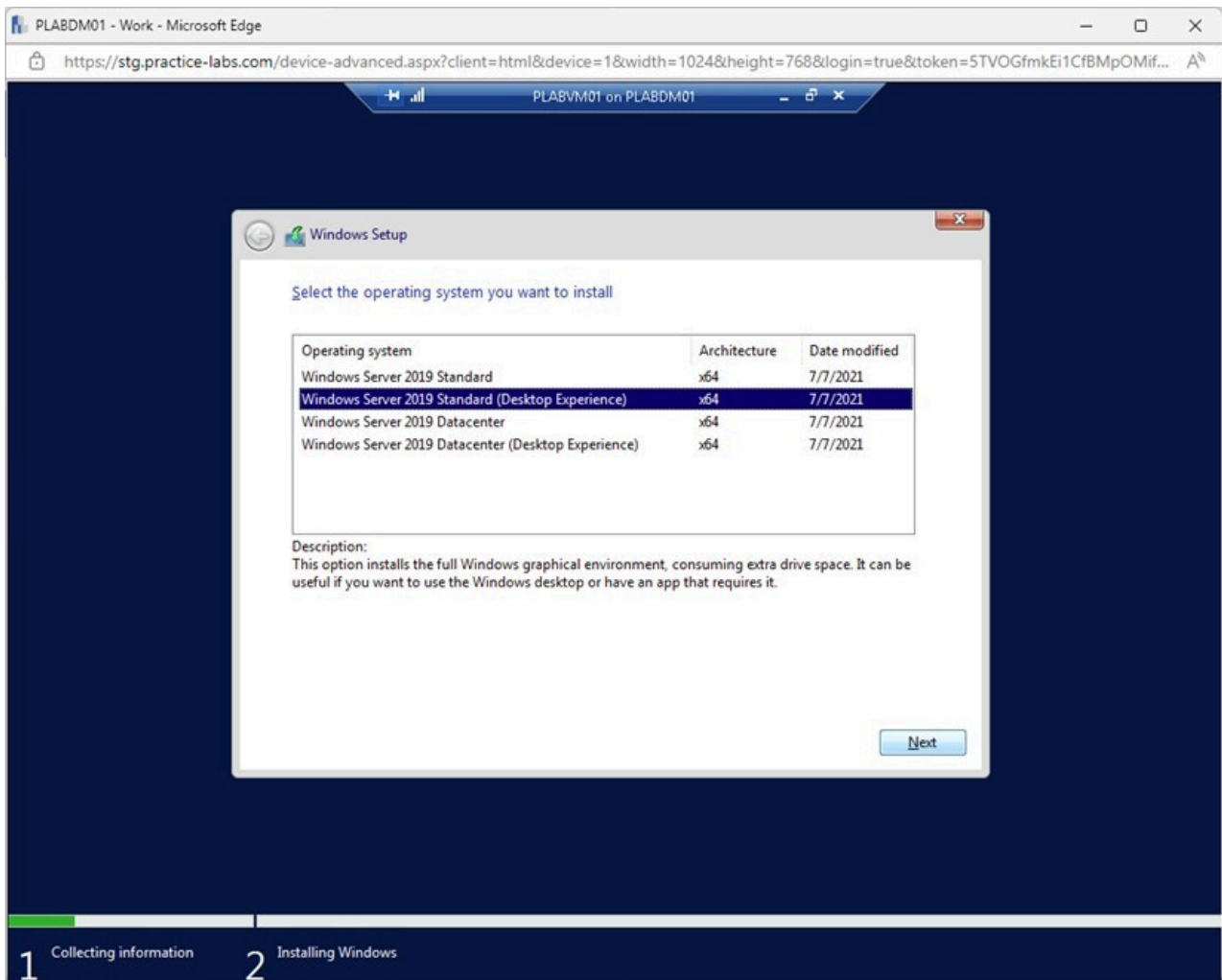


Figure 1.56 Screenshot of PLABDM01: Displaying the Windows Setup - Select the operating system you want to install page with the required option selected and the Next button highlighted.

## Step 19

In the **Applicable notices and license terms** page, select the checkbox next to **I accept the license terms**.

Click **Next**.

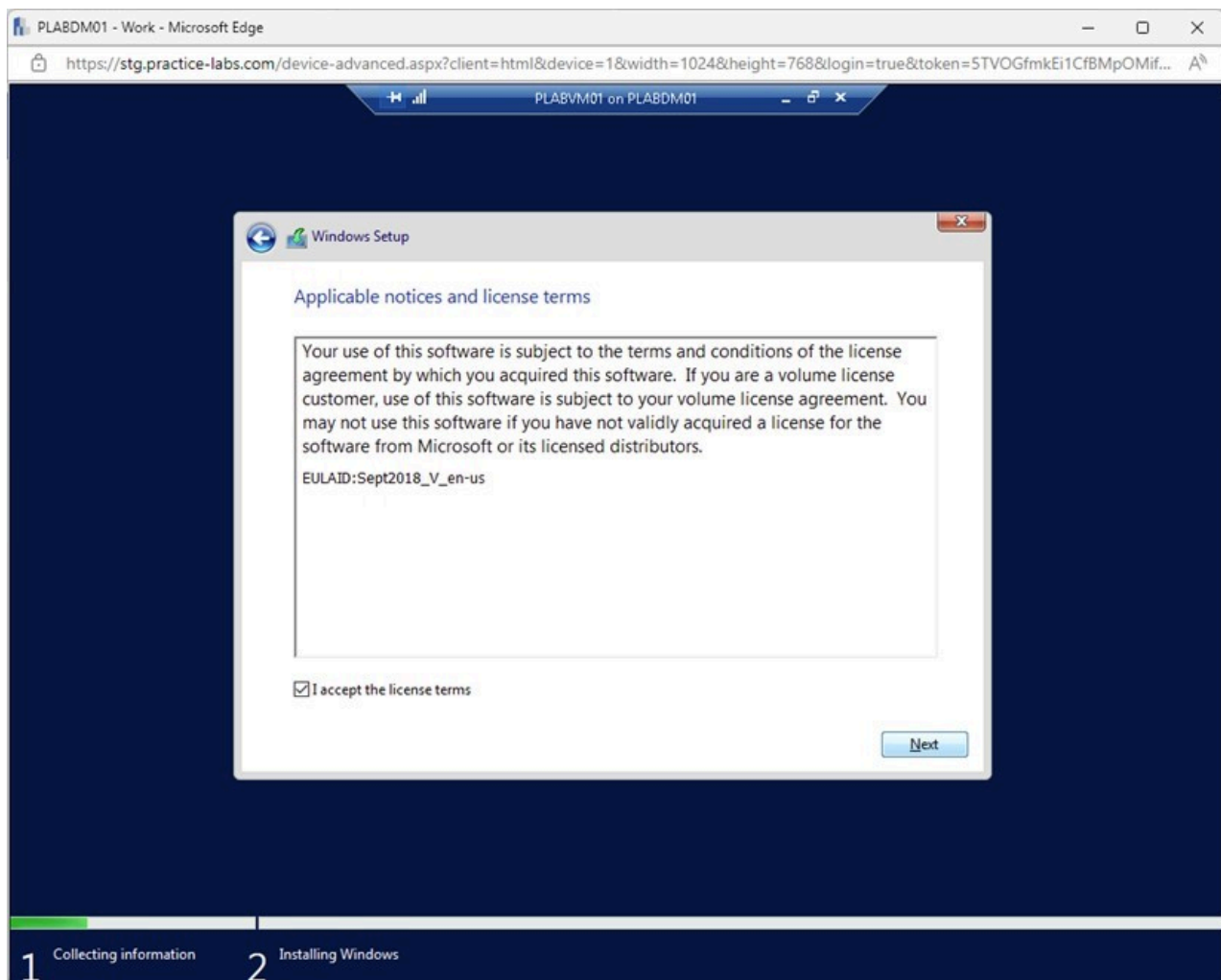


Figure 1.57 Screenshot of PLABDM01: Displaying the Applicable notices and license terms page with the Next button highlighted.

## Step 20

In the **Which type of installation do you want?** page, select **Custom: Install Windows only (advanced)**.

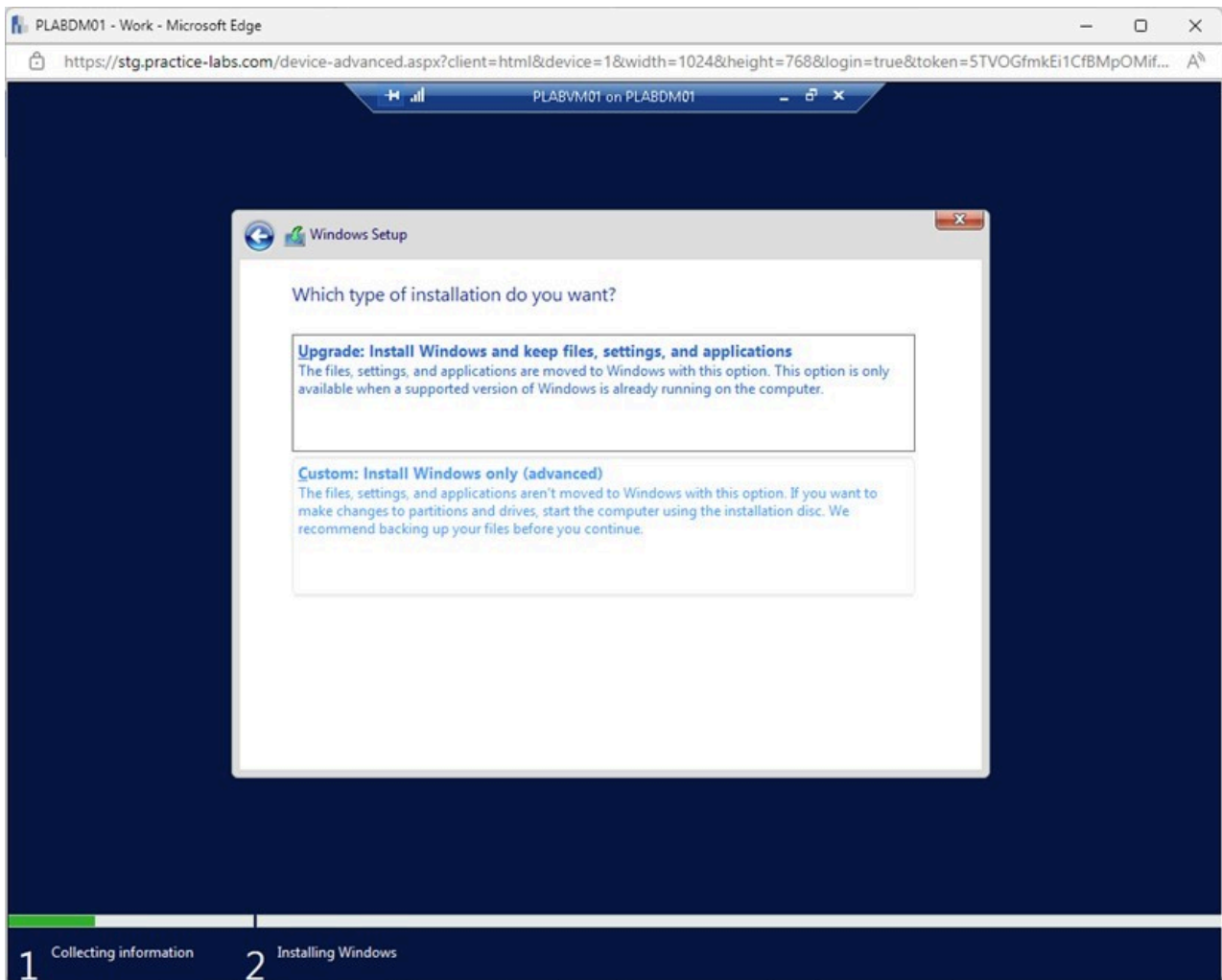


Figure 1.58 Screenshot of PLABDM01: Displaying the Windows Setup - Which type of installation do you want? page with the required option selected.

## Step 21

In the **Where do you want to install Windows?** page, select **Drive 0 Unallocated Space**.

Click **Next**.

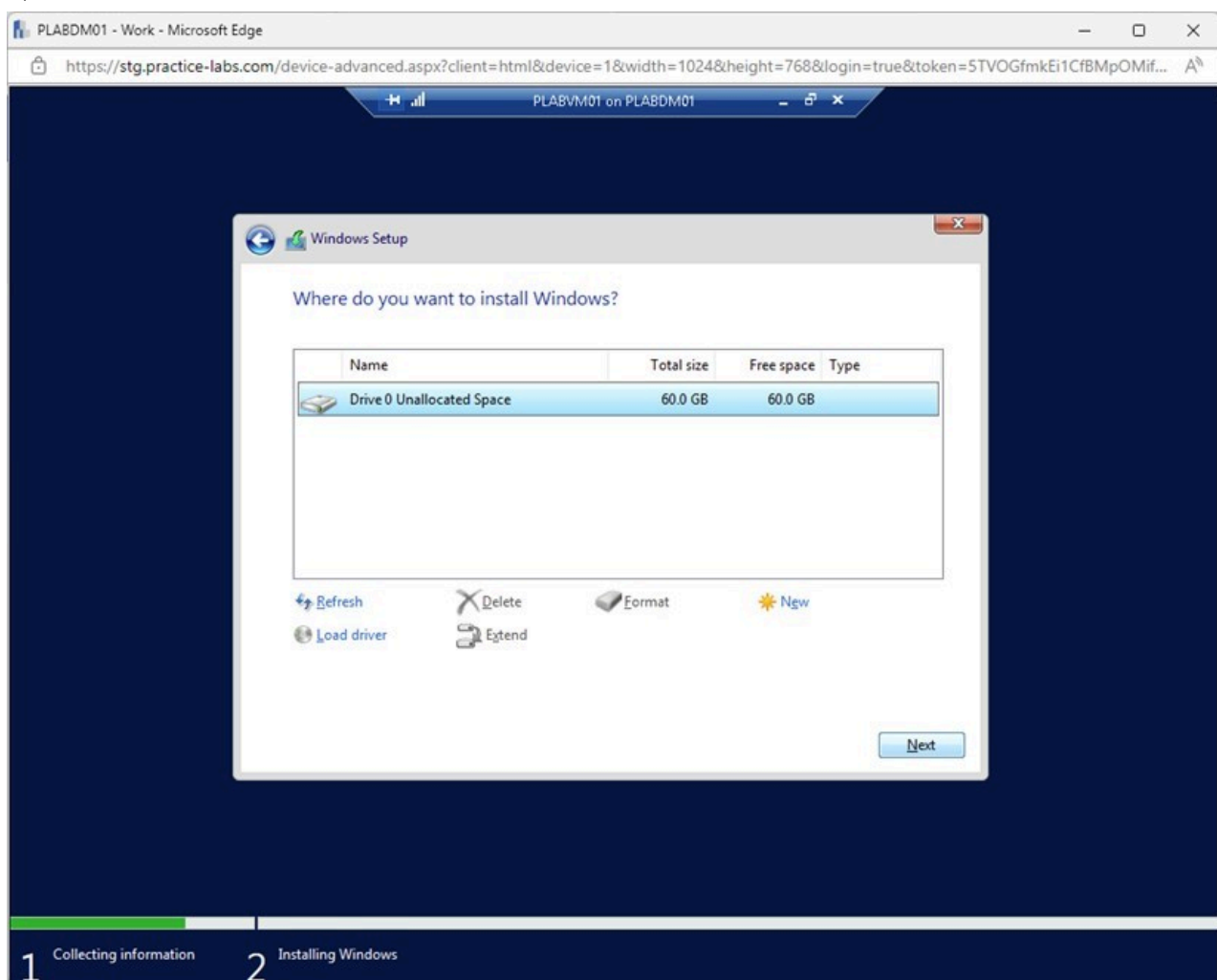


Figure 1.59 Screenshot of PLABDM01: Displaying the Where do you want to install Windows? page with the required option selected and the Next button highlighted.

## Step 22

In the **Windows Setup** installation window, under **Installing Windows**, observe the installation process.

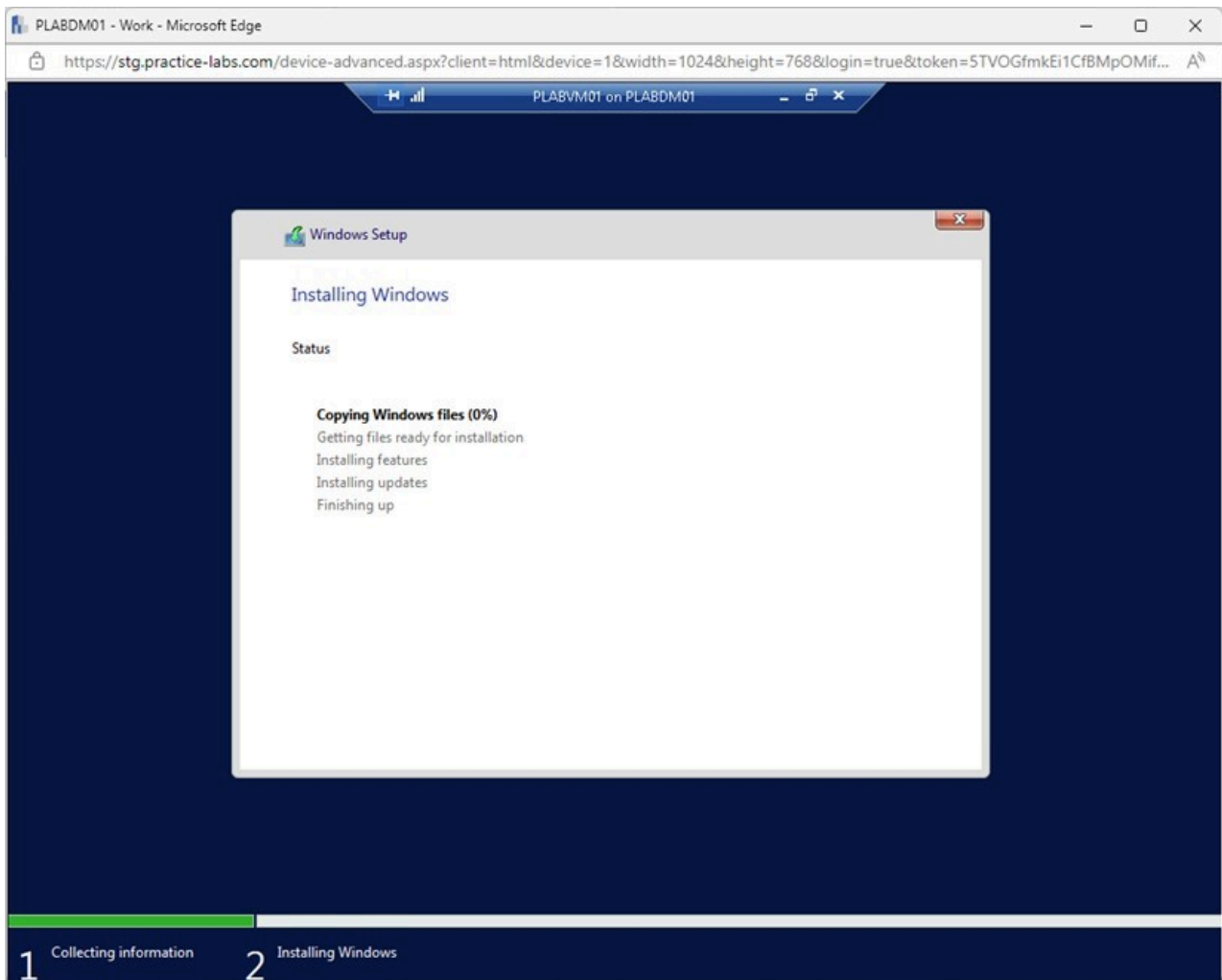


Figure 1.60 Screenshot of PLABDM01: Displaying the Windows - Setup Installing Windows window.

**Note:** This process could take a few minutes, and the virtual machine may reboot a few times. The system will restart automatically after the **Windows needs to restart to continue** page.

## Step 23

In the **PLABVM01 on PLABDM01** window, observe the continued installation process.

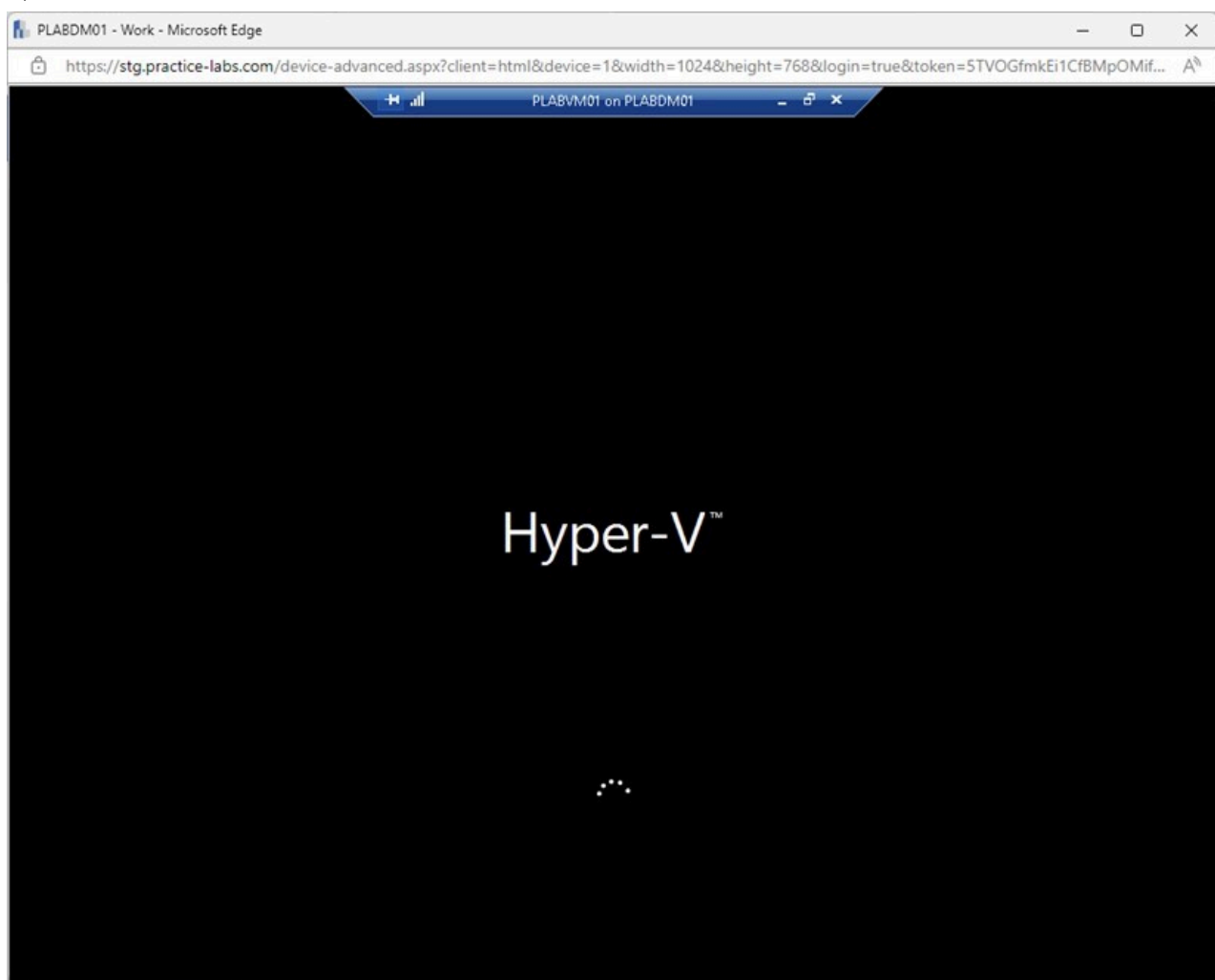


Figure 1.61 Screenshot of PLABDM01: Displaying the installation process in the PLABVM01 on PLABDM01 window.

## Step 24

In the **Customize settings** window, type the following for the **Password** and **Reenter password** field:

**Passw0rd**

Click **Finish**.

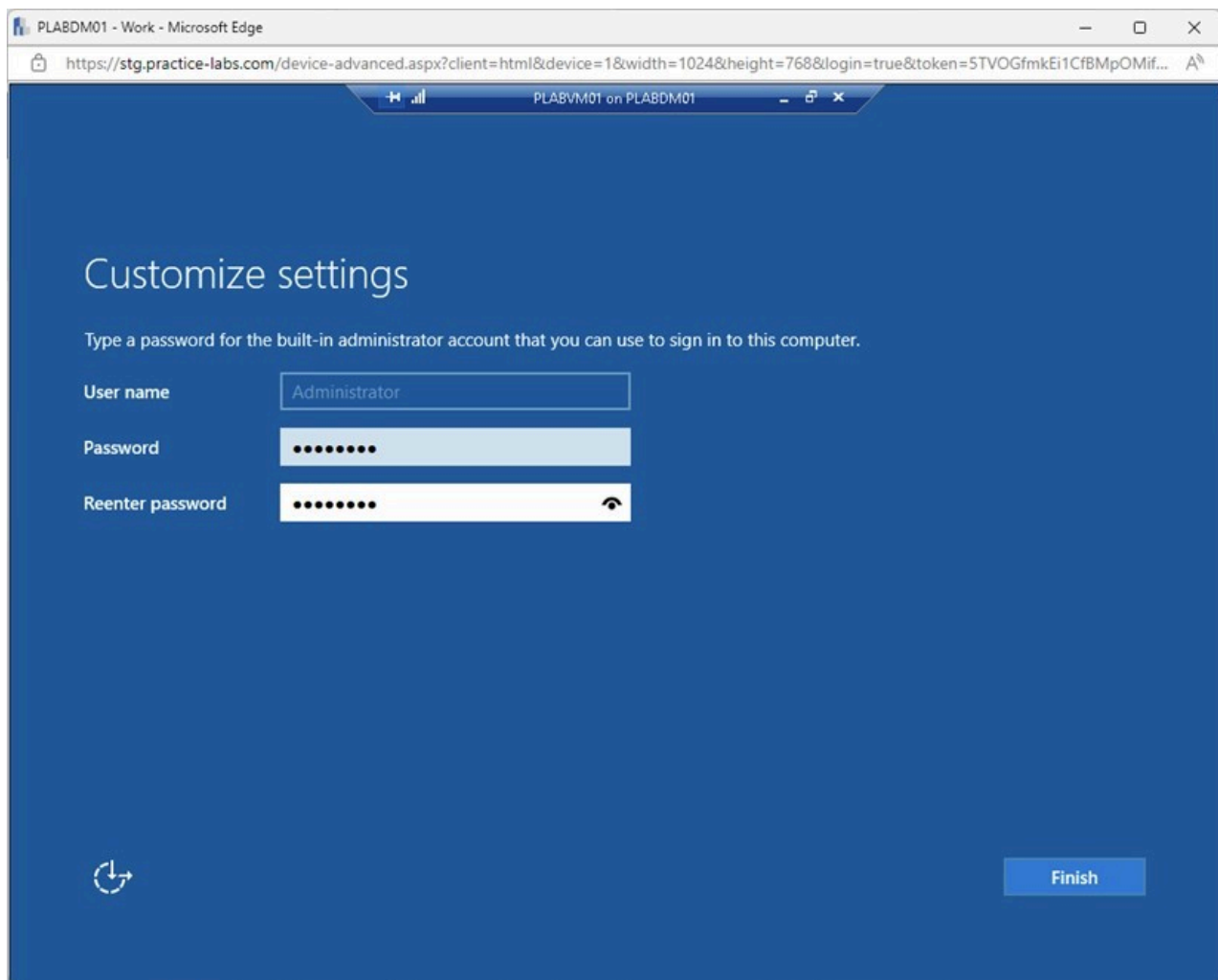


Figure 1.62 Screenshot of PLABDM01: Displaying the Customize settings window with the required password entered and the Finish button selected.

## Step 25

In the **PLABVM01 on PLABDM01** window, click the **Restore Down** icon.



Figure 1.63 Screenshot of PLABDM01: Displaying the PLABVM01 on PLABDM01 window with the Restore Down icon selected.

## Step 26

In the **PLABVM01 on PLABDM01 - Virtual Machine Connection** window, click the **Ctrl+Alt+Delete** actions icon.

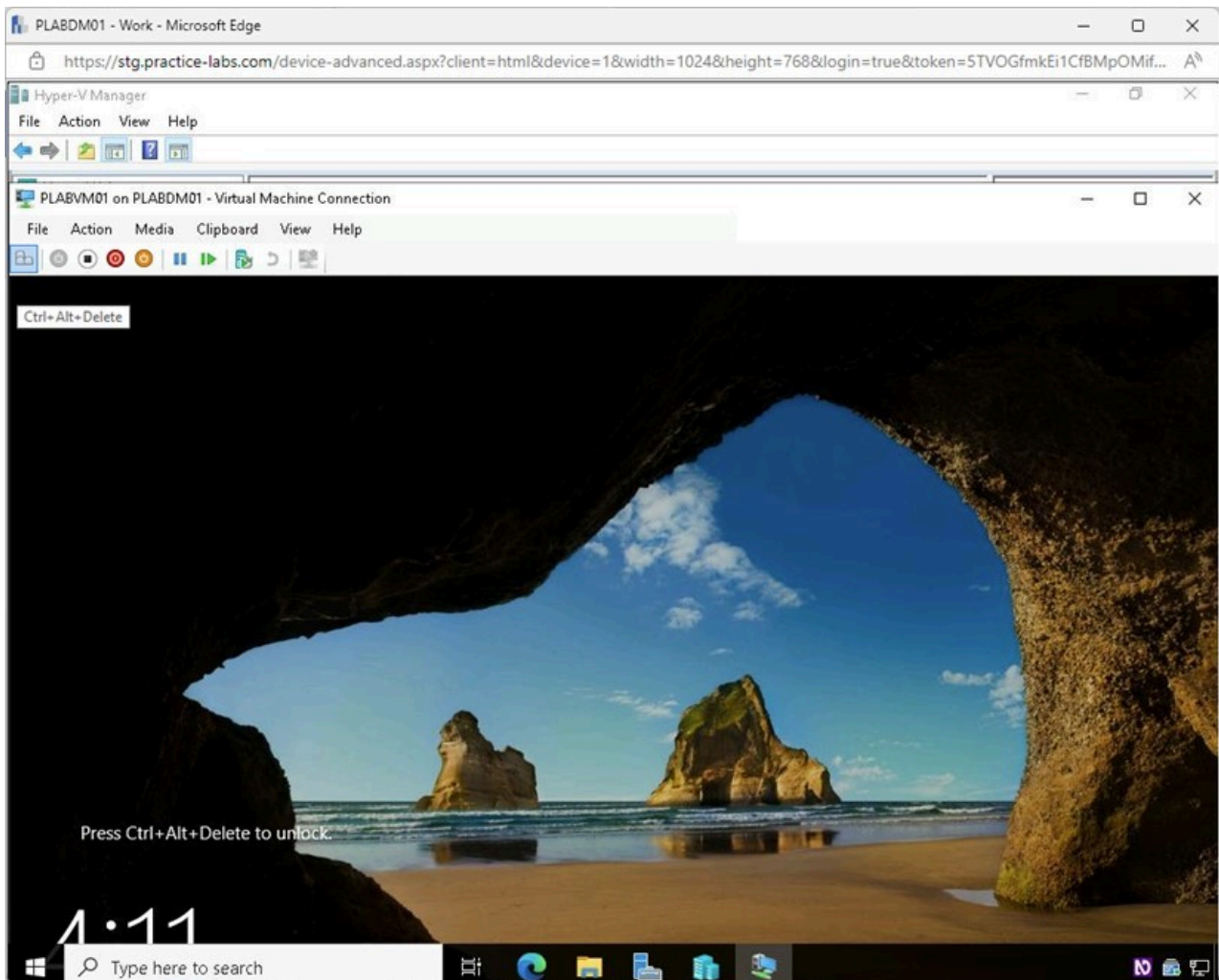


Figure 1.64 Screenshot of PLABDM01: Displaying selecting the Ctrl+Alt+Delete actions icon in the PLABVM01 on PLABDM01 - Virtual Machine Connection window.

## Step 27

In the **PLABVM01 on PLABDM01 - Virtual Machine Connection** window, log in as **Administrator**.

Type the following password and press **Enter**:

Passw0rd

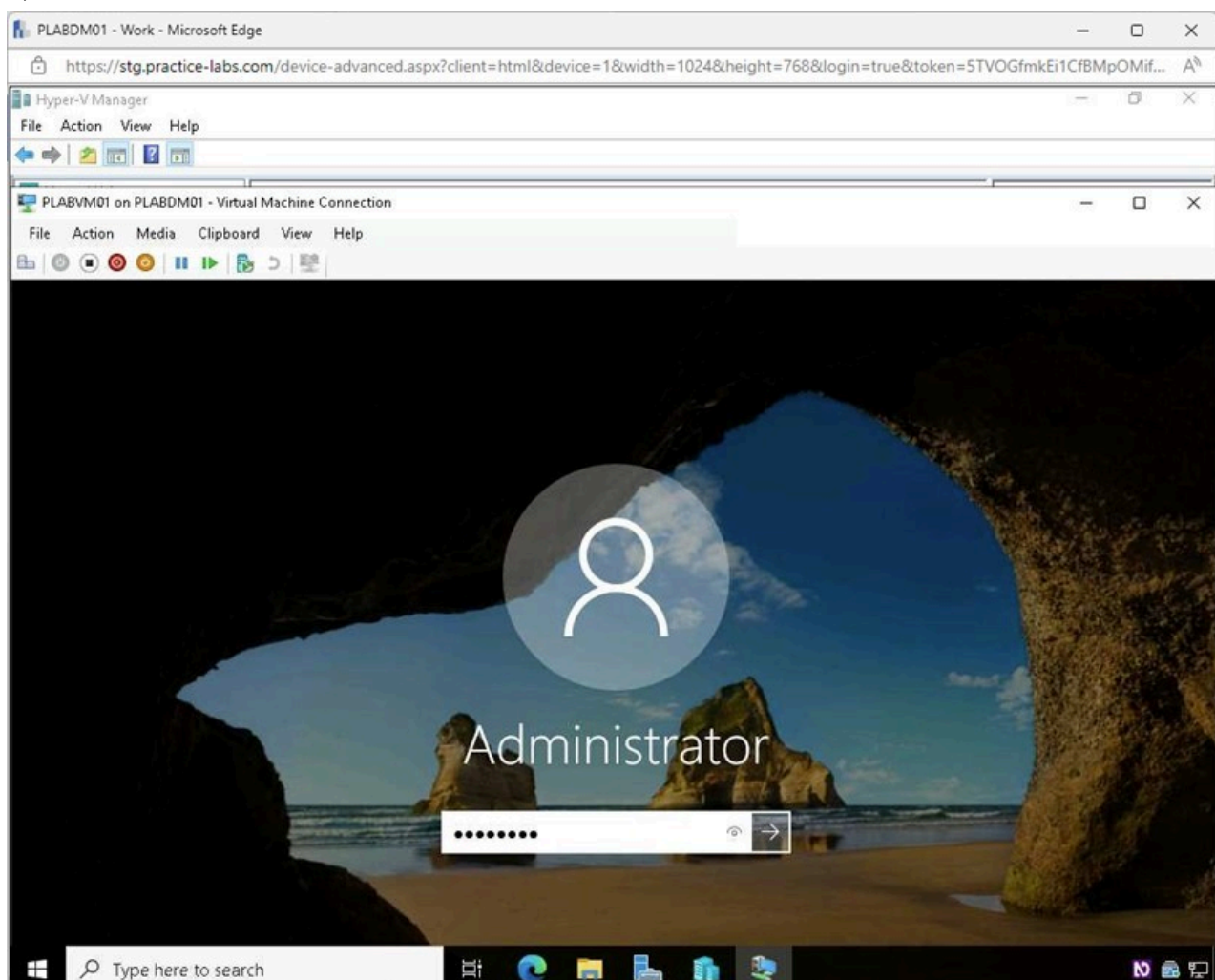


Figure 1.65 Screenshot of PLABDM01: Displaying typing the required password for the Administrator account in the PLABVM01 on PLABDM01 - Virtual Machine Connection window.

## Step 28

In the **PLABVM01 on PLABDM01 - Virtual Machine Connection** window, click the **Maximize** icon.

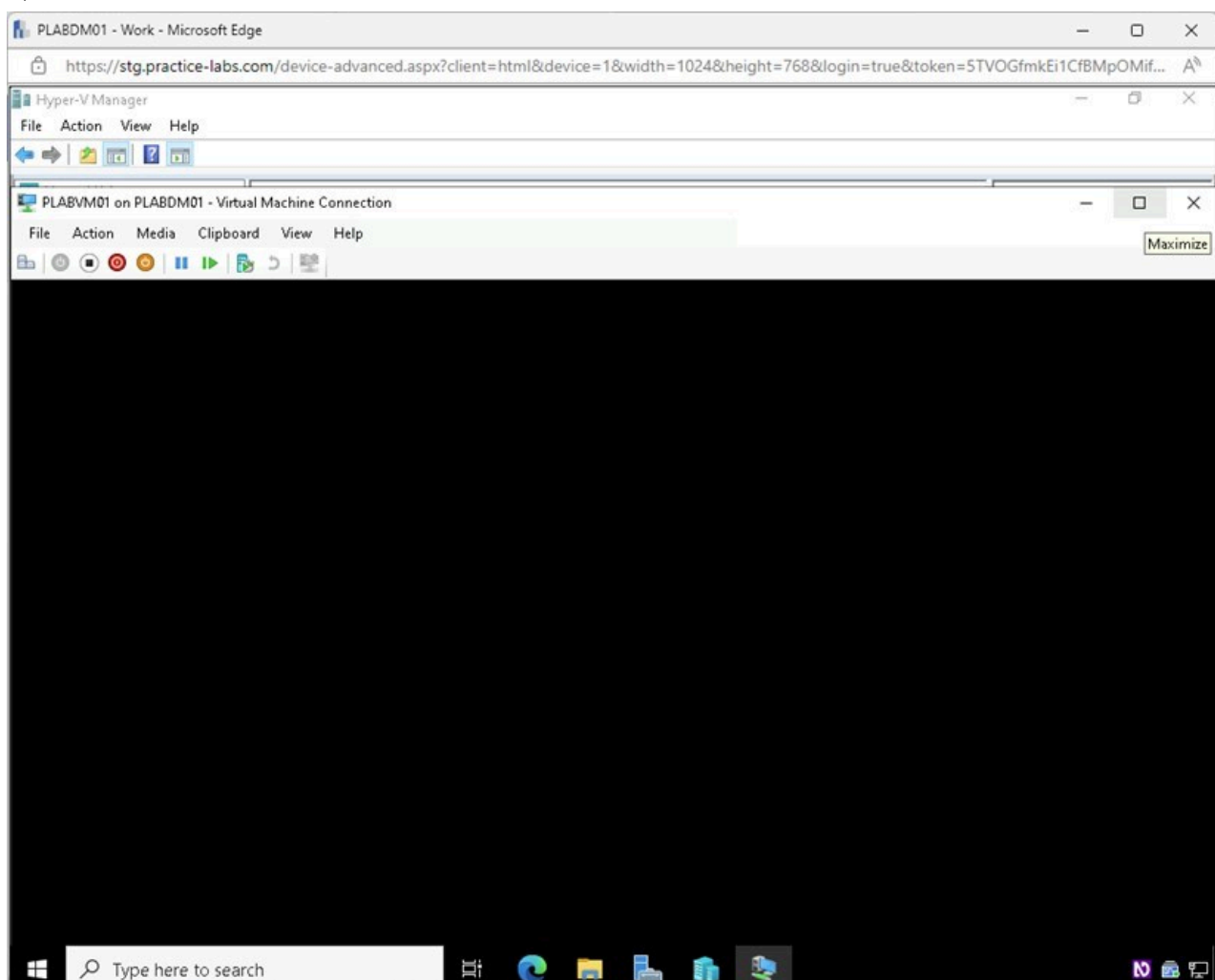


Figure 1.66 Screenshot of PLABDM01: Displaying selecting the Maximize icon in the PLABVM01 on PLABDM01 - Virtual Machine Connection window.

## Step 29

In the **PLABVM01 on PLABDM01** window, close the **Server Manager** pop-up box.

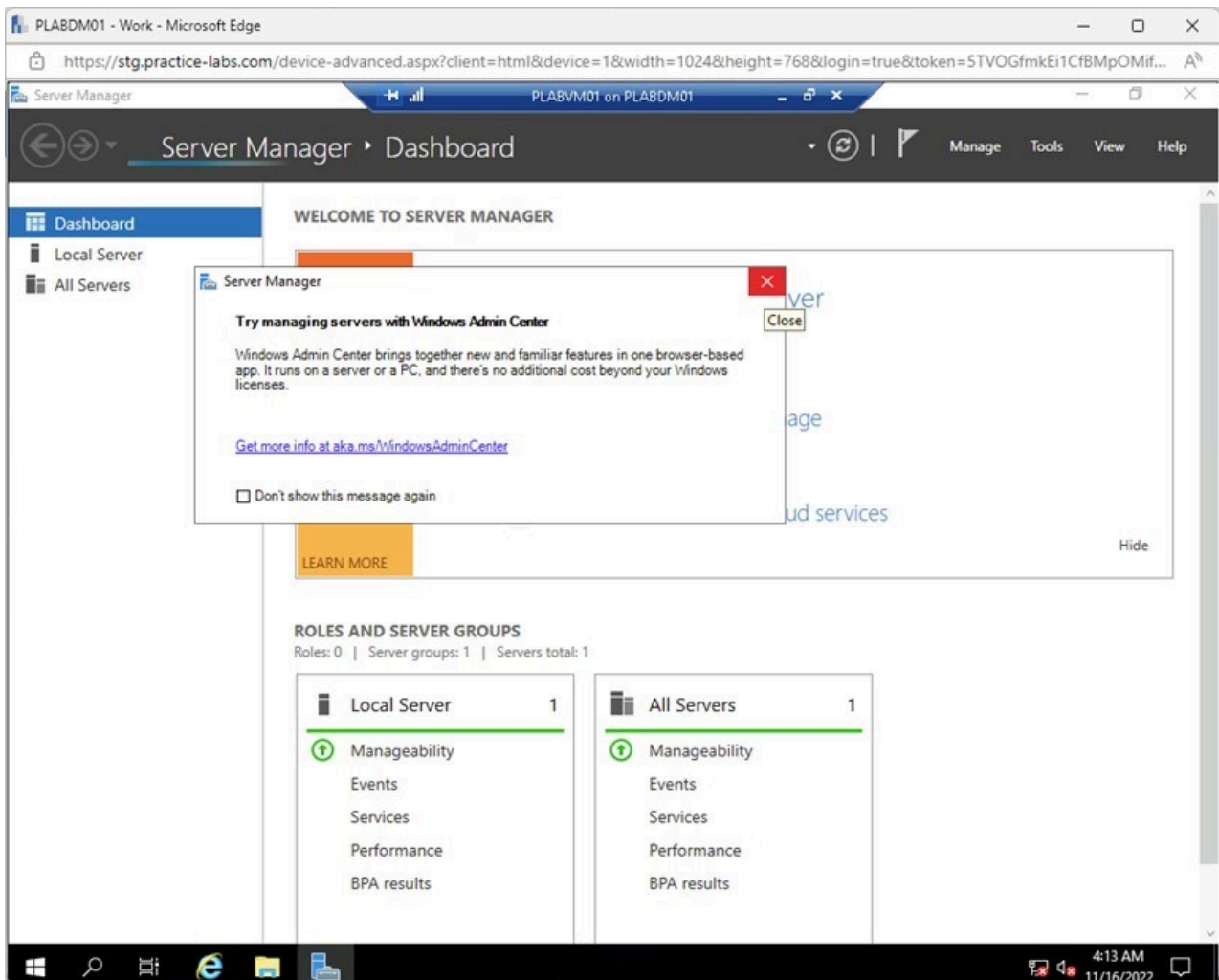


Figure 1.67 Screenshot of PLABDM01: Displaying closing the Server Manager pop-up box in the PLABVM01 on PLABDM01 window.

**Note:** You may need to click and drag the windows down to gain access to the Close icon.

## Step 30

In the **PLABVM01 on PLABDM01** window, select **Local Server** in the left pane on the **Server Manager** screen.

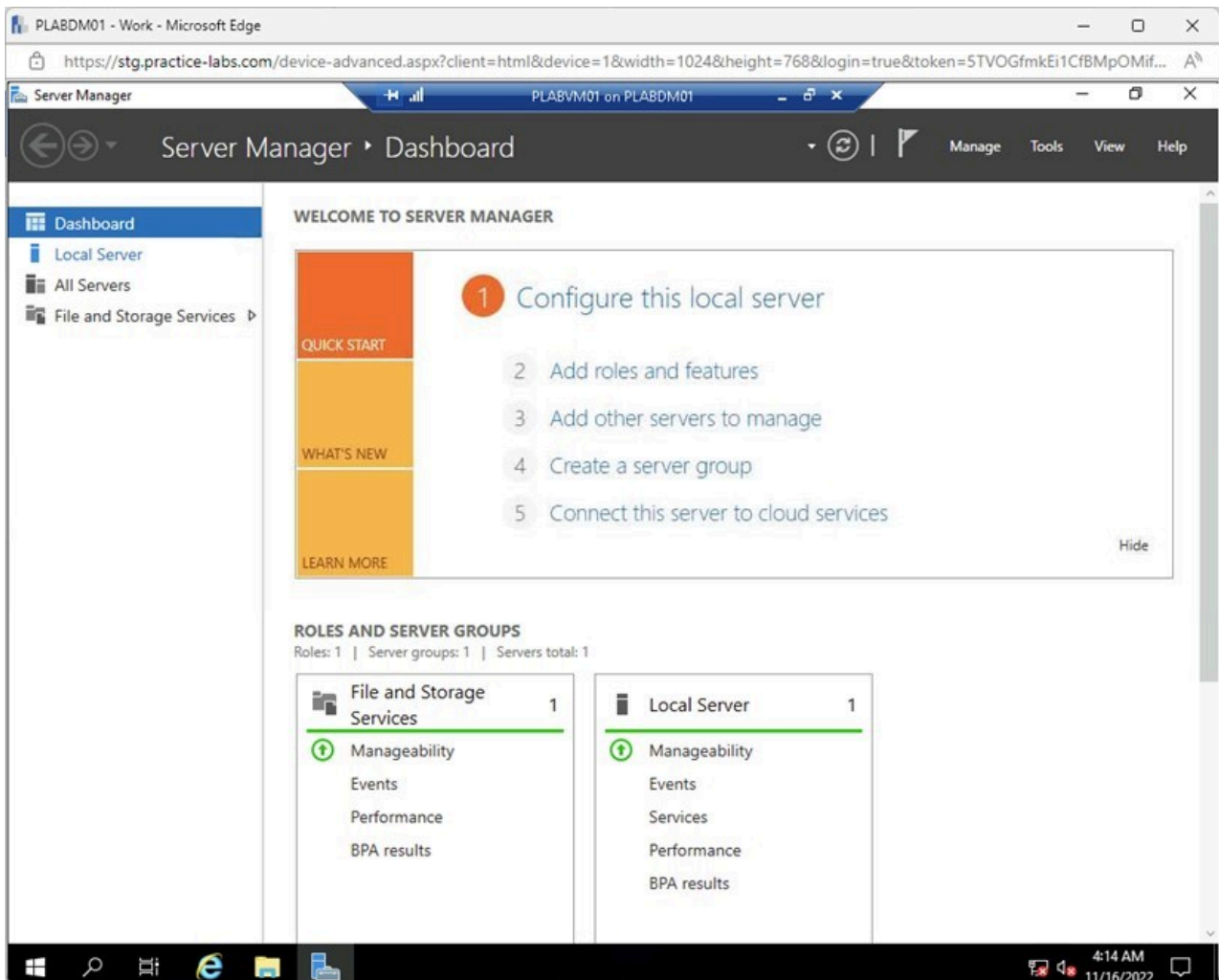


Figure 1.68 Screenshot of PLABDM01: Displaying the PLABVM01 on PLABDM01 window. Local Server is selected on the Server Manager screen.

## Step 31

In the **Server Manager > Local Server** screen, under the **PROPERTIES** of the Local Server next to **Hardware information**, verify that **Microsoft Corporation Virtual Machine** is present.

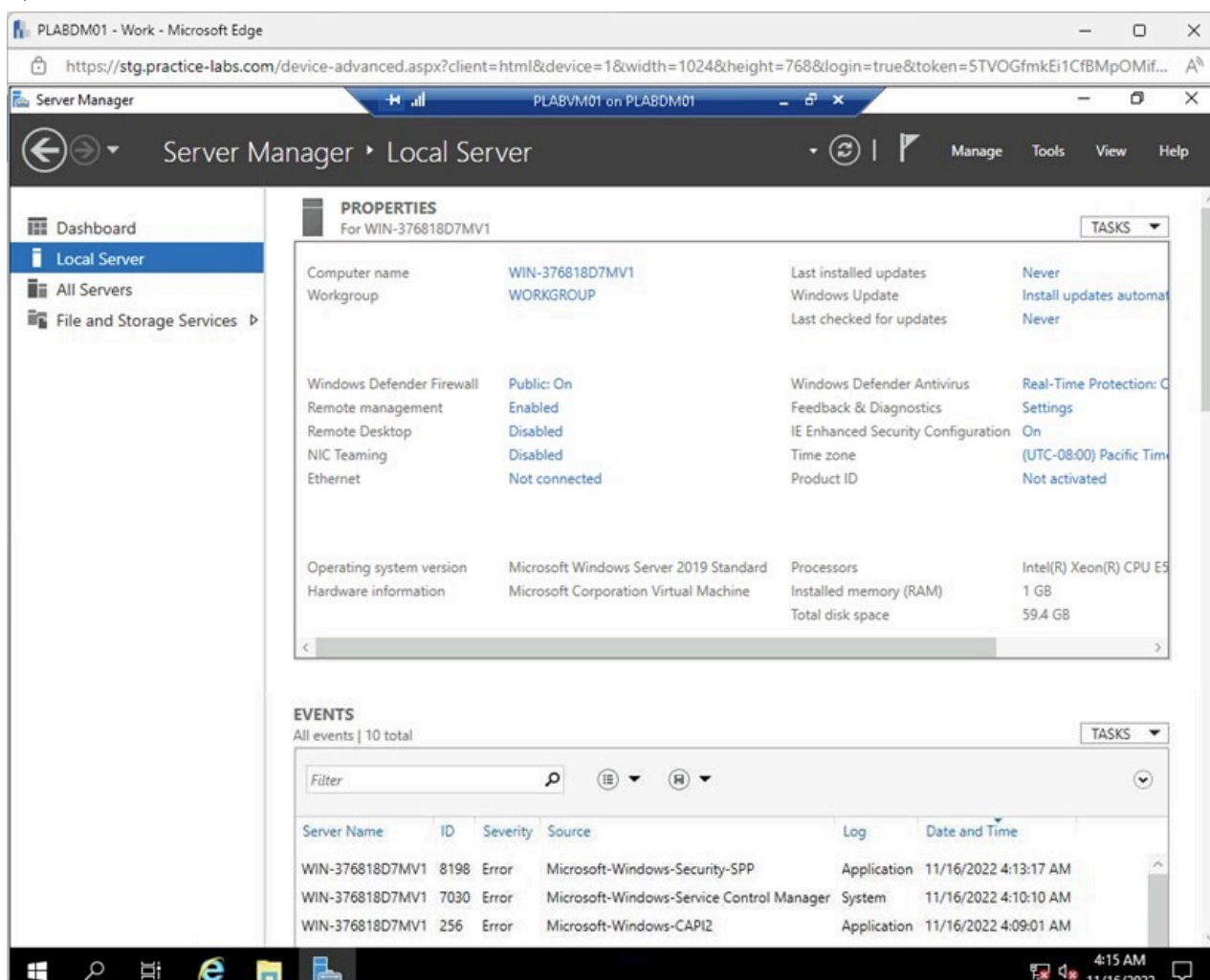


Figure 1.69 Screenshot of PLABDM01: Displaying the PLABVM01 on PLABDM01 window. Microsoft Corporation Virtual Machine Hardware information is present on the Server Manager > Local Server screen.

## Step 32

In the **Server Manager > Local Server** screen, under the **PROPERTIES** of the Local Server next to **Computer name**, select the **WIN-\*\*\*\*** hyperlink.

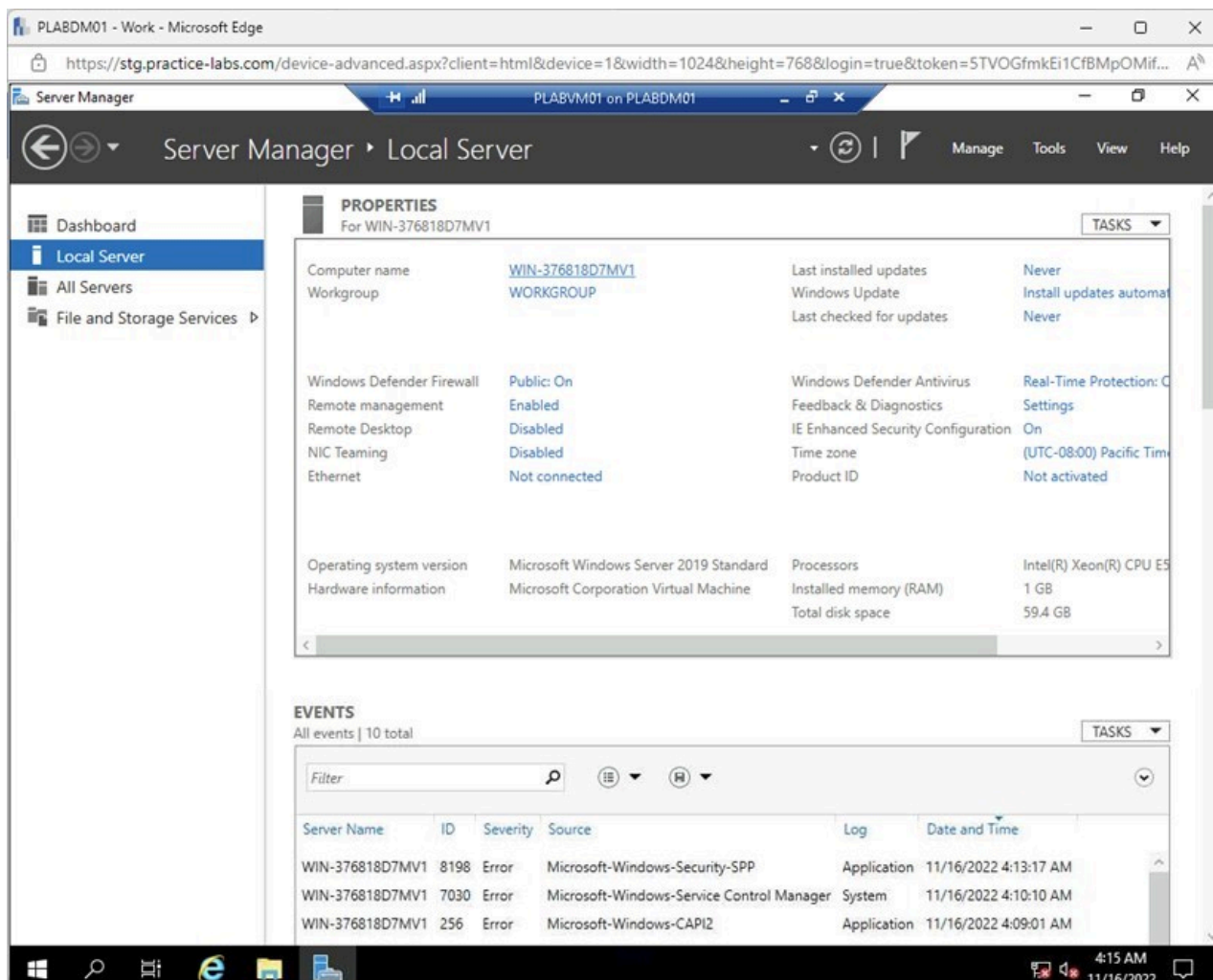


Figure 1.70 Screenshot of PLABDM01: Displaying the PLABVM01 on PLABDM01 window. WIN-\*\*\*\* Computer name is selected under the PROPERTIES of the Local Server.

**Note:** The default computer name will differ from the one shown in the screenshot.

## Step 33

In the **System Properties** dialog box, click **Change**.

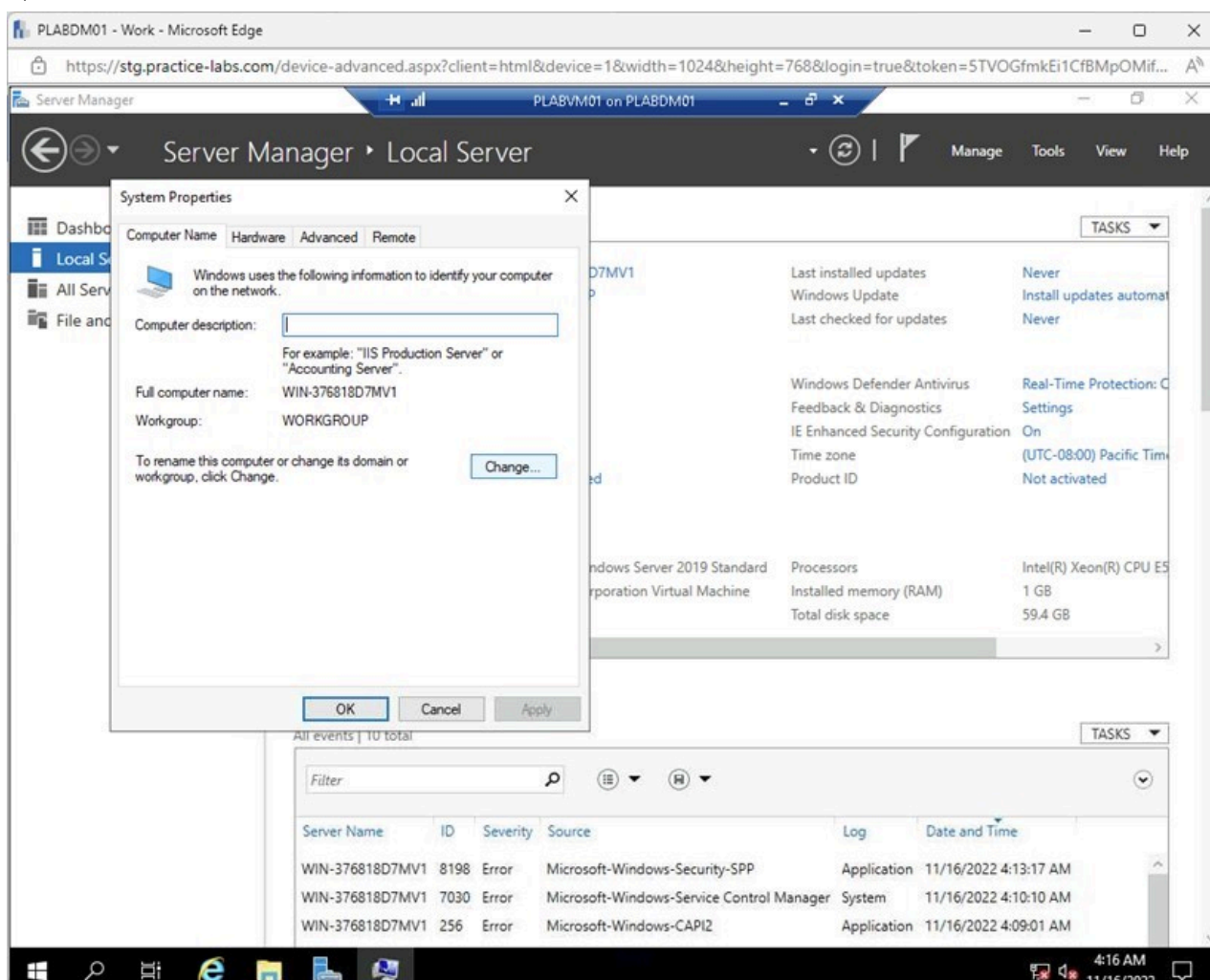


Figure 1.71 Screenshot of PLABDM01: Displaying the PLABVM01 on PLABDM01 window. Change button on the System Properties dialog box is selected.

## Step 34

In the **Computer Name/Domain Changes** dialog box, type the following under **Computer name**:

PLABVM01

Click **OK**.

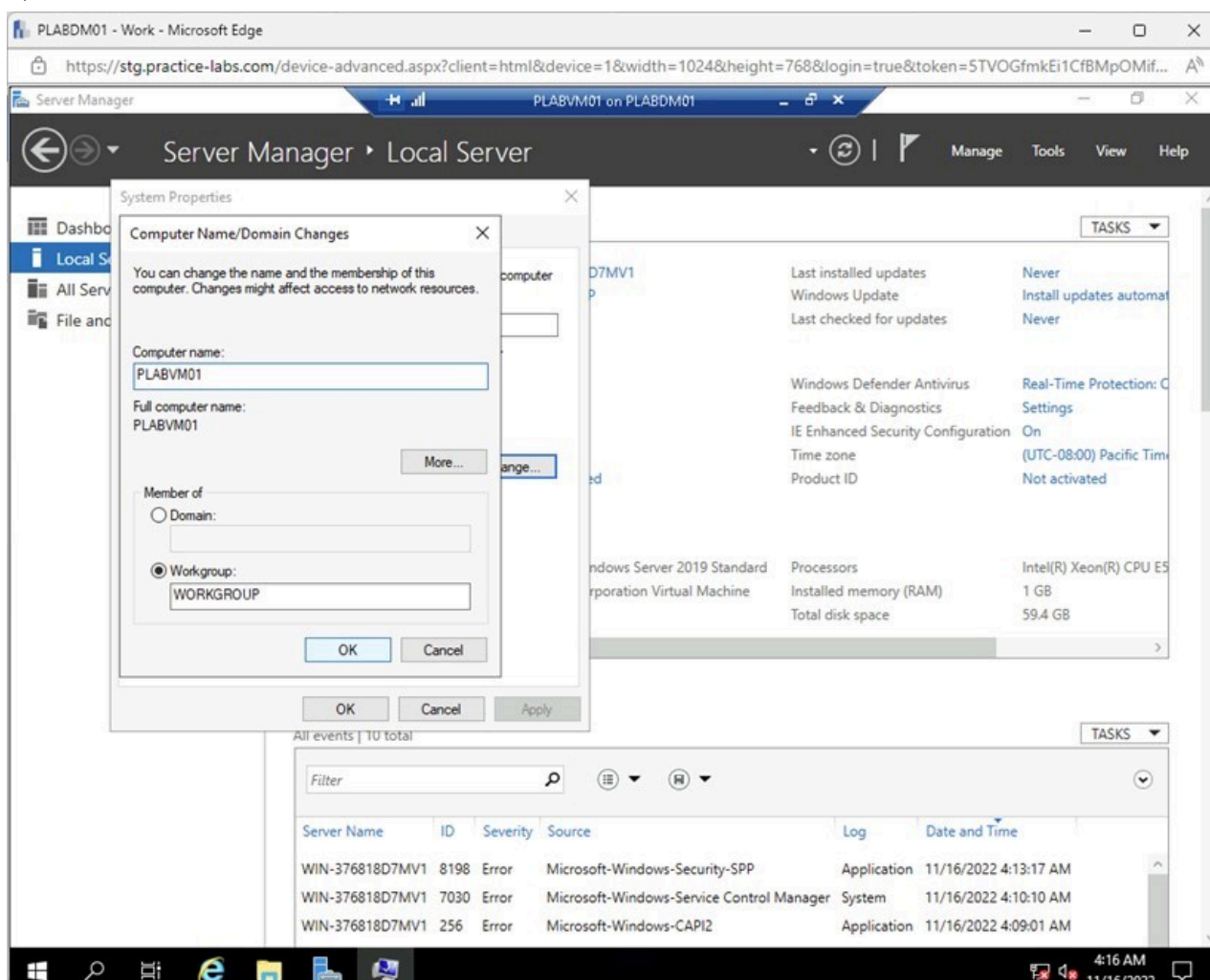


Figure 1.72 Screenshot of PLABDM01: Displaying the PLABVM01 on PLABDM01 window. The required Computer name is entered in the Computer Name/Domain Changes dialog box, and the OK button is highlighted.

## Step 35

In the **Computer Name/Domain Changes** pop-up window, click **OK**.

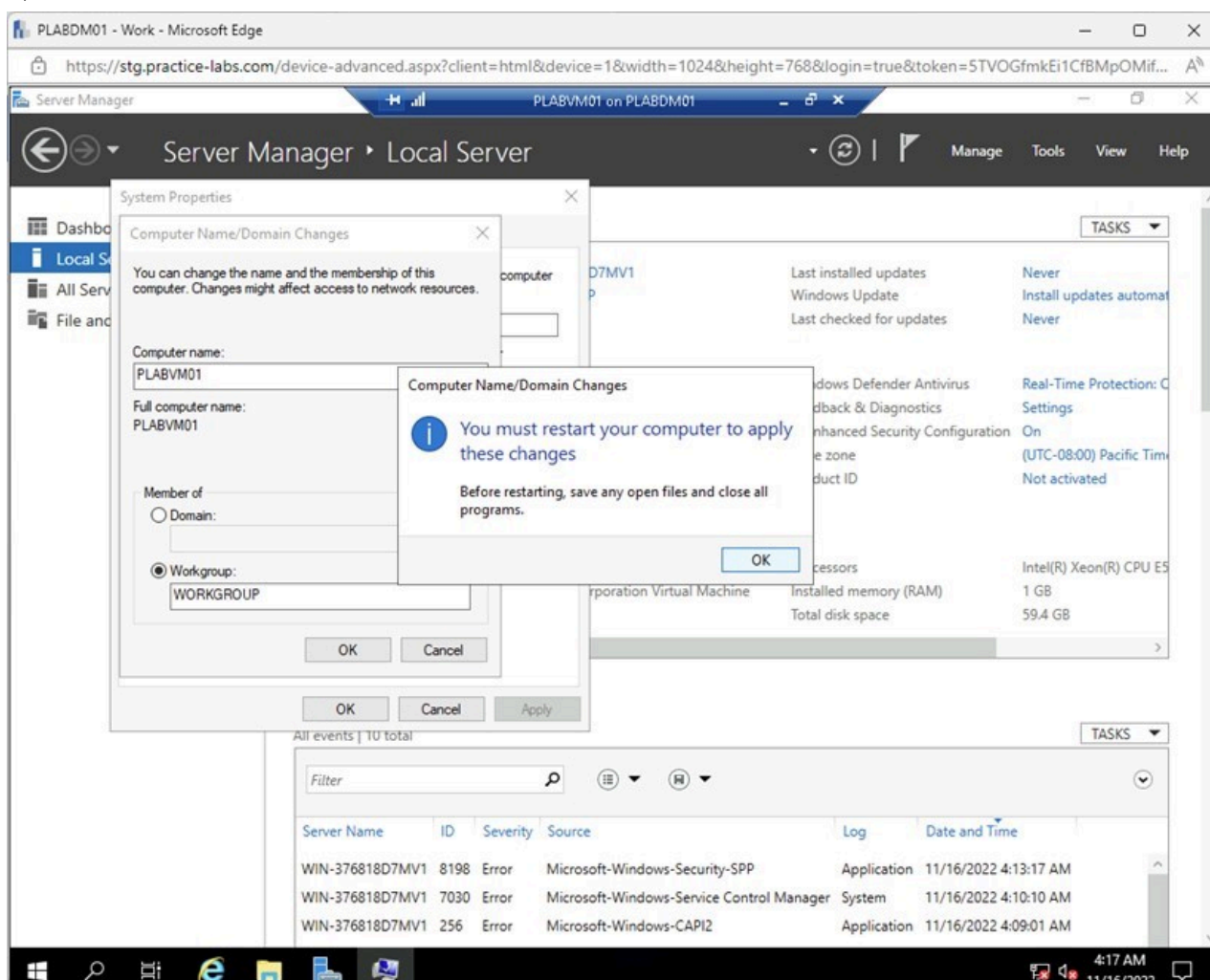


Figure 1.73 Screenshot of PLABDM01: Displaying the PLABVM01 on PLABDM01 window. OK is selected on the Computer Name/Domain Changes pop-up window.

## Step 36

In the **System Properties** dialog box, click **Close**.

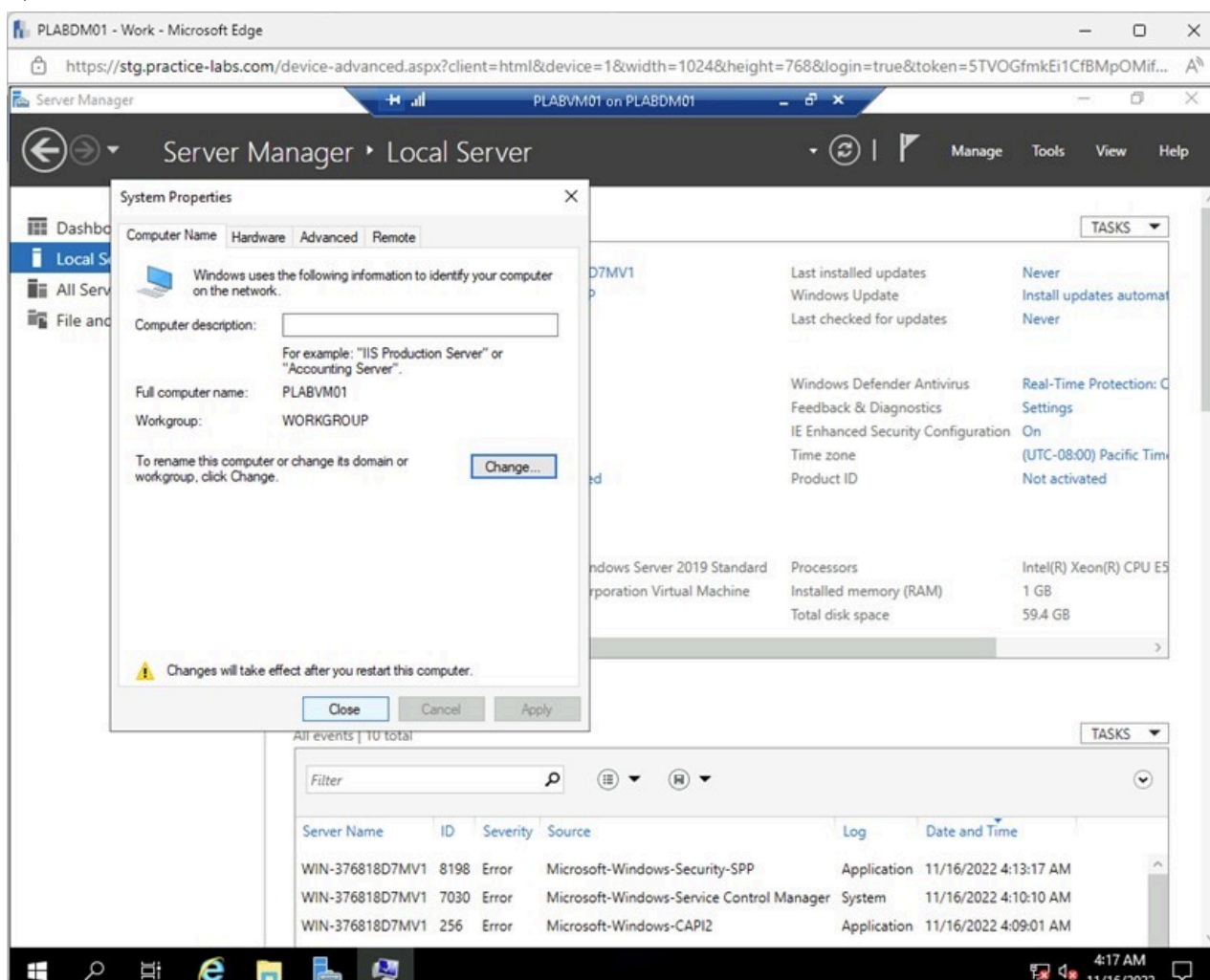


Figure 1.74 Screenshot of PLABDM01: Displaying the PLABVM01 on PLABDM01 window. Close button on the System Properties dialog box is selected.

## Step 37

In the **Microsoft Windows** pop-up box, click **Restart Now**.

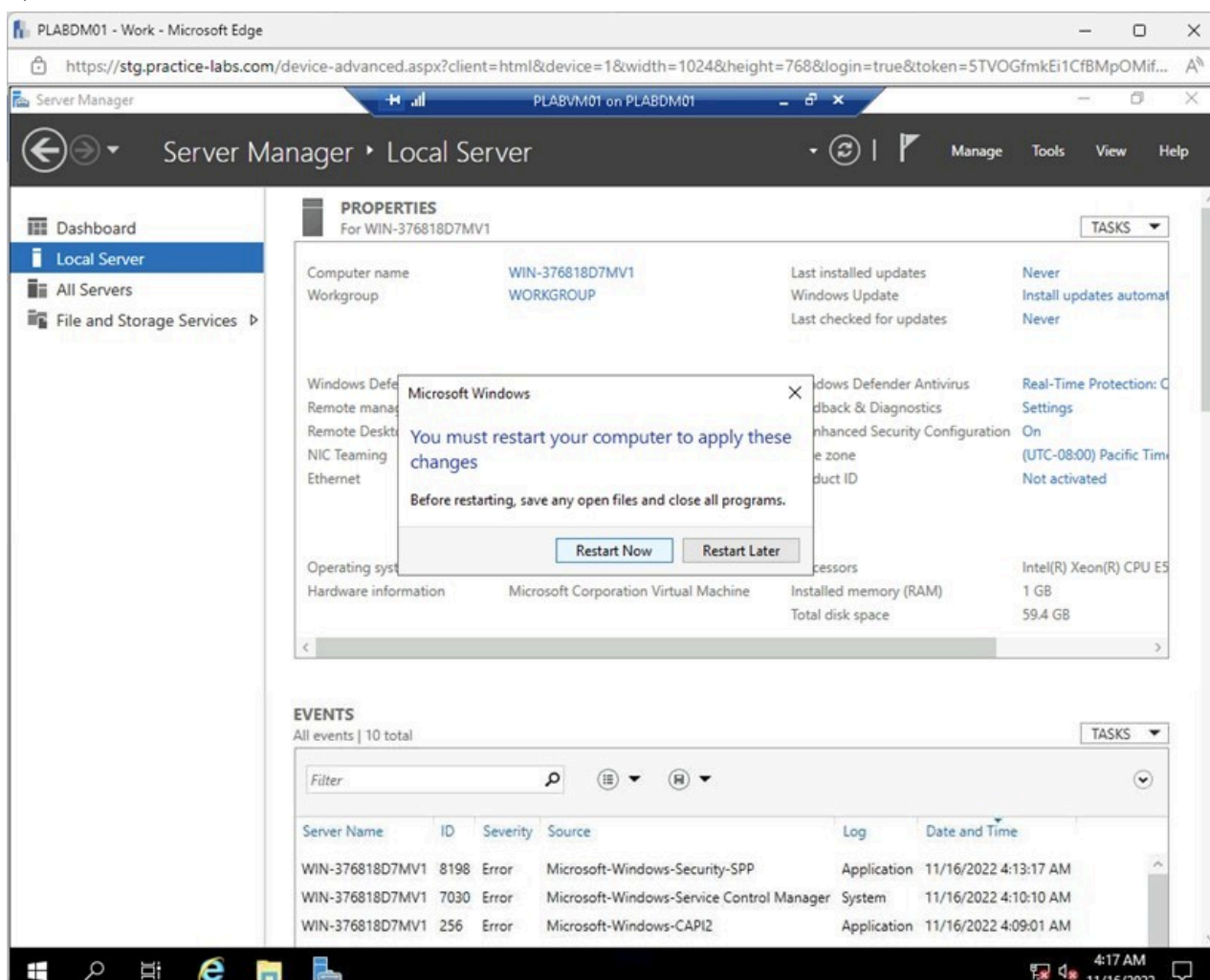


Figure 1.75 Screenshot of PLABDM01: Displaying the PLABVM01 on PLABDM01 - Virtual Machine Connection window. Restart Now button on the Microsoft Windows pop-up box is selected.

## Step 38

In the **PLABVM01 on PLABDM01** window, click the **Restore Down** icon.

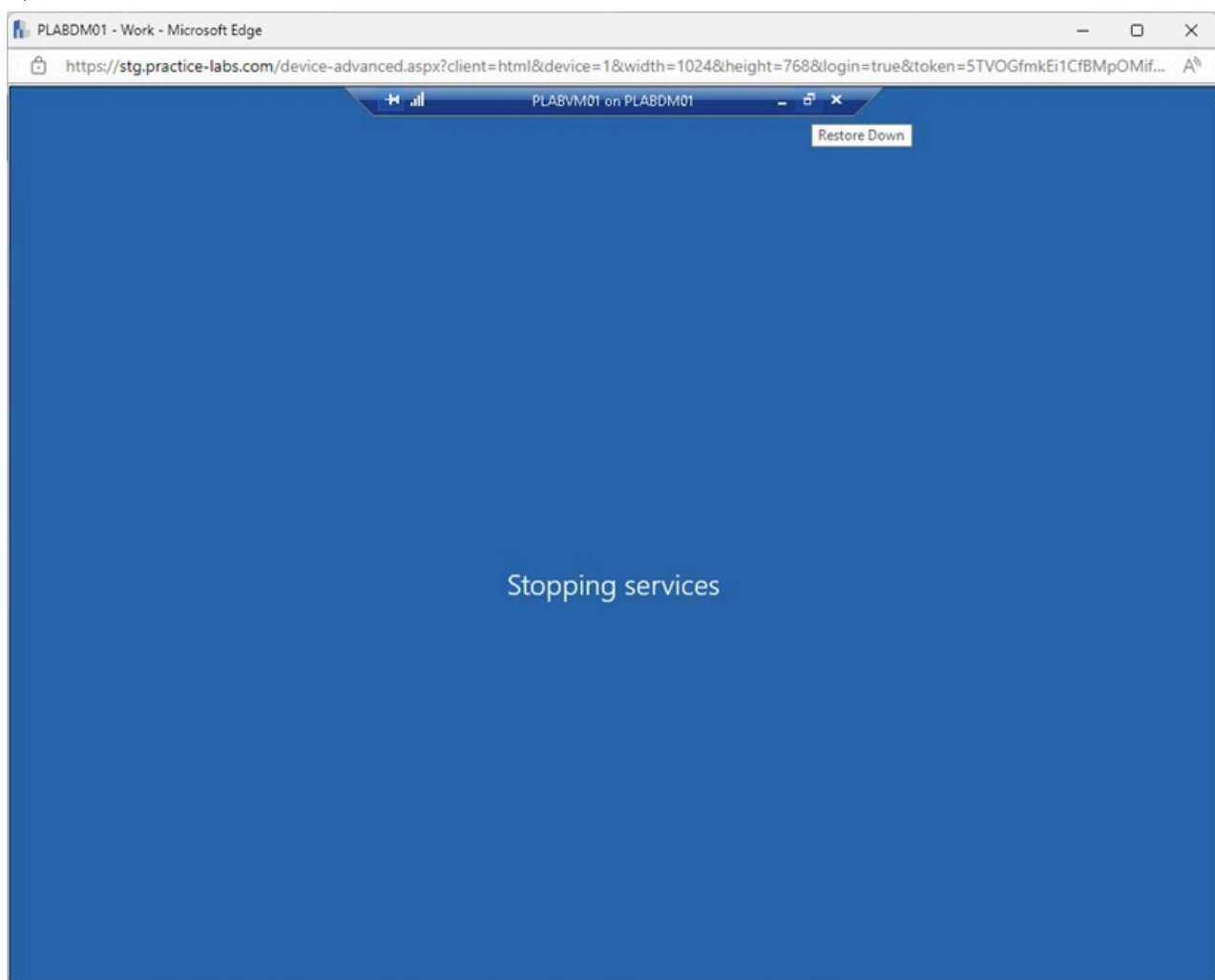


Figure 1.76 Screenshot of PLABDM01: Displaying clicking the Restore Down icon in the PLABVM01 on PLABDM01 -window.

## Step 39

In the **PLABVM01 on PLABDM01 - Virtual Machine Connection** window, click the **Ctrl+Alt+Delete** actions icon.

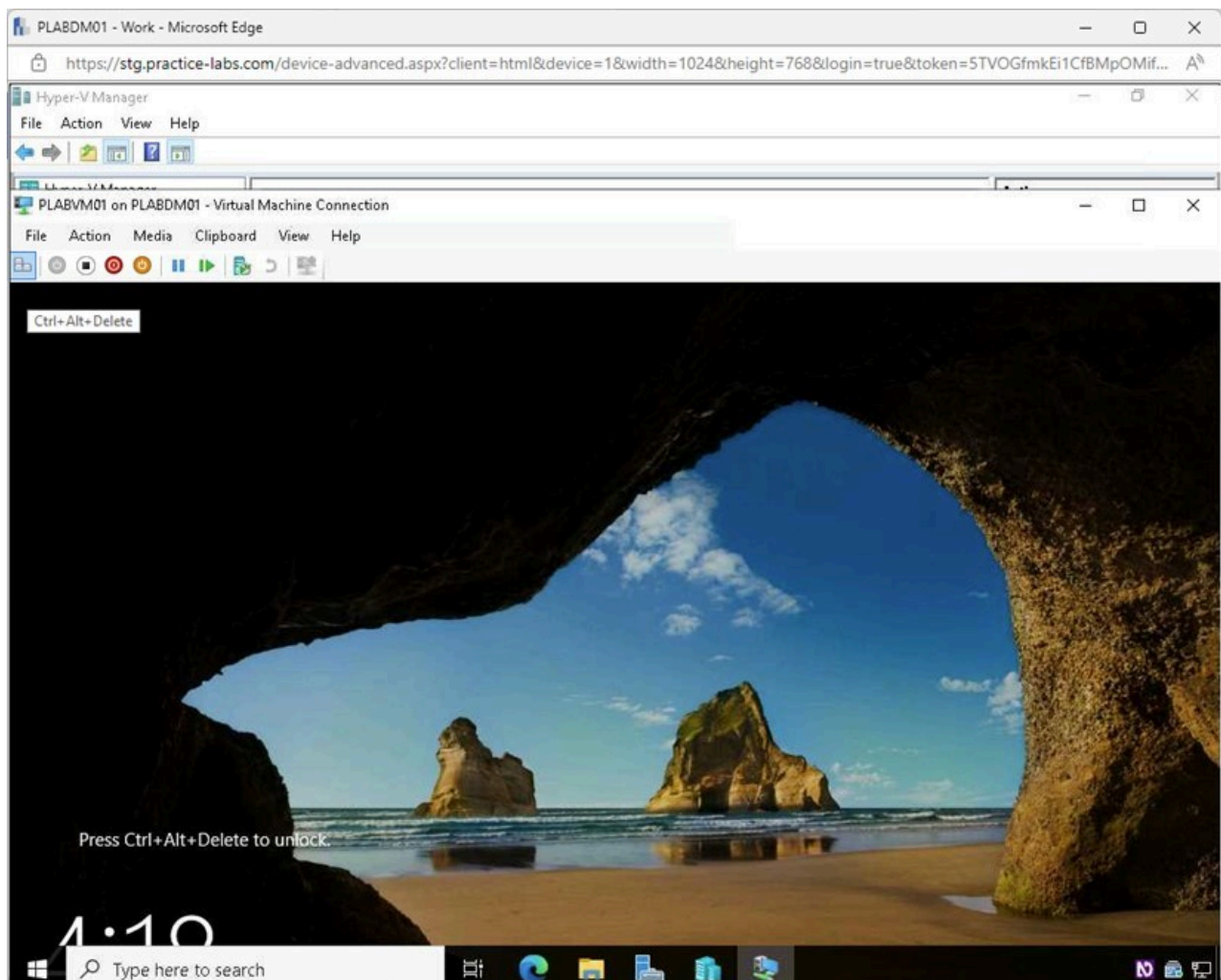


Figure 1.77 Screenshot of PLABDM01: Displaying selecting the Ctrl+Alt+Delete actions icon in the PLABVM01 on PLABDM01 - Virtual Machine Connection window.

## Step 40

In the **PLABVM01 on PLABDM01 - Virtual Machine Connection** window, log in as **Administrator**.

Type the following password and press **Enter**:

Passw0rd

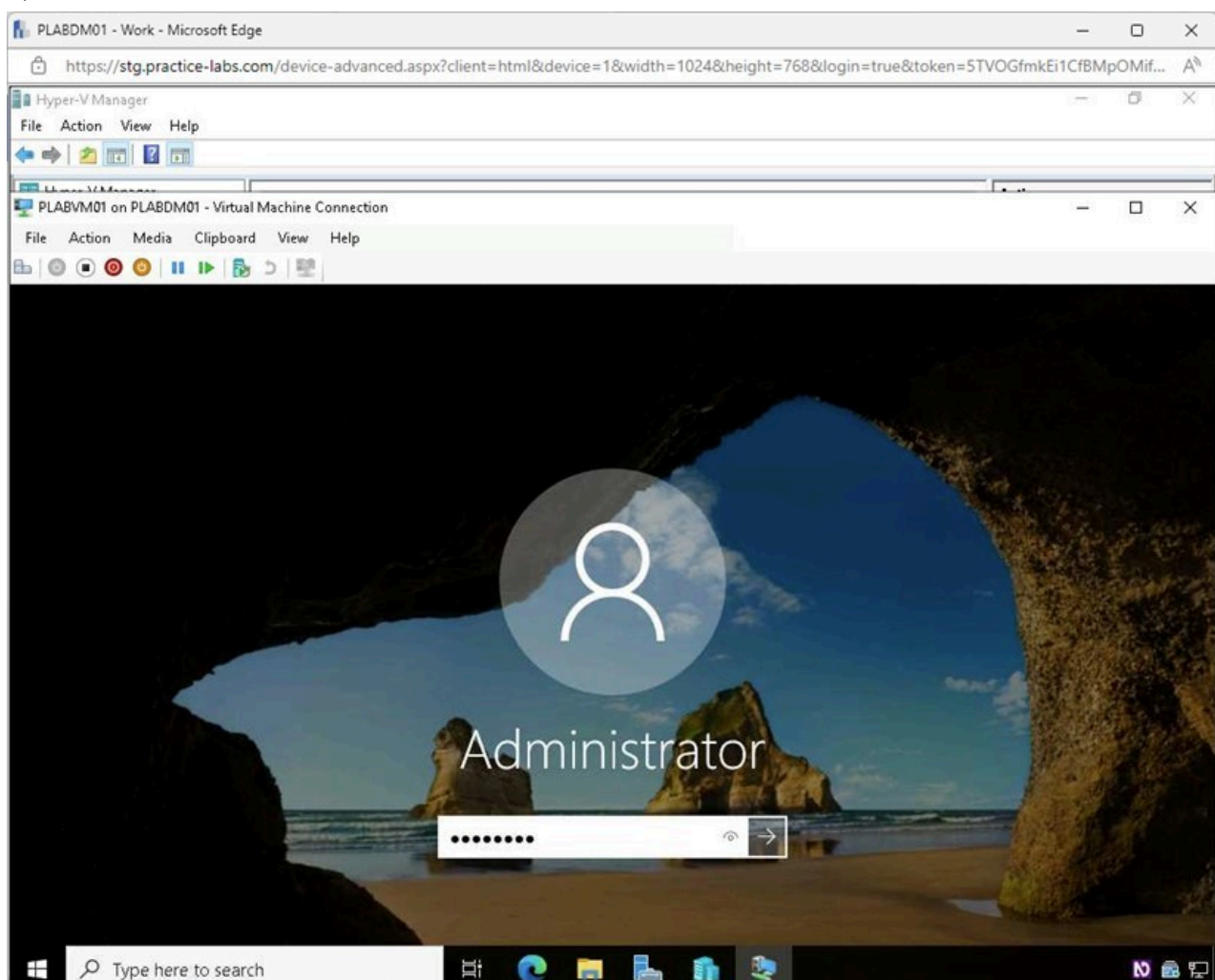


Figure 1.78 Screenshot of PLABDM01: Displaying typing the required password for the Administrator account in the PLABVM01 on PLABDM01 - Virtual Machine Connection window.

## Step 41

In the **PLABVM01 on PLABDM01 - Virtual Machine Connection** window, click the **Maximize** icon.

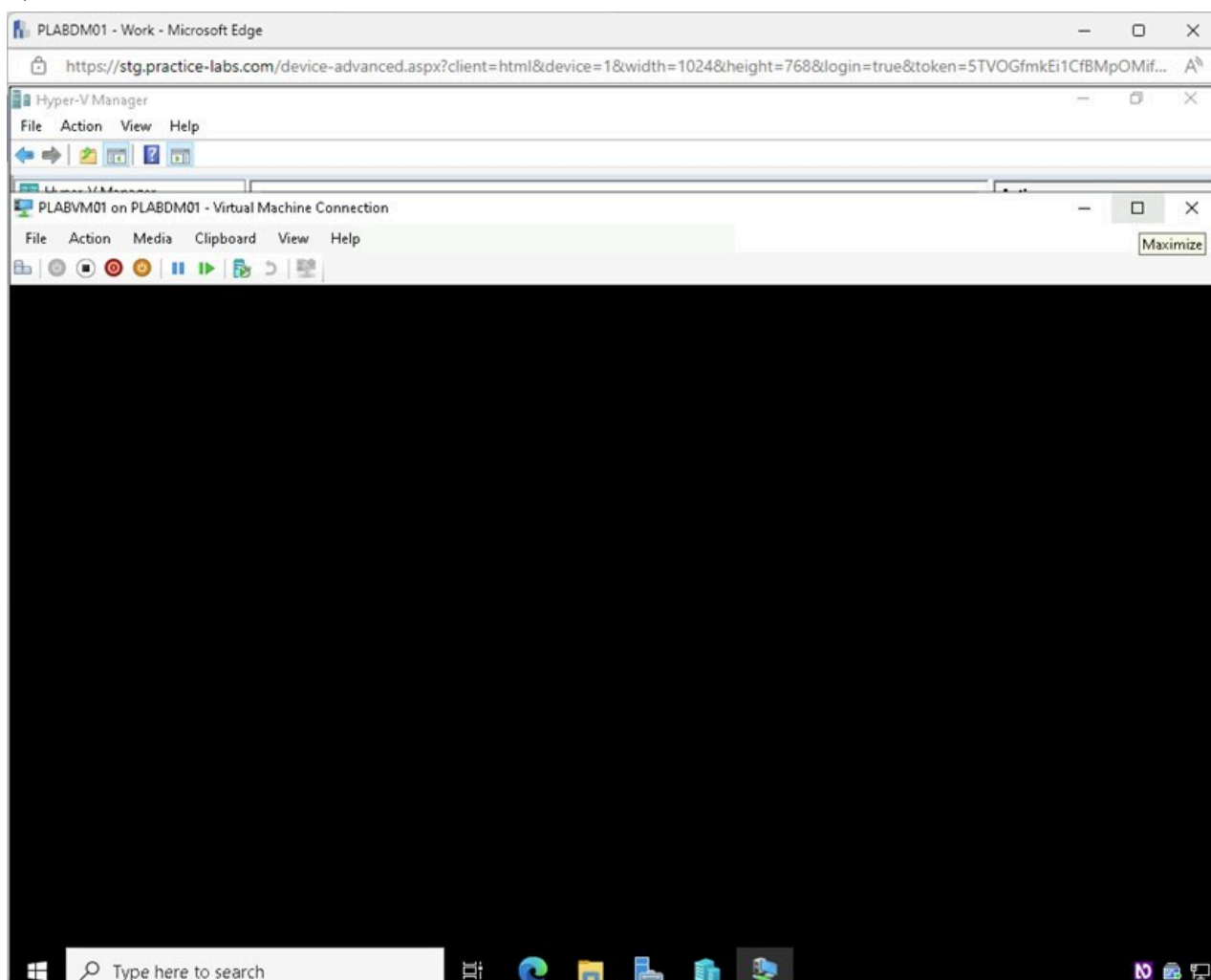


Figure 1.79 Screenshot of PLABDM01: Displaying selecting the Maximize icon in the PLABVM01 on PLABDM01 - Virtual Machine Connection window.

## Step 42

In the **PLABVM01 on PLABDM01** window, close the **Server Manager** pop-up box.

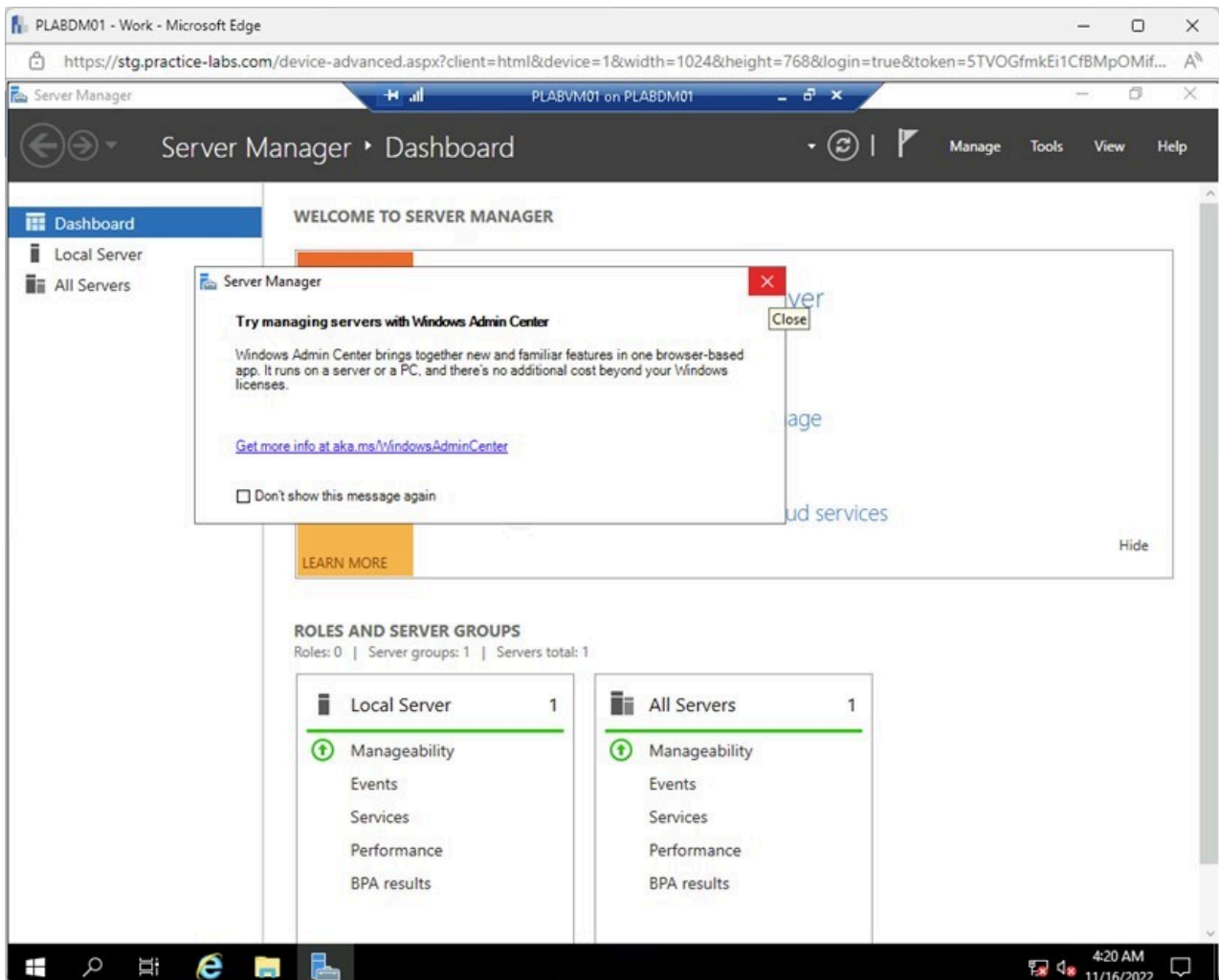


Figure 1.80 Screenshot of PLABDM01: Displaying closing the Server Manager pop-up box in the PLABVM01 on PLABDM01 window.

**Note:** You may need to click and drag the windows down to gain access to the Close icon.

## Step 43

In the **PLABVM01 on PLABDM01** window, select **Local Server** in the left pane on the **Server Manager** screen.

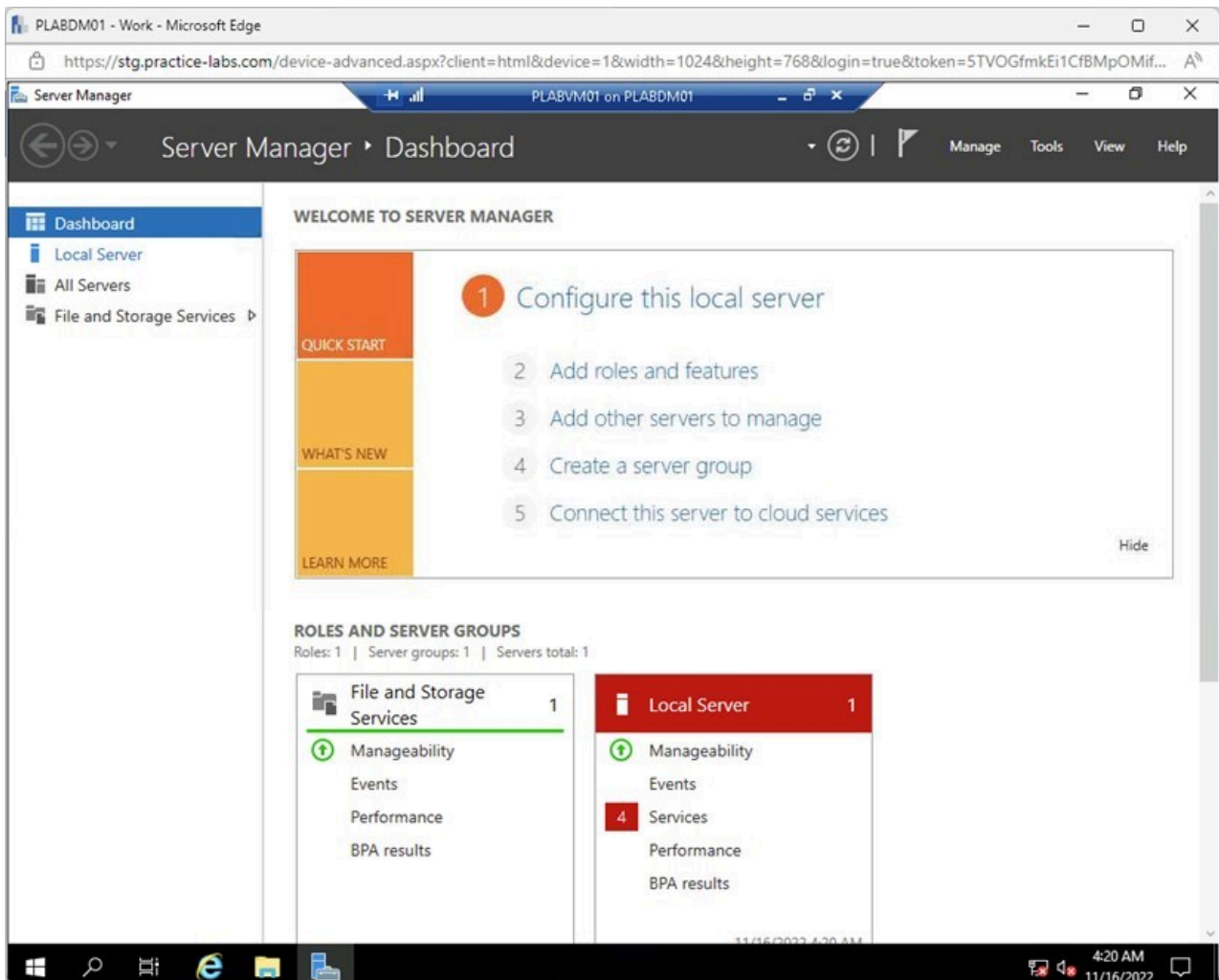


Figure 1.81 Screenshot of PLABDM01: Displaying the PLABVM01 on PLABDM01 window. Local Server is selected on the Server Manager screen.

## Step 44

In the **Server Manager > Local Server** screen, under the **PROPERTIES** of the Local Server next to **Computer name**, verify that the name is **PLABVM01**.

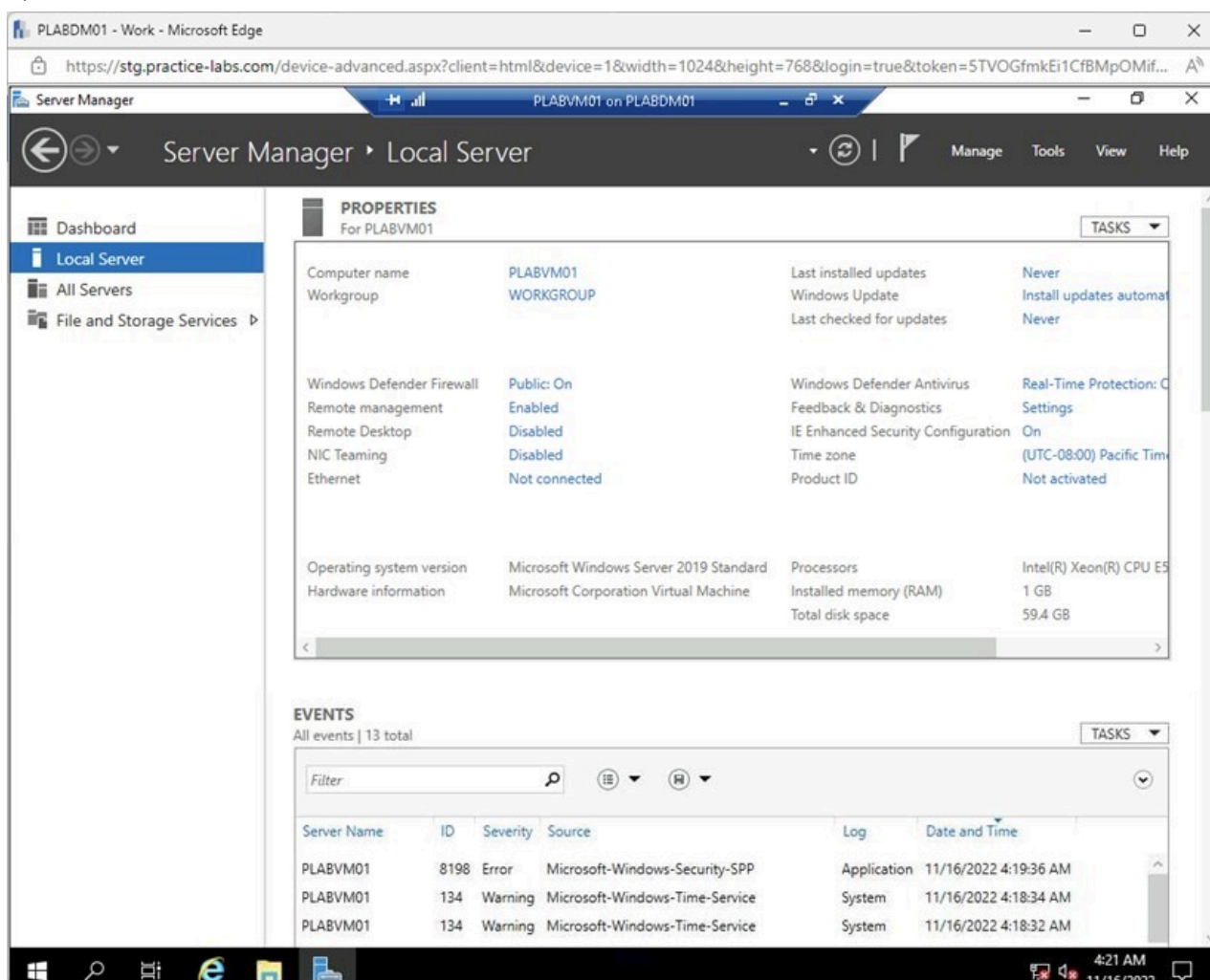


Figure 1.82 Screenshot of PLABDM01: Displaying the PLABVM01 on PLABDM01 window. Computer Name is shown as PLABVM01 on the Server Manager > Local Server screen.

## Step 45

Click the **Close** icon on the **PLABVM01 on PLABDM01** connection bar.

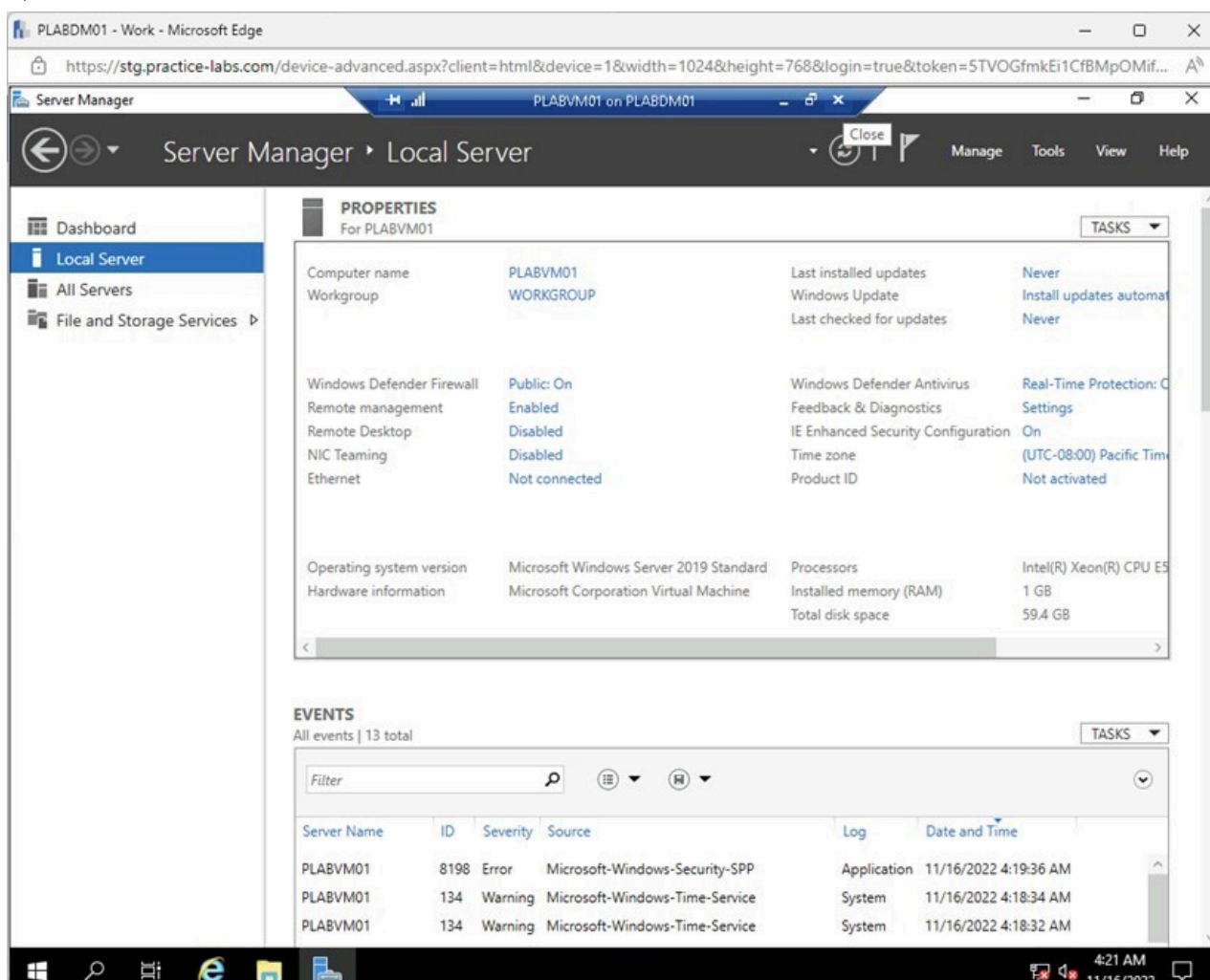


Figure 1.83 Screenshot of PLABDM01: Displaying clicking the Close icon on the PLABVM01 on PLABDM01 connection bar.

## Step 46

In the **Hyper-V Manager** window, under the **Virtual Machines** pane, select and right-click on **PLABVM01** and select **Turn Off**.

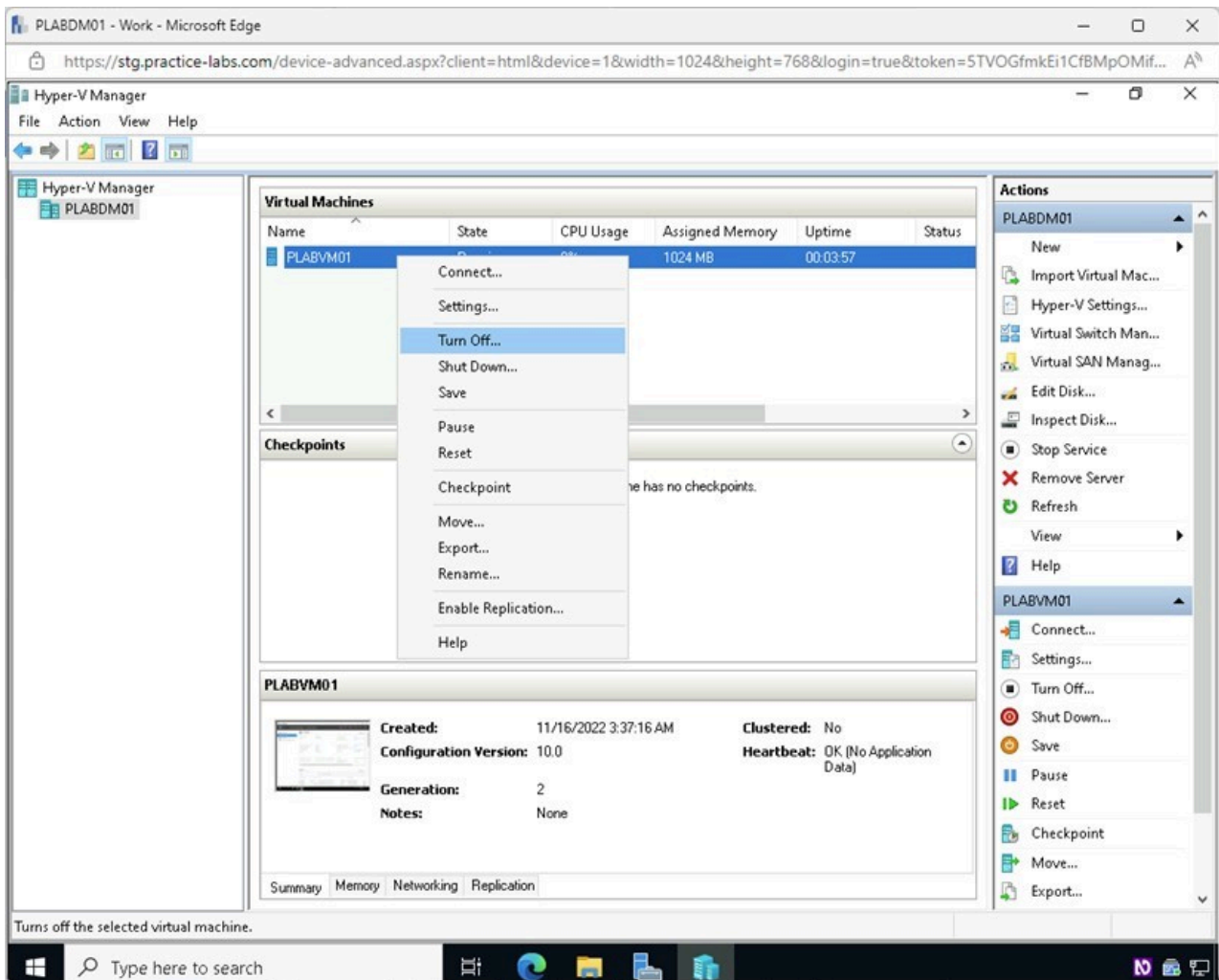


Figure 1.84 Screenshot of PLABDM01: Displaying the Hyper-V Manager window. PLABVM01 virtual machine is right-clicked and Turn Off selected from the context menu.

Close all open windows.

## Exercise 2 - Cloud Models

Cloud computing consists of different cloud models depending on the requirements of the company.

In this exercise, the different types of cloud models will be discussed and how they can be applied in an IT infrastructure. In addition, the benefits of these different models will also be covered.

## Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Different Cloud Computing Models

## Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

## Different Cloud Computing Models

### *Private Cloud*

A **Private Cloud** Infrastructure is implemented by a company using its own hardware. The specific computer hardware needs to be purchased, implemented, and maintained by the company.

The computer hardware that needs to be procured by a company will typically consist of several components, including servers, switches, and routers. This may have a massive financial impact on the company.

Hardware can be situated locally on the company premise or in a Data center of a third-party provider.

A Private Cloud is used to create a pool of computer resources that can then be accessed and utilized by the company.

Different types of Private Cloud software are available from different vendors, including Microsoft, Red Hat, and Suse. This software is designed to implement a private cloud infrastructure and provide the needed services to access the resources allocated to it.

### *Public Cloud*

With a Public Cloud Infrastructure, the computer resources are procured, implemented, and maintained by a third-party provider, for example, Microsoft. The computing hardware is situated in Data centers across the world and is accessible through the internet and can be accessed by anybody who has purchased these specific services or resources.

The resources that are provided include computing, networking, and storage.

Cloud services are provided, which can then be utilized by companies for specific purposes. Companies that access these services pay for the computing resources being utilized but does not need to maintain the hardware in the backend.

Below is an image of the Azure portal web interface. This interface is used to create and manage cloud services.

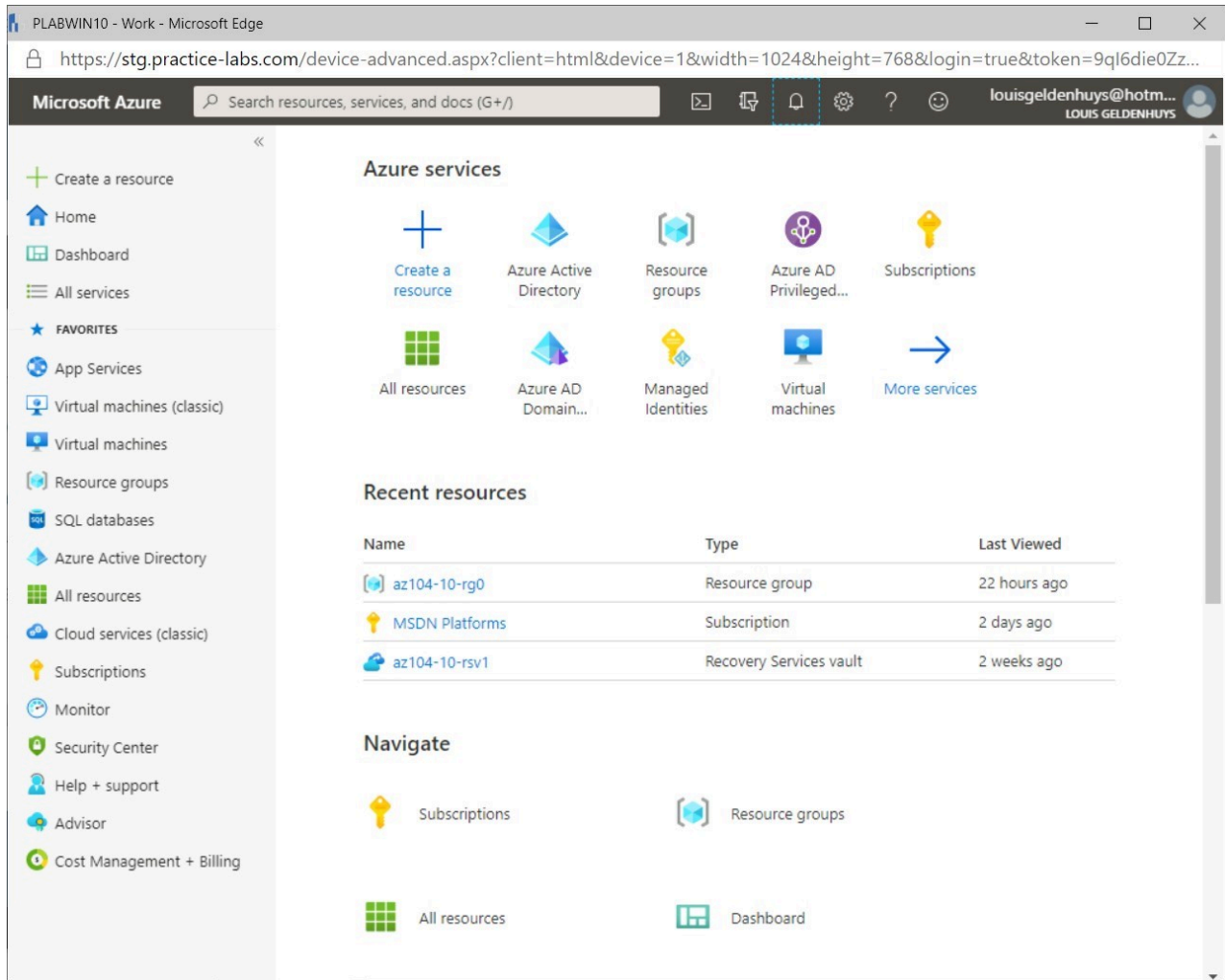


Figure 2.1 Screenshot of PLABWIN10: Displaying the Azure portal web interface.

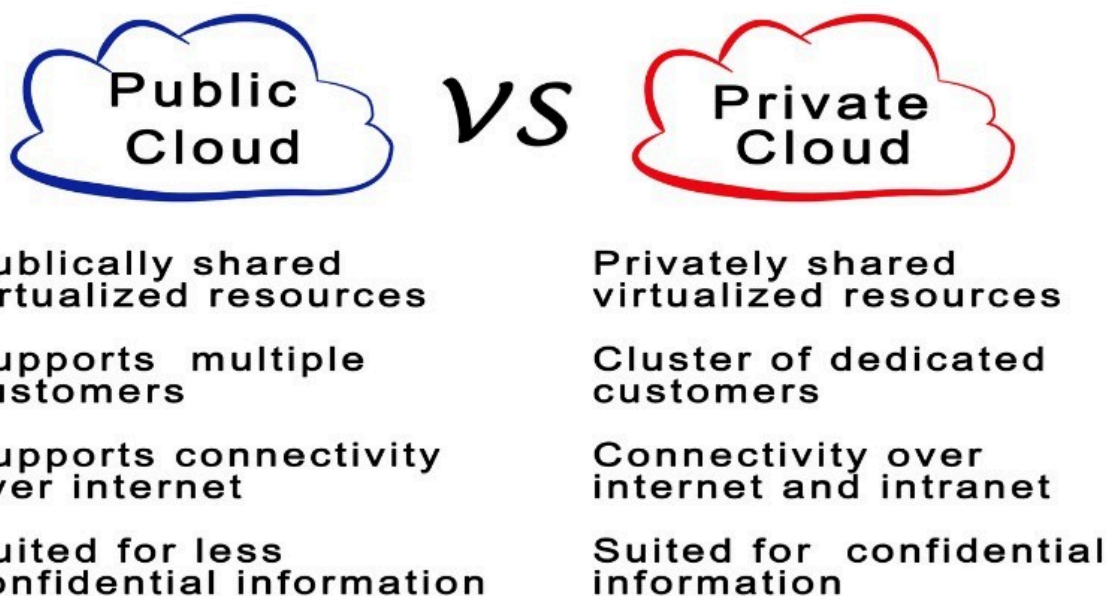


Figure 2.2 Screenshot of PLABWIN10: Displaying the differences between Public and Private Cloud.

## Hybrid Cloud

A Hybrid Cloud Infrastructure is a combination of a public and private cloud where the services that are provided are shared between the different infrastructures. The public and private infrastructures are connected to each other through the internet to ensure the resources are available.

This type of cloud infrastructure is used when the demand for resources fluctuates, and the Private Cloud cannot meet these demands. The Public Cloud's Infrastructure is then used to seamlessly scale the resources to meet these demands.

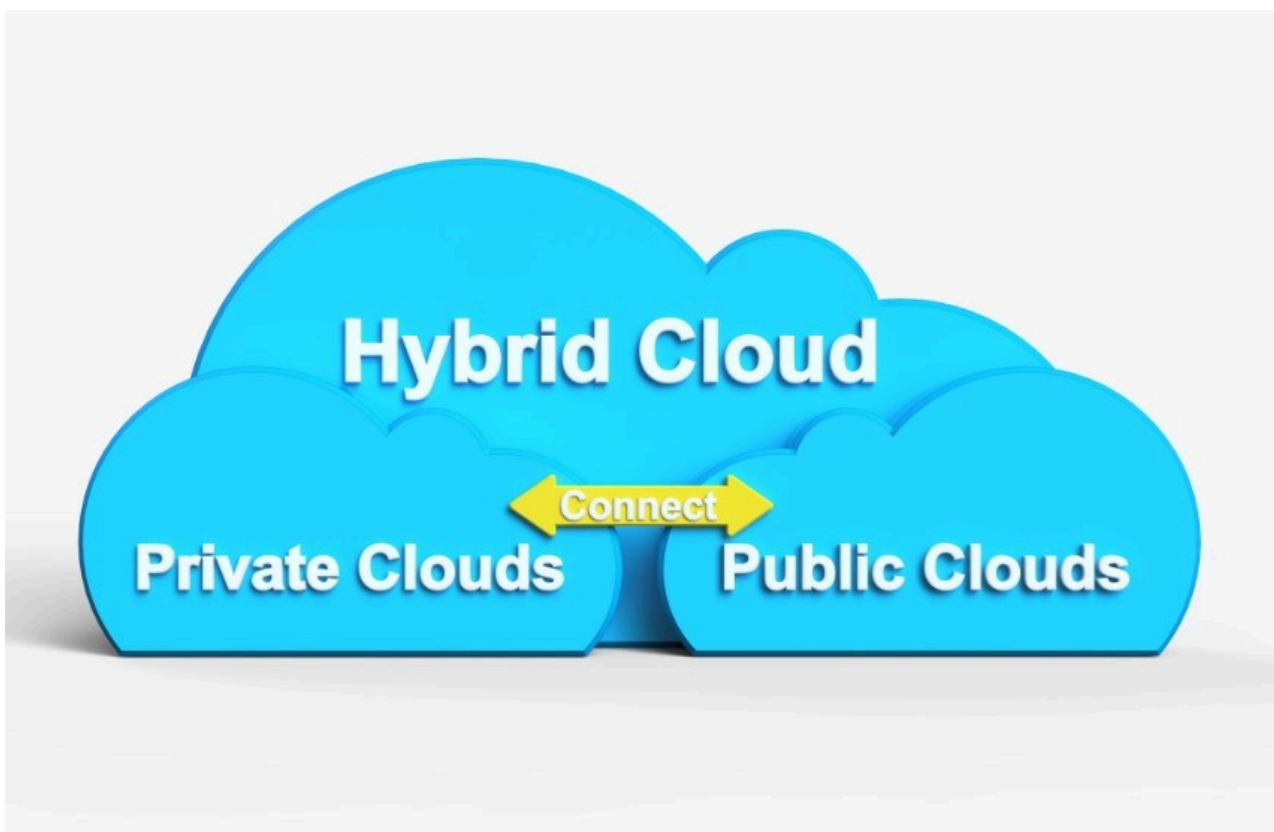


Figure 2.3: Displaying the Hybrid Cloud model, showing it is linked to Private and Public Clouds.

Keep all devices that you have powered on in their current state and proceed to the review section.

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## Review

Well done, you have completed the **Server Virtualization Concepts** Practice Lab.

# Summary

You completed the following exercises:

- Exercise 1 - Install and Configure a Server-side Virtualization
- Exercise 2 - Cloud Models

You should now be able to:

- Install the Microsoft Hyper-V
- Virtual Switching Concepts
- Create a Virtual Machine
- Install Windows Server on a Virtual Machine

You should now have further knowledge of:

- Different Cloud Computing Models

# Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.